

# Building Knowledge Together: How Open Curation is Transforming Natural Products Research

Natural Products in the 21st Century, Provinciehuis Conference Center, Leuven, Belgium

Adriano Rutz,  2026-02-11

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## Descriptive & Botanical Era

★ 1804: Friedrich Wilhelm Adam Sertürner publishes about the first “*vegetal alcali*” or *alkaloid*, Morphine

★ 1813: Augustin Pyramus de Candolle publishes *Théorie Élémentaire de la Botanique*, where he introduces the word taxonomy and writes: “*Plant taxonomy would be the most useful guide to man in his search for new industrial and medicinal plants*”

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## Classical Isolation & Structural Elucidation Era

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★ 1945: Dorothy Hodgkin structural elucidation of *penicillin*, *first structure of a steroid*, *vitamin B<sub>12</sub>*, *insulin* (1964 Nobel prize)



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## Digital & Web Era

★ 1804: Friedrich Wilhelm Adam Sertürner publishes about the first “*vegetal alcali*” or *alkaloid*, Morphine

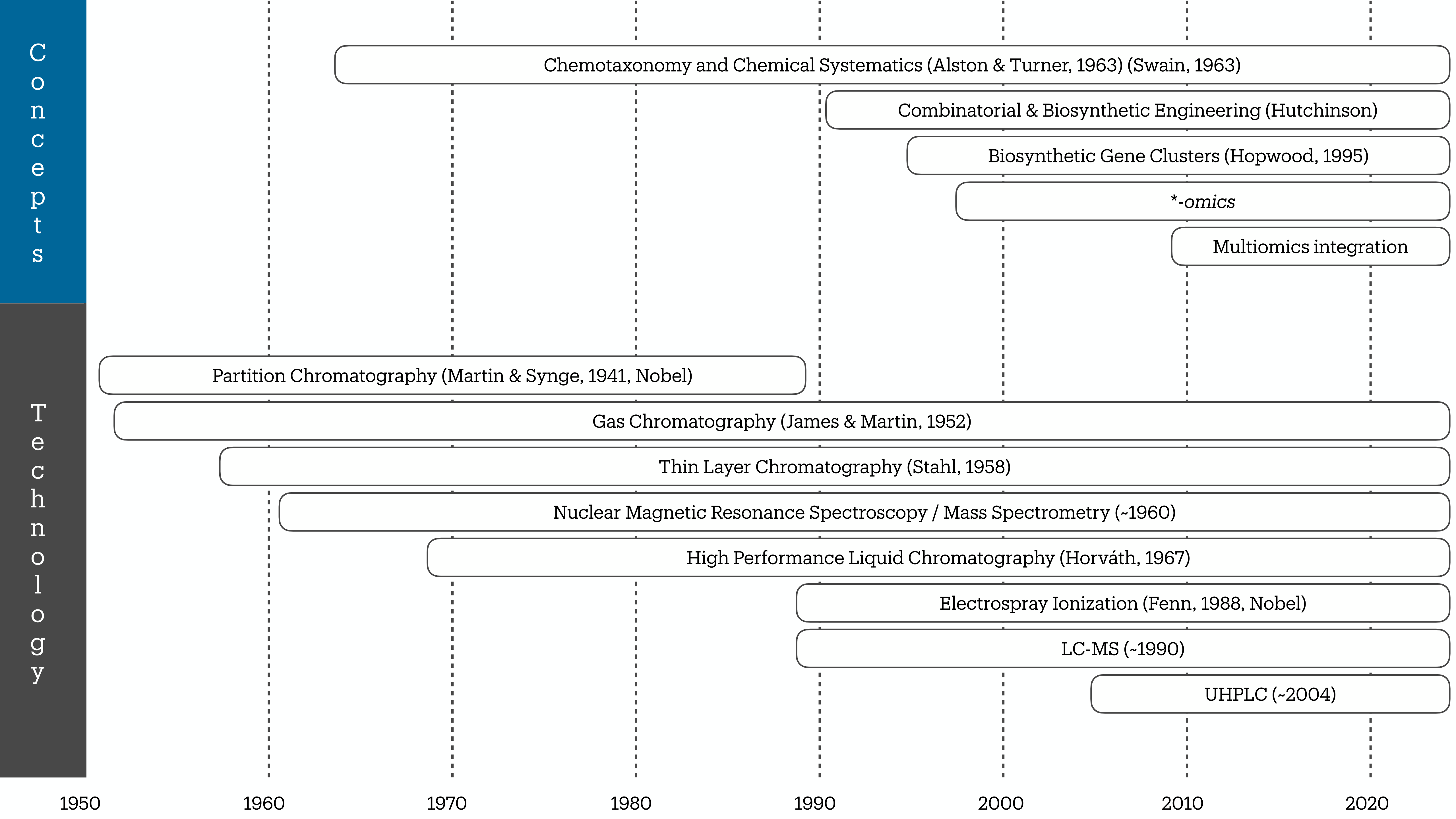
★ 1813: Augustin Pyramus de Candolle publishes *Théorie Élémentaire de la Botanique*, where he introduces the word taxonomy and writes: “*Plant taxonomy would be the most useful guide to man in his search for new industrial and medicinal plants*”

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★ 1993: Tim Berners-Lee opens the World Wide Web to the public

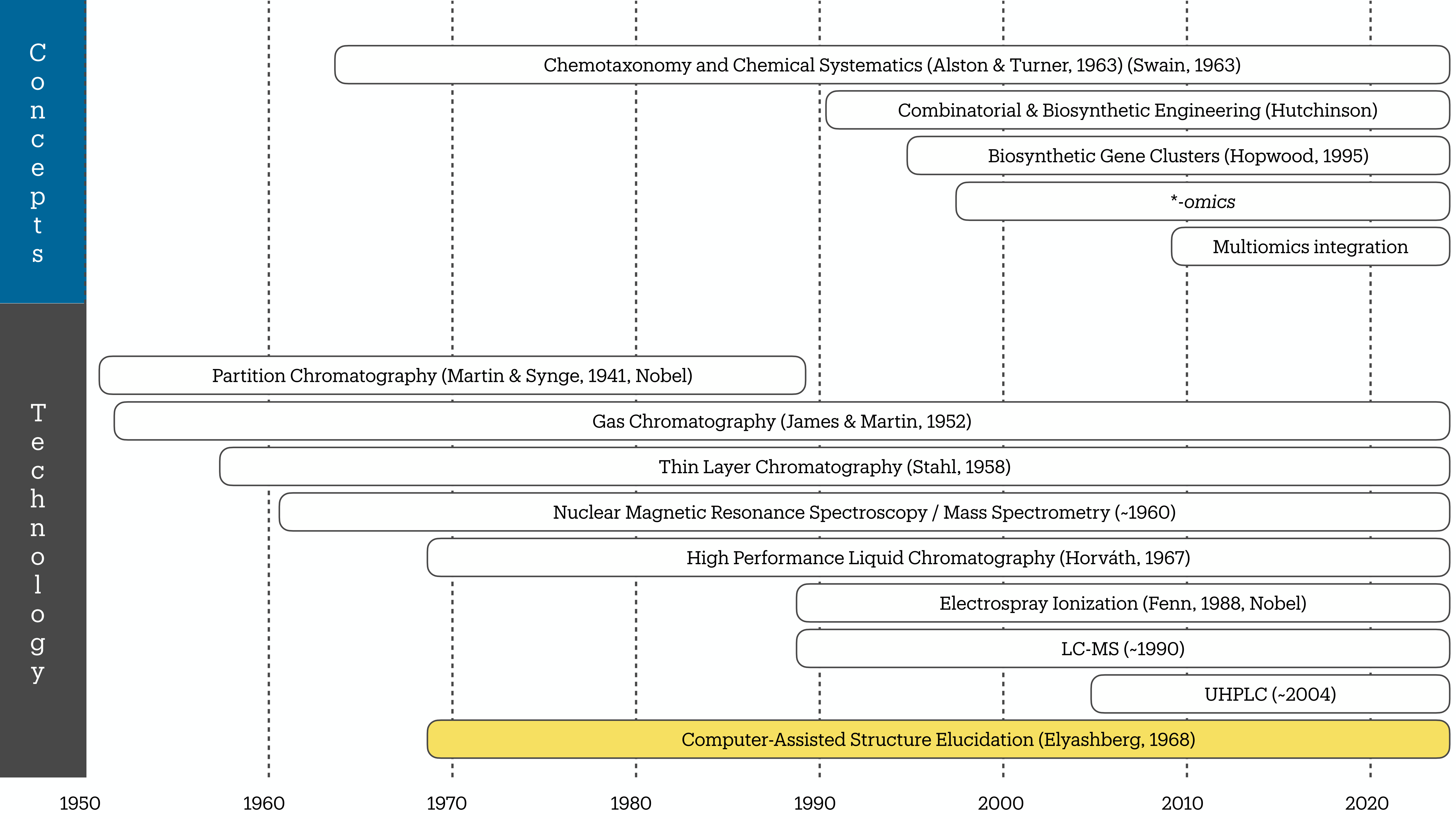


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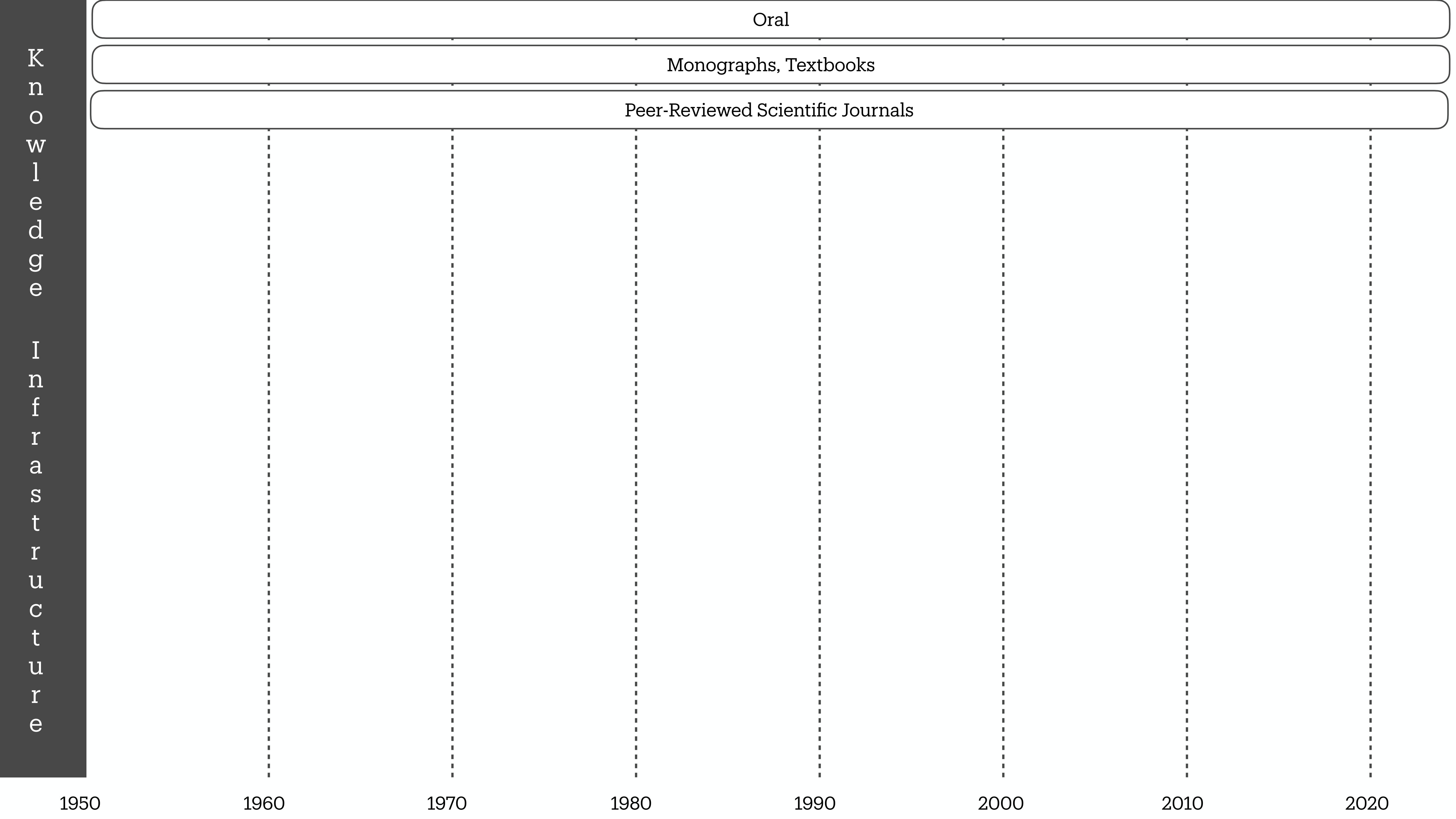
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1990

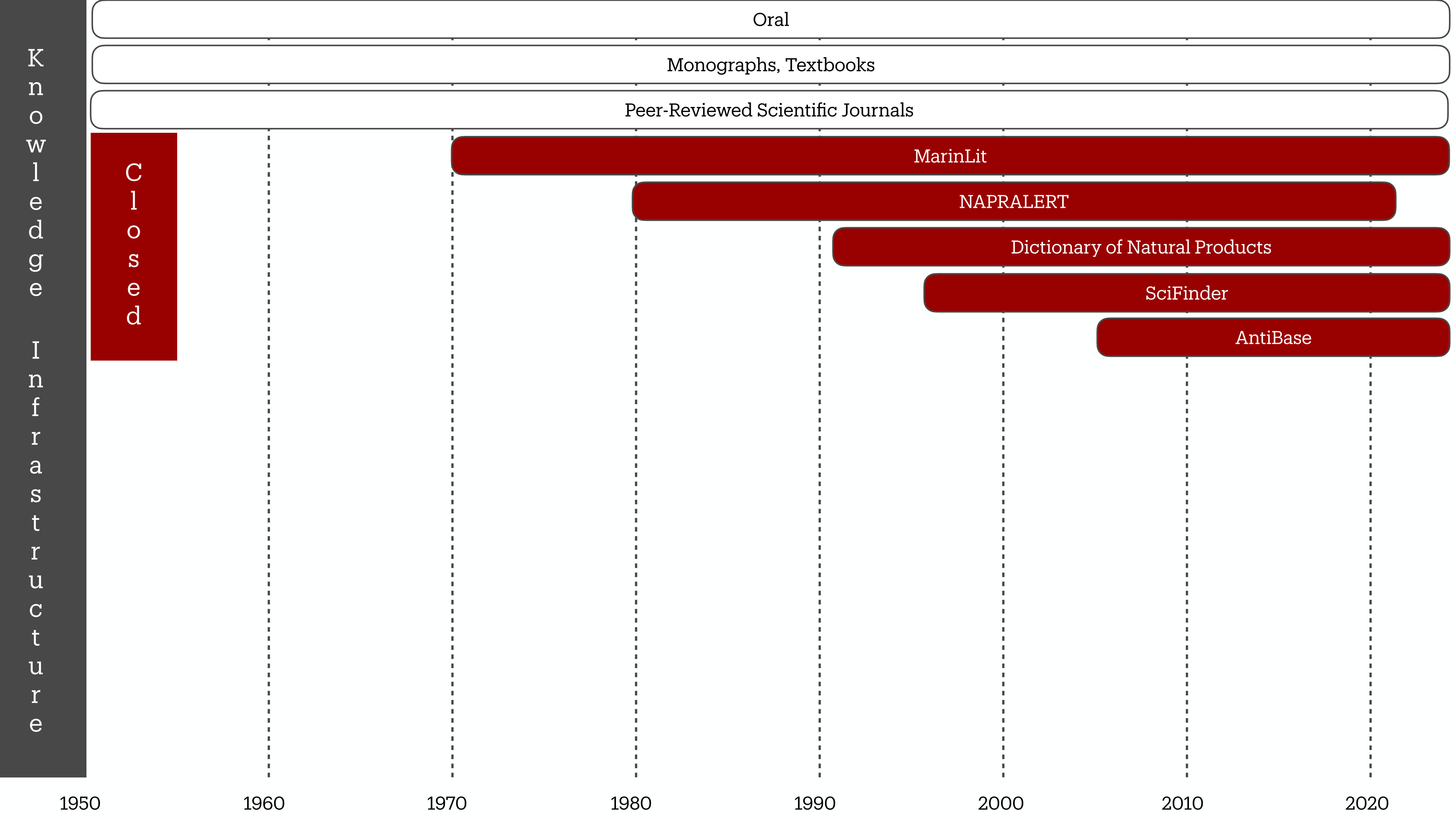
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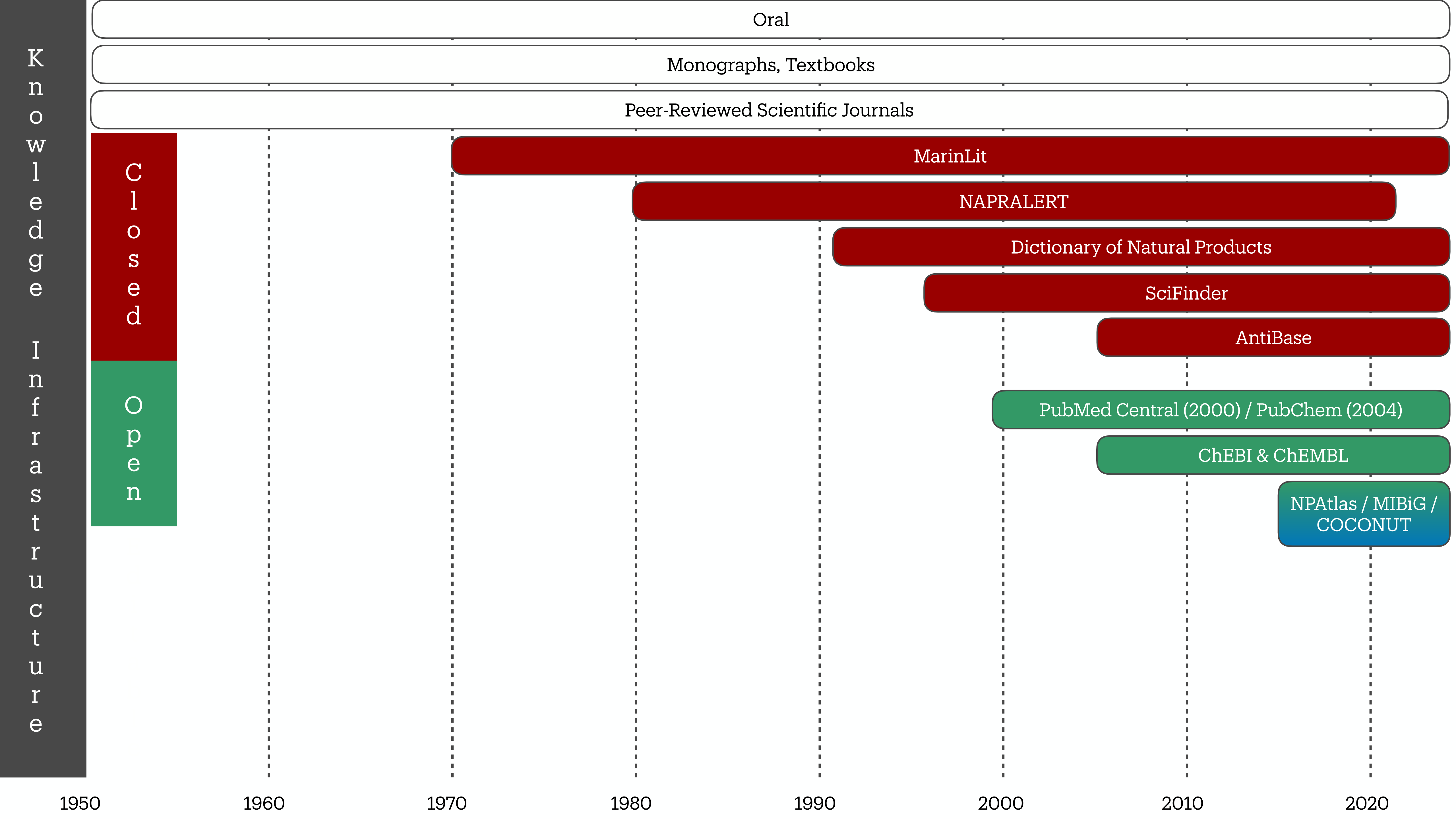
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Monographs, Textbooks

Peer-Reviewed Scientific Journals

MarinLit

NAPRALERT

Dictionary of Natural Products

SciFinder

AntiBase

PubMed Central (2000) / PubChem (2004)

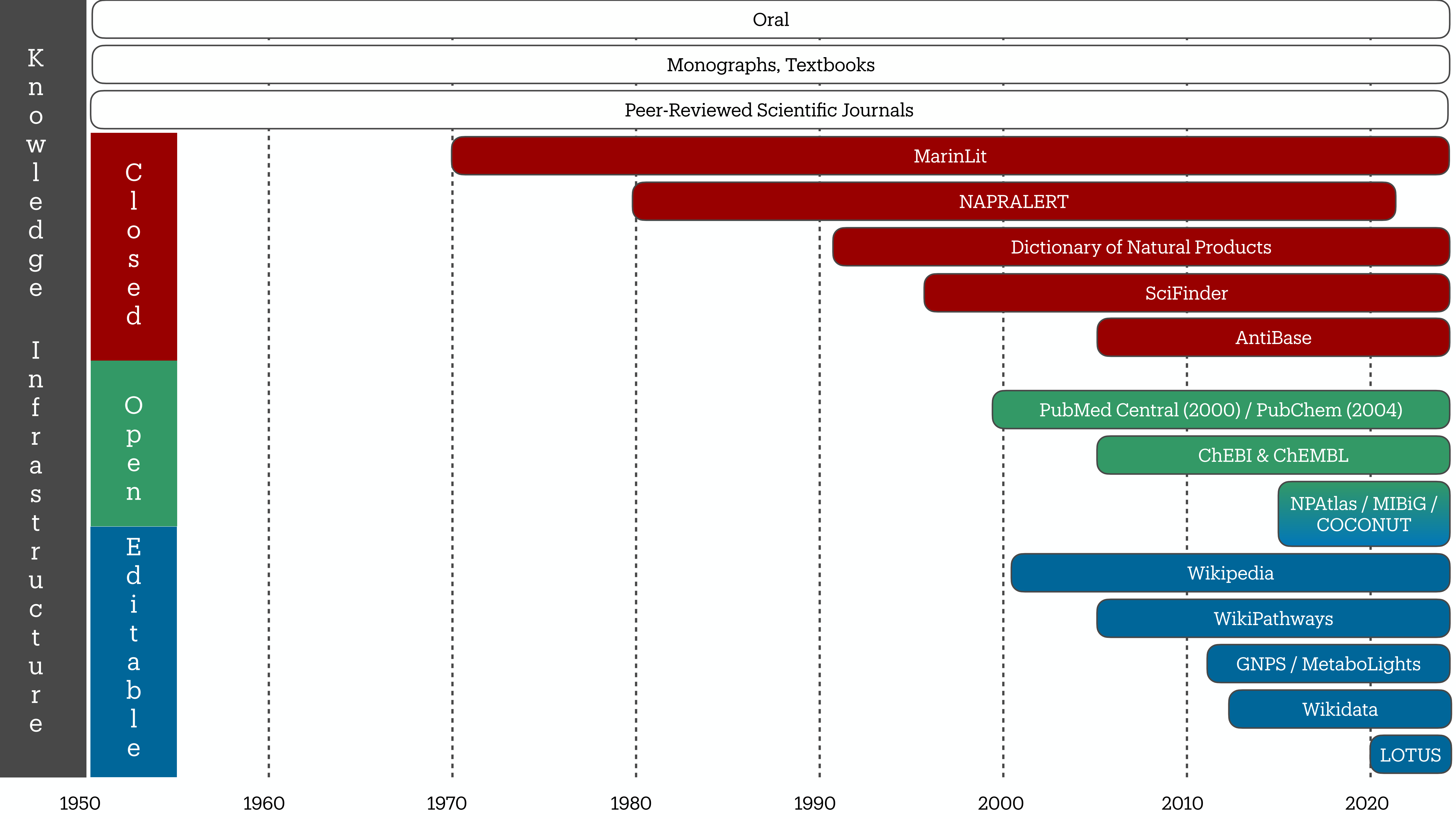
ChEBI & ChEMBL

NPAAtlas / MIBiG /  
COCONUT

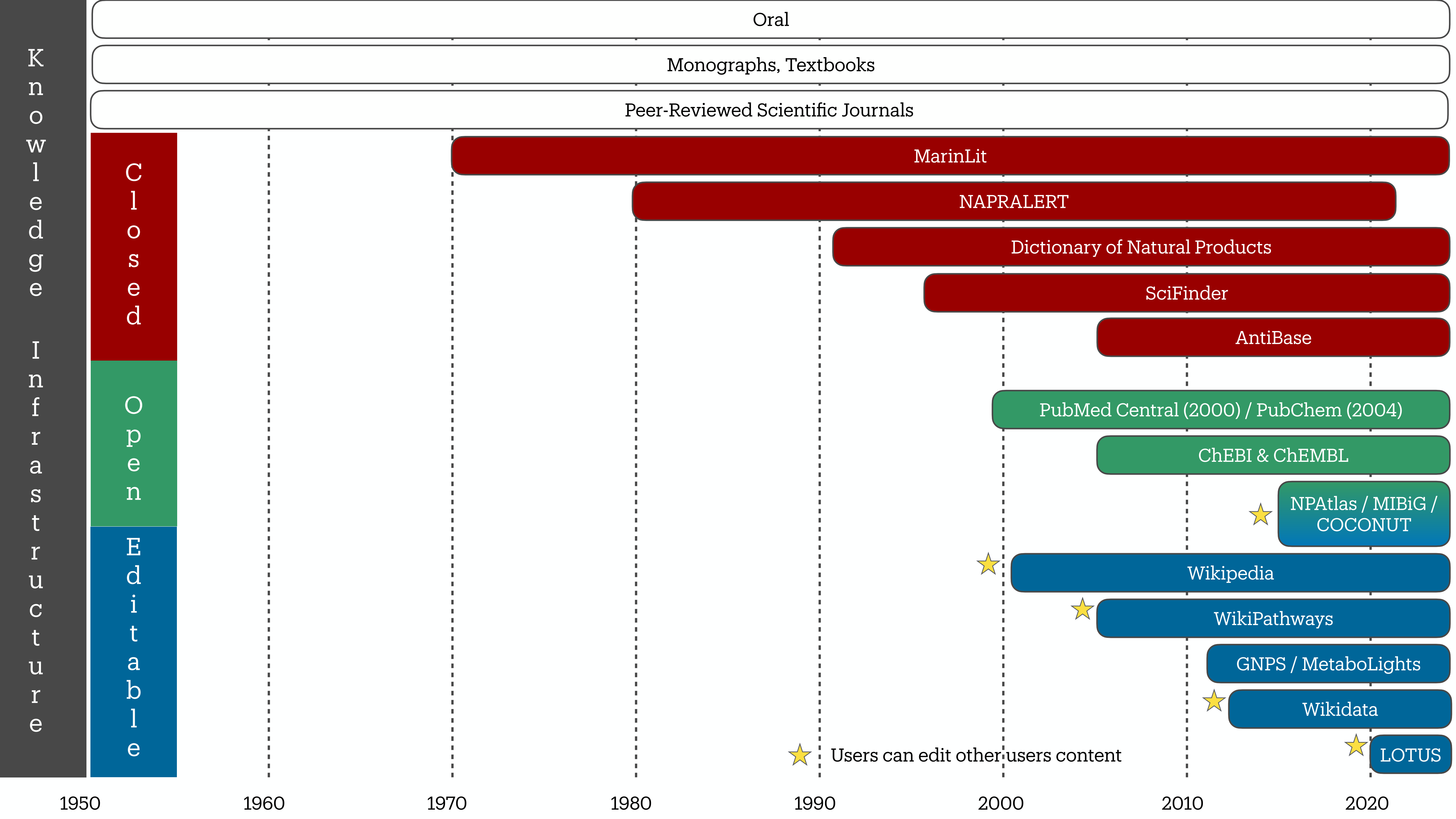
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1950 1960 1970 1980 1990 2000 2010 2020









# The Paradigm Shift

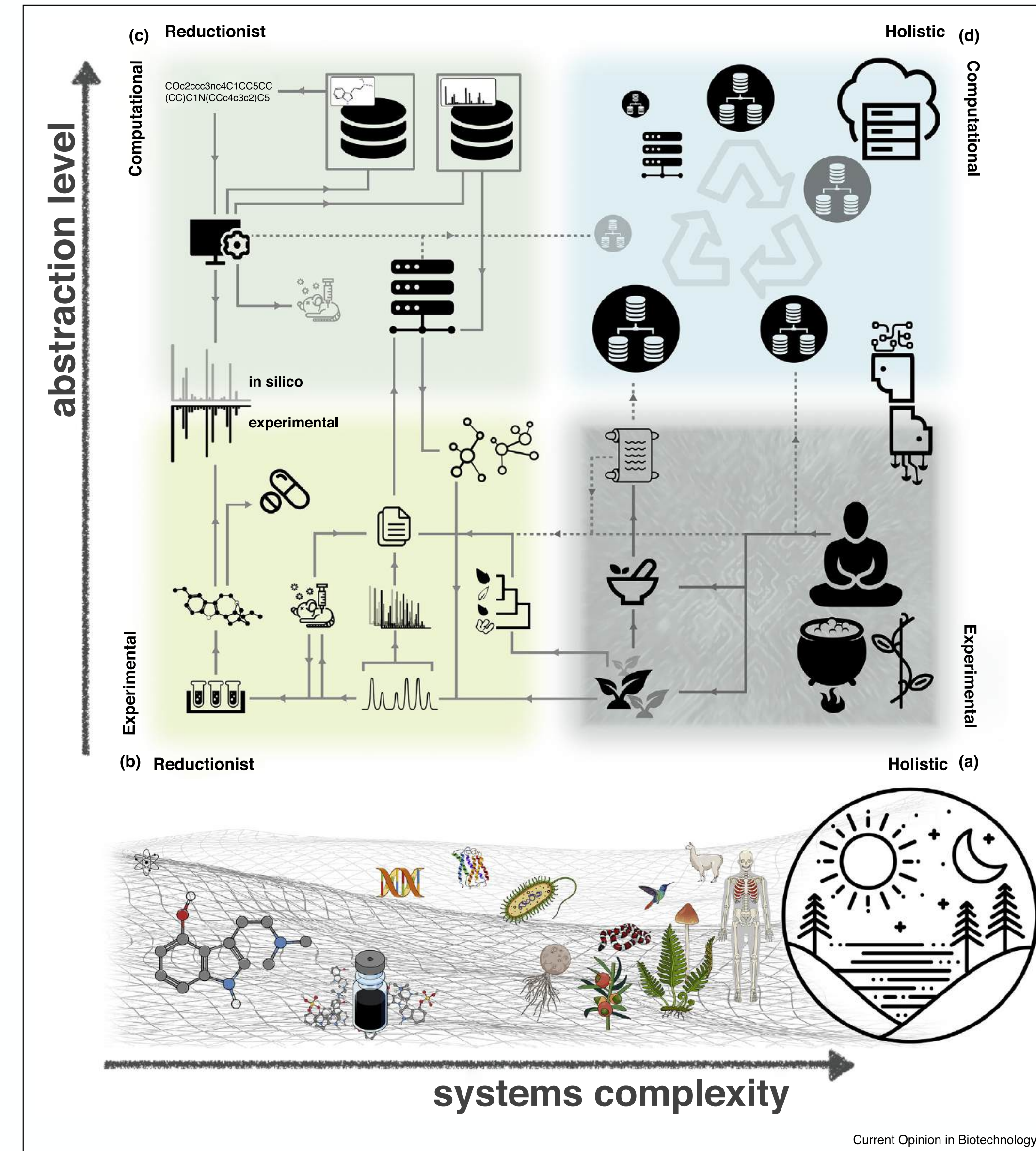


Figure from Allard et al. 2018. “Pharmacognosy in the Digital Era: Shifting to Contextualized Metabolomics.” Current Opinion in Biotechnology 54 (December): 57–64. <https://doi.org/10.1016/j.copbio.2018.02.010>



# The Paradigm Shift

Natural products research evolved into a data-intensive, interconnected science:

- From isolated compounds to integrated knowledge: pathways, biological activities, evolutionary relationships
- Questions now span multiple domains, requiring cross-referenced, machine-readable data

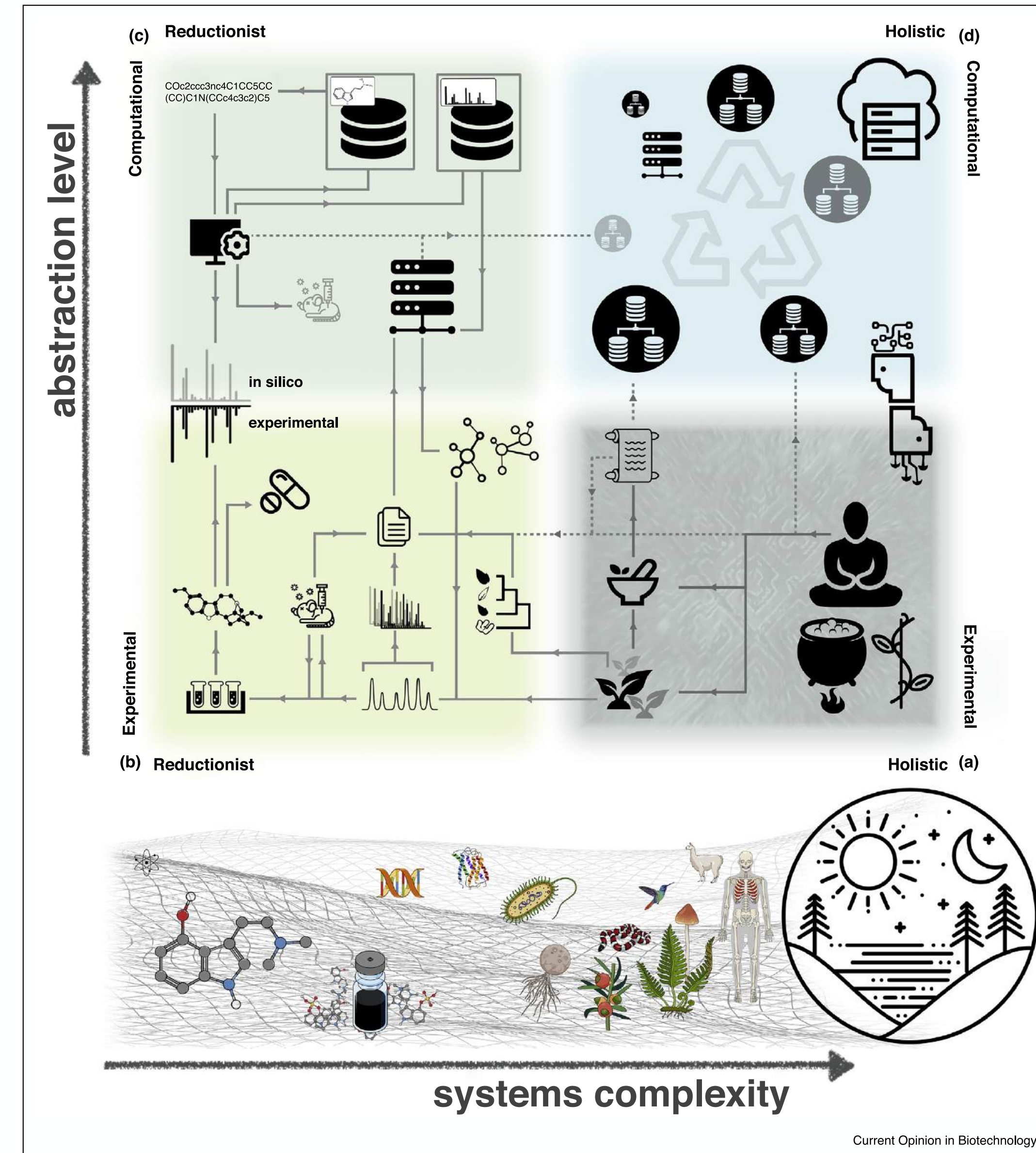


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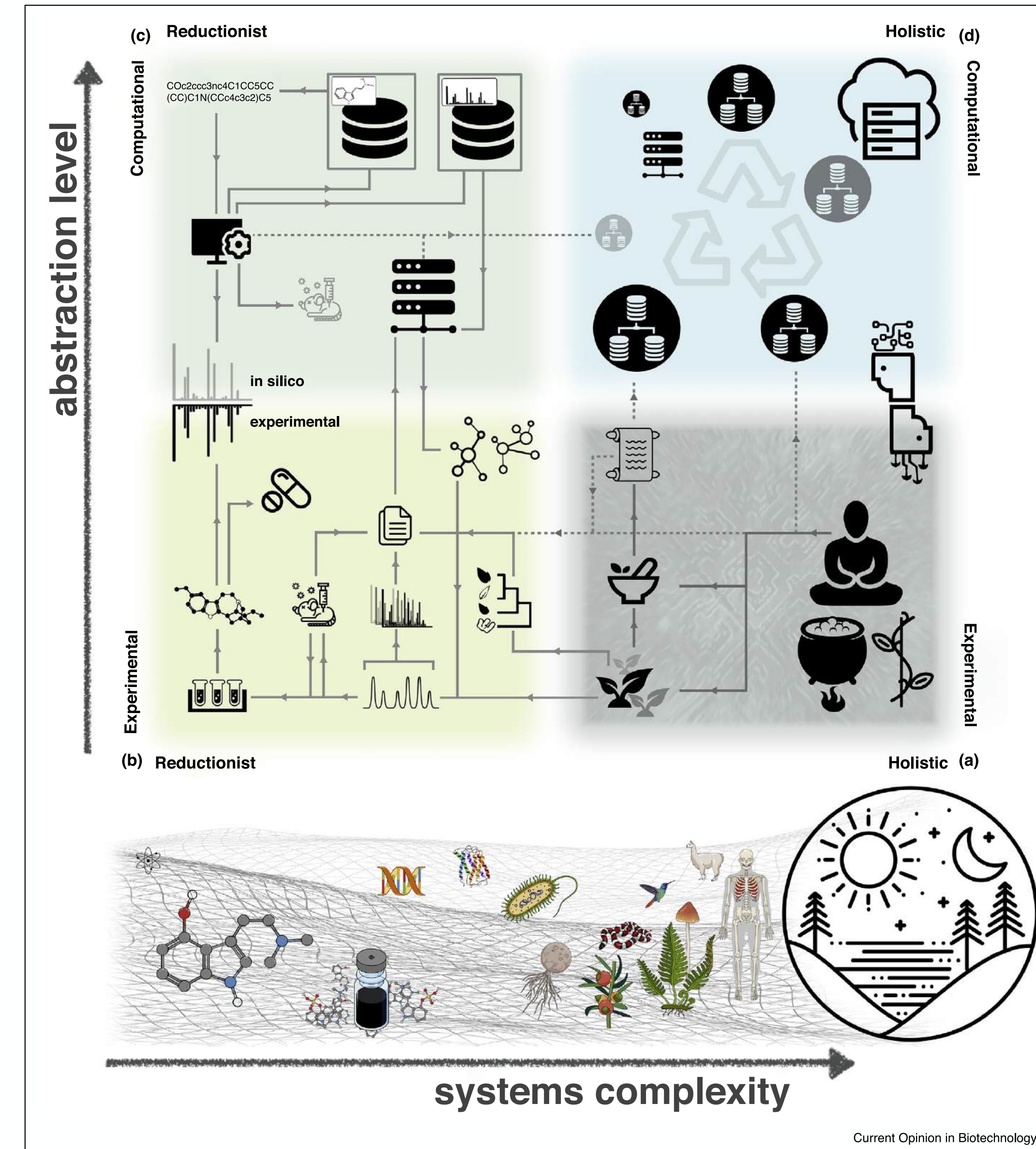
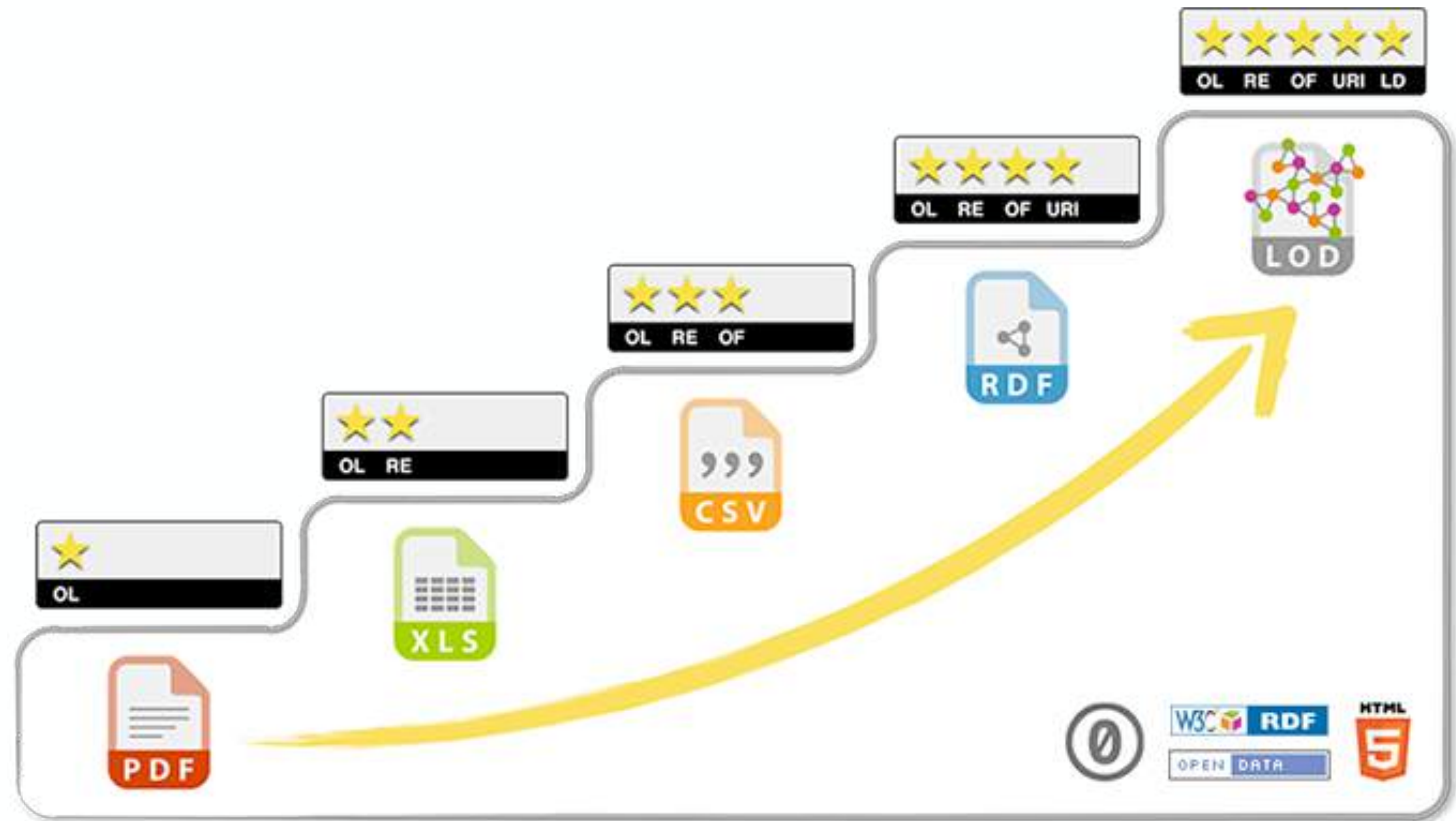


Figure from Allard et al. 2018. “*Pharmacognosy in the Digital Era: Shifting to Contextualized Metabolomics.*” Current Opinion in Biotechnology 54 (December): 57–64. <https://doi.org/10.1016/j.copbio.2018.02.010>



# Linked Open Data



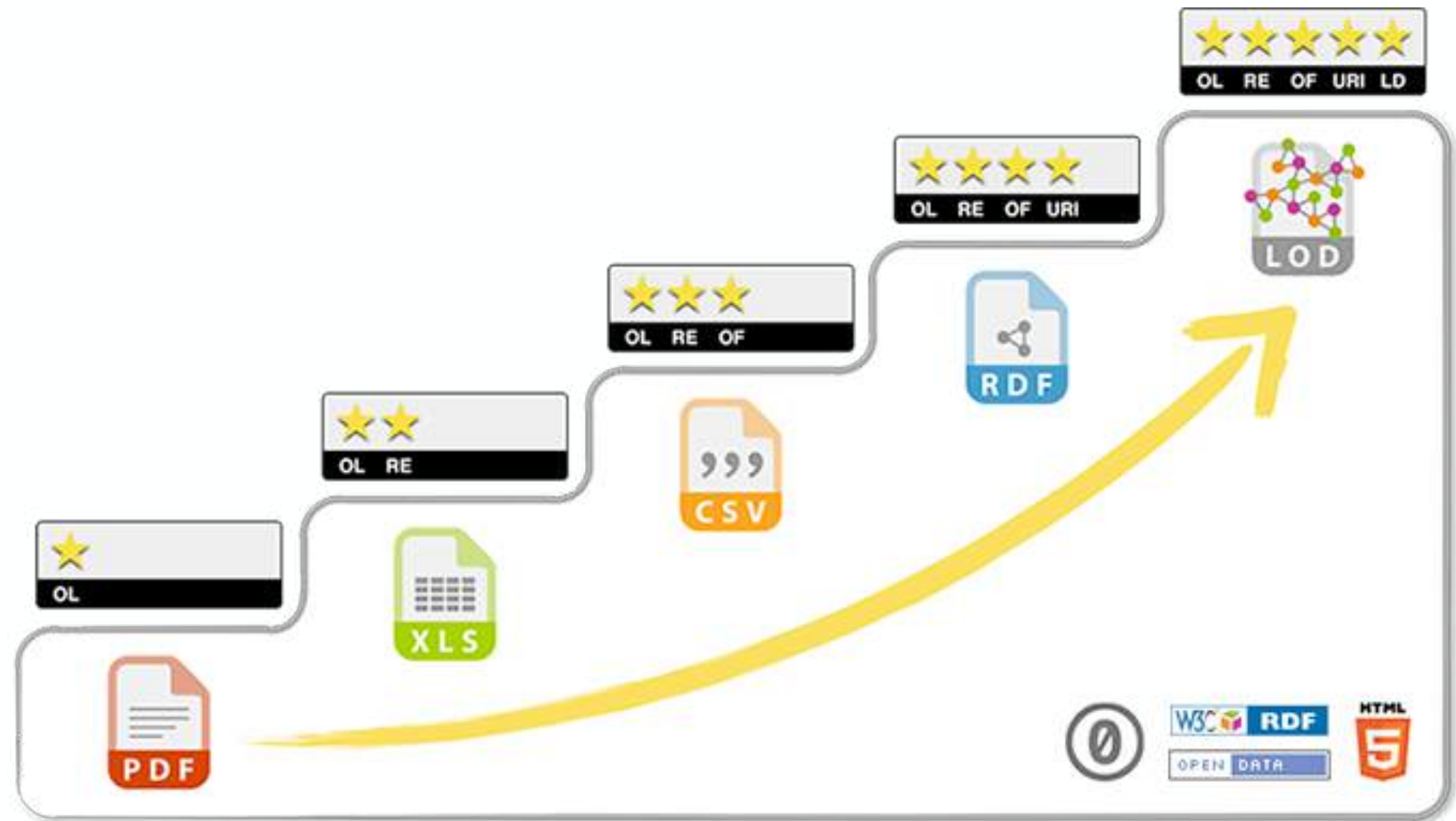
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<https://5stardata.info>



# Linked Open Data

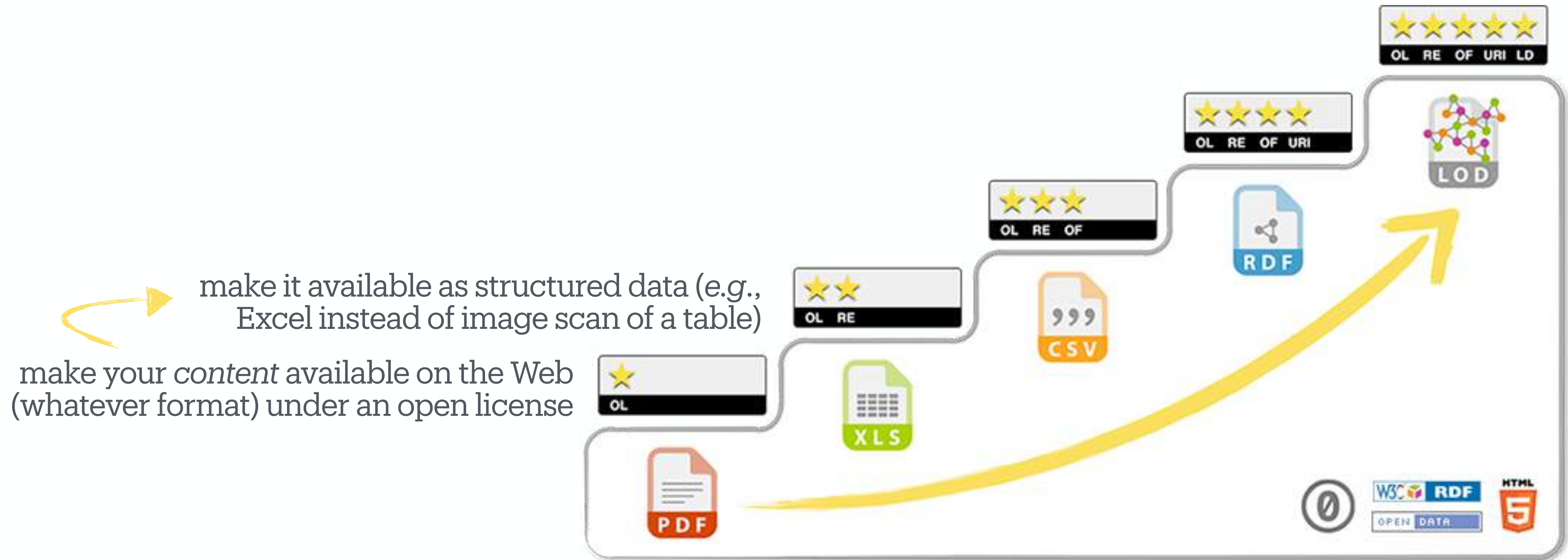
make your *content* available on the Web  
(whatever format) under an open license



[https://commons.wikimedia.org/wiki/File:5-star\\_deployment\\_scheme\\_for\\_Open\\_Data.png](https://commons.wikimedia.org/wiki/File:5-star_deployment_scheme_for_Open_Data.png)

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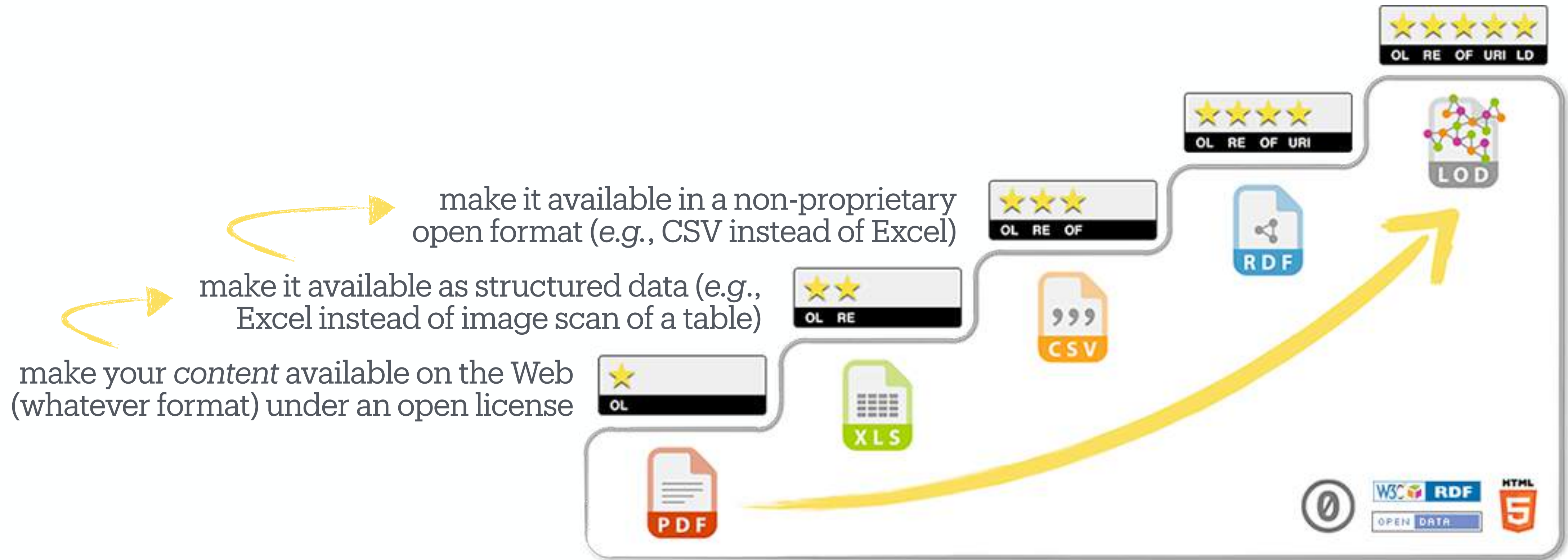
# Linked Open Data



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# Linked Open Data

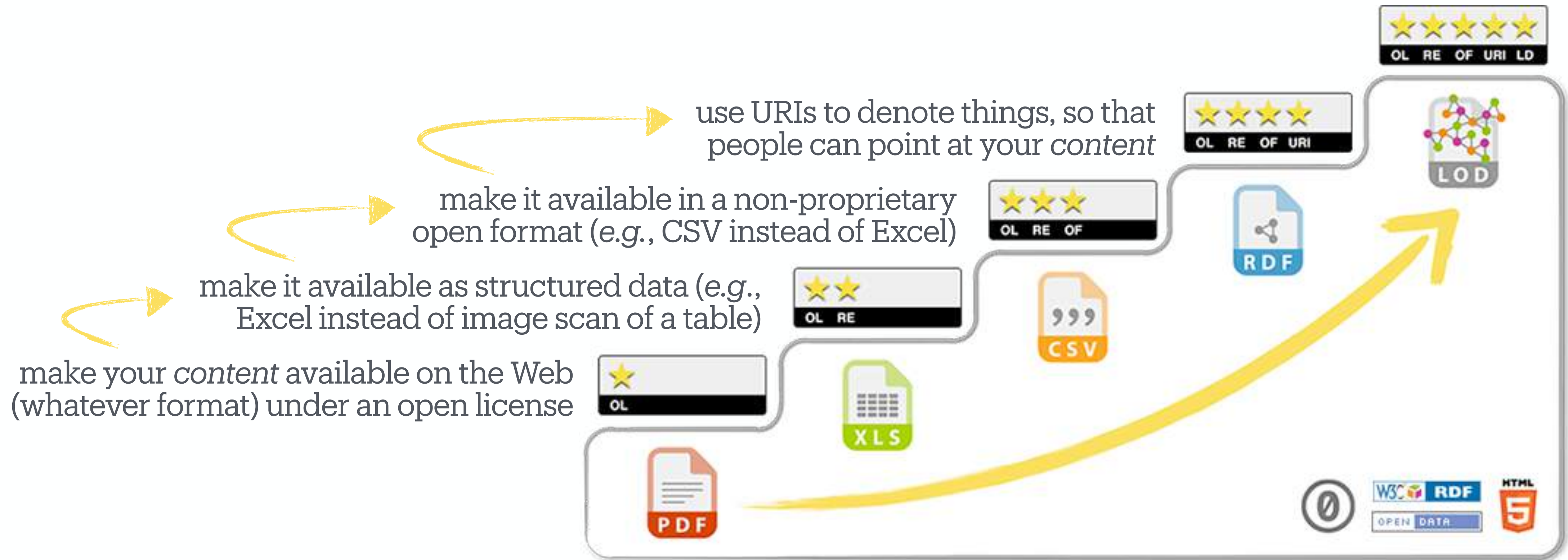


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# Linked Open Data

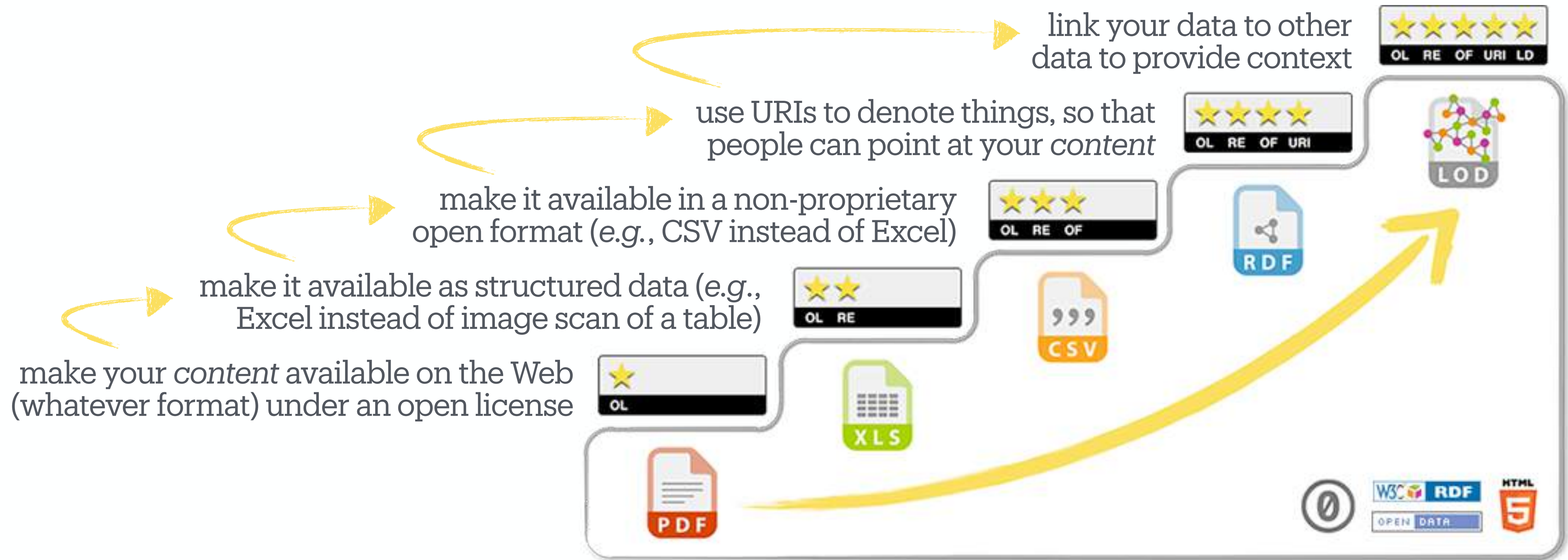


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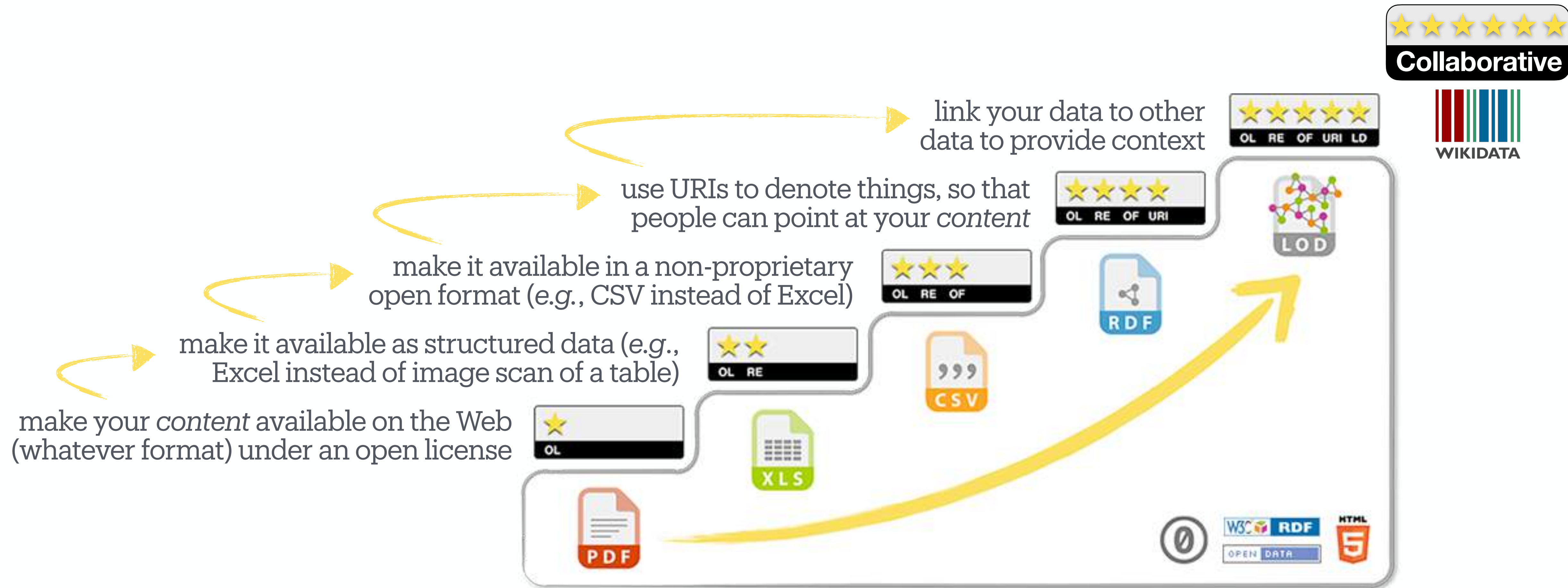
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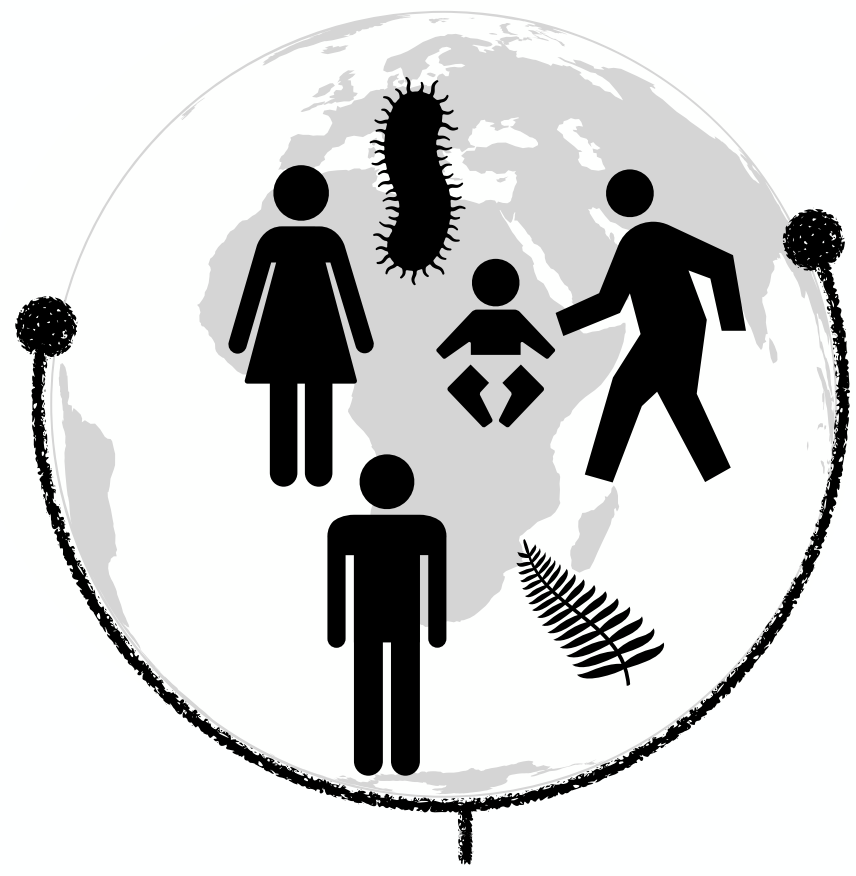
# Linked Open Data



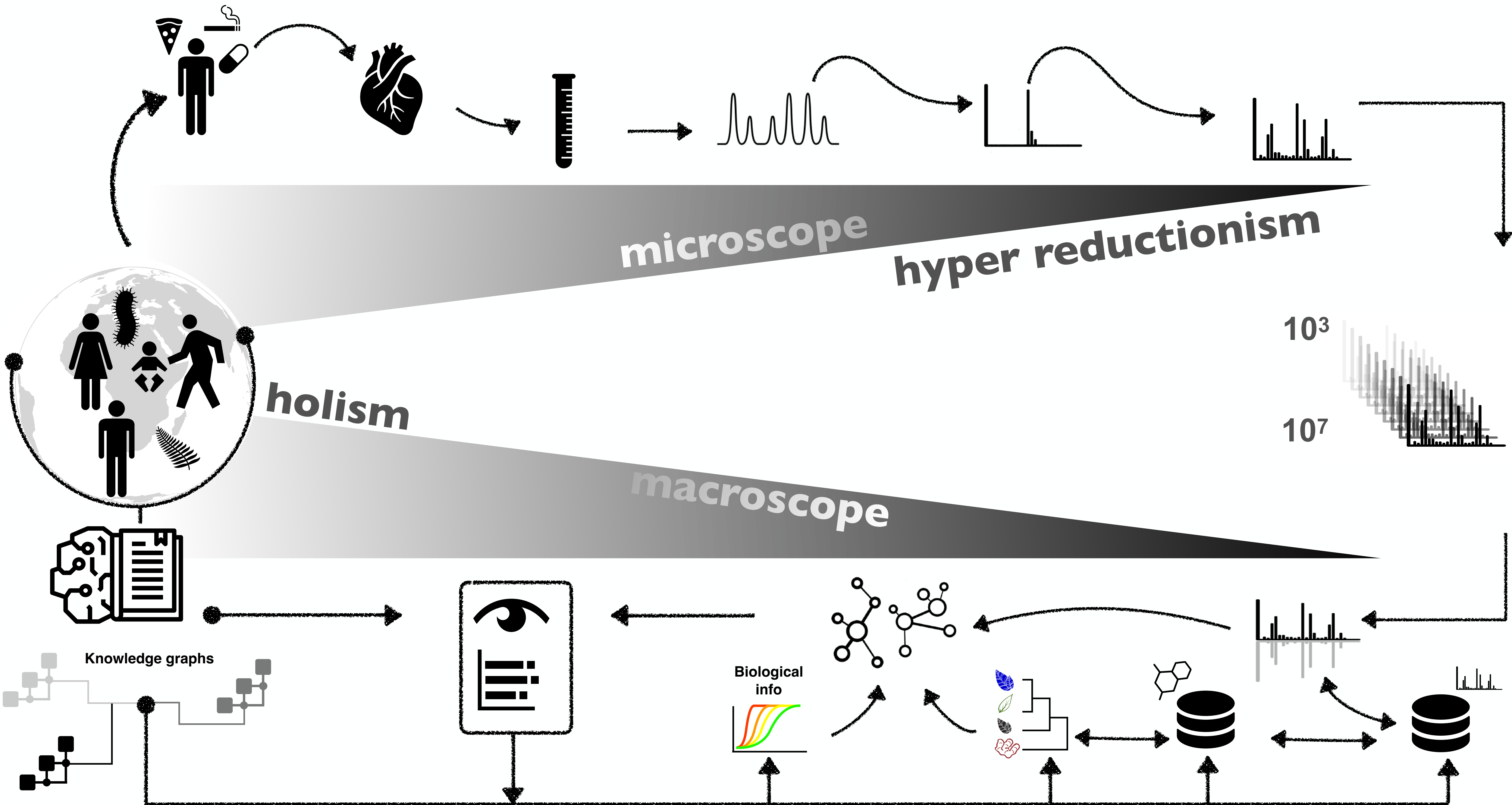
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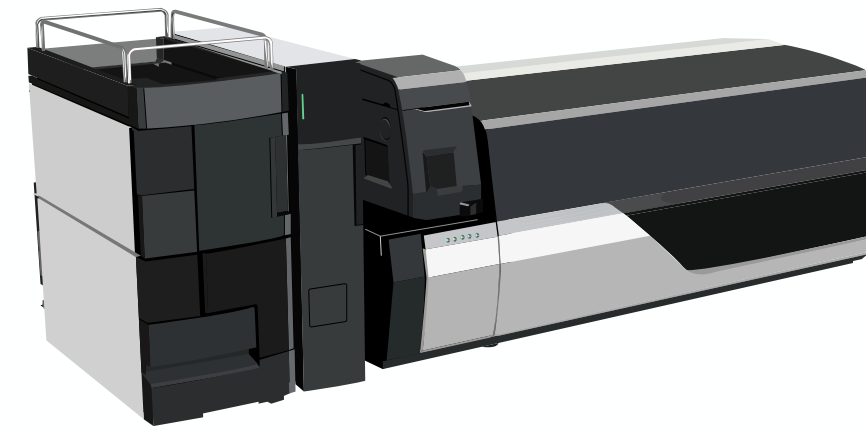
# Linked Open Data as a *macroscope* for natural products research







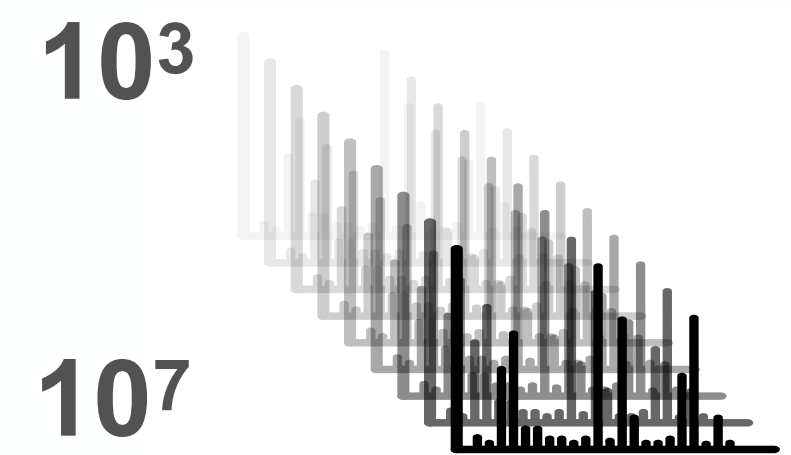
Slide adapted from Pierre-Marie Allard

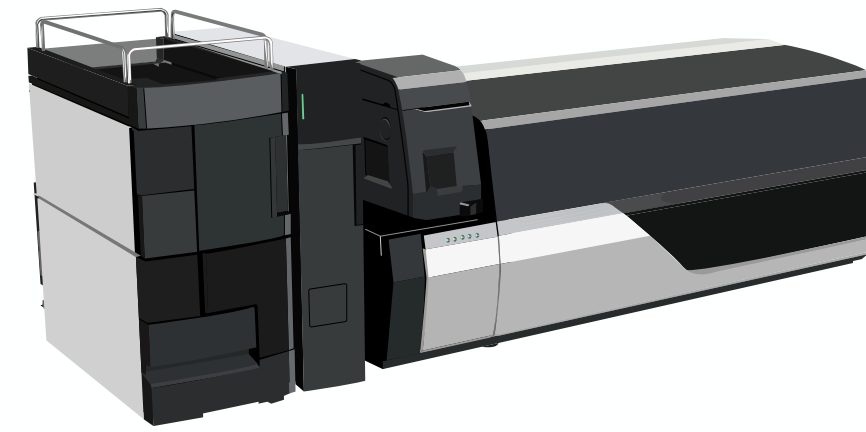


microscope



macroscope

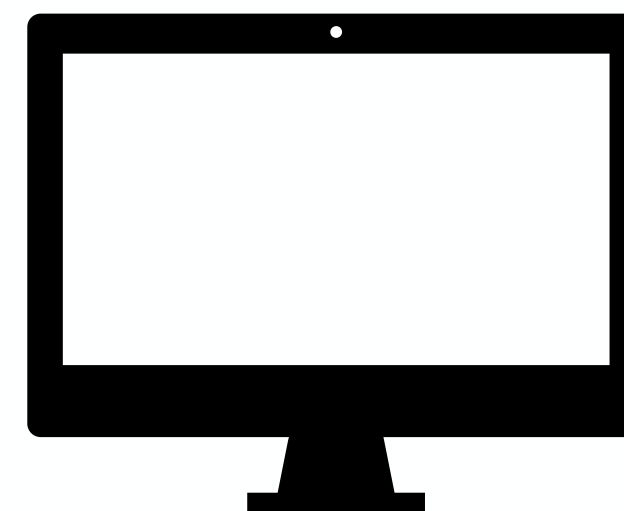
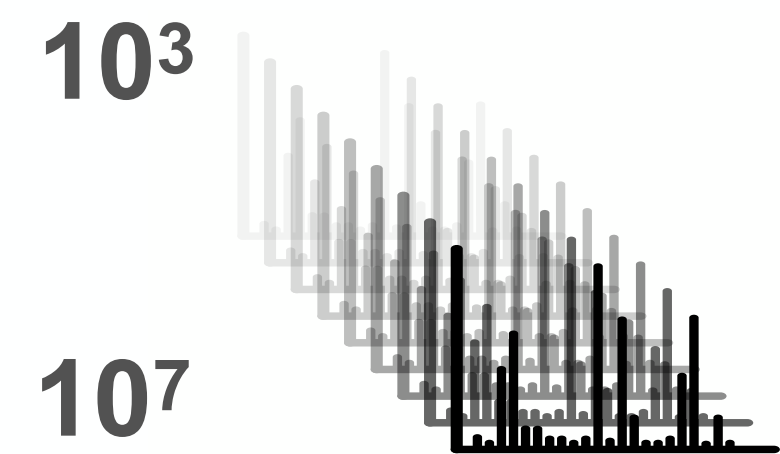
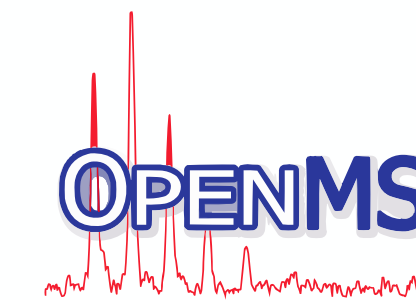




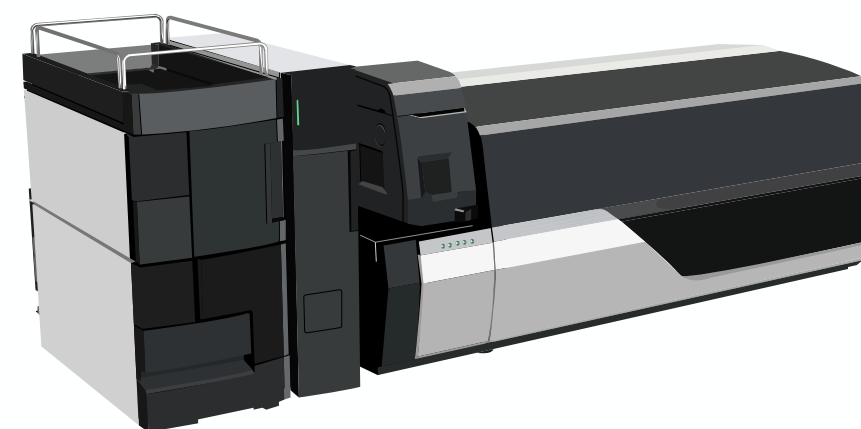
microscope



macroscope







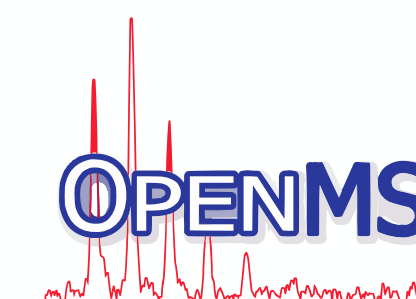
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macroscope

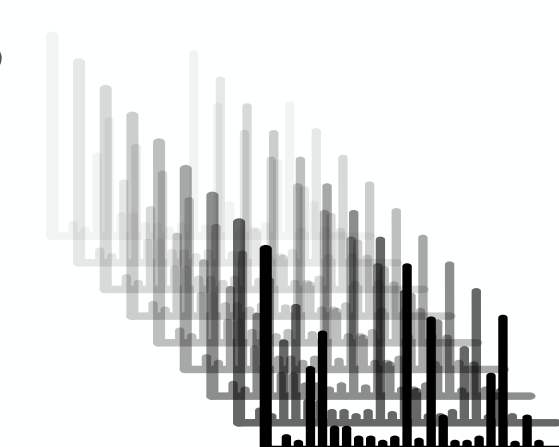


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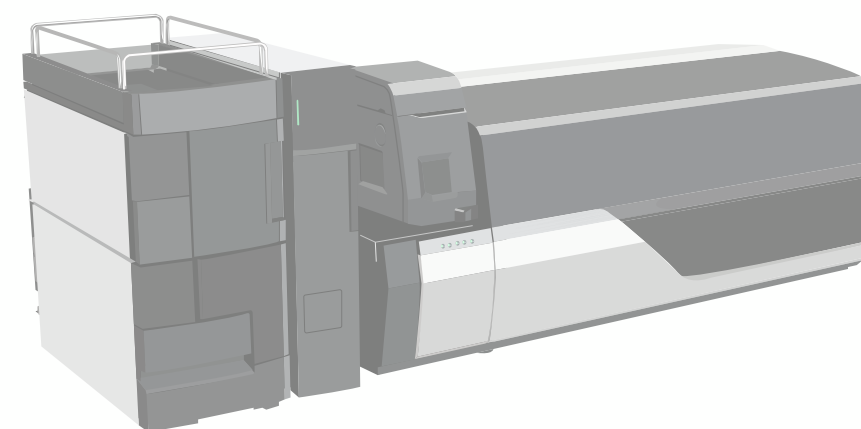


$10^3$

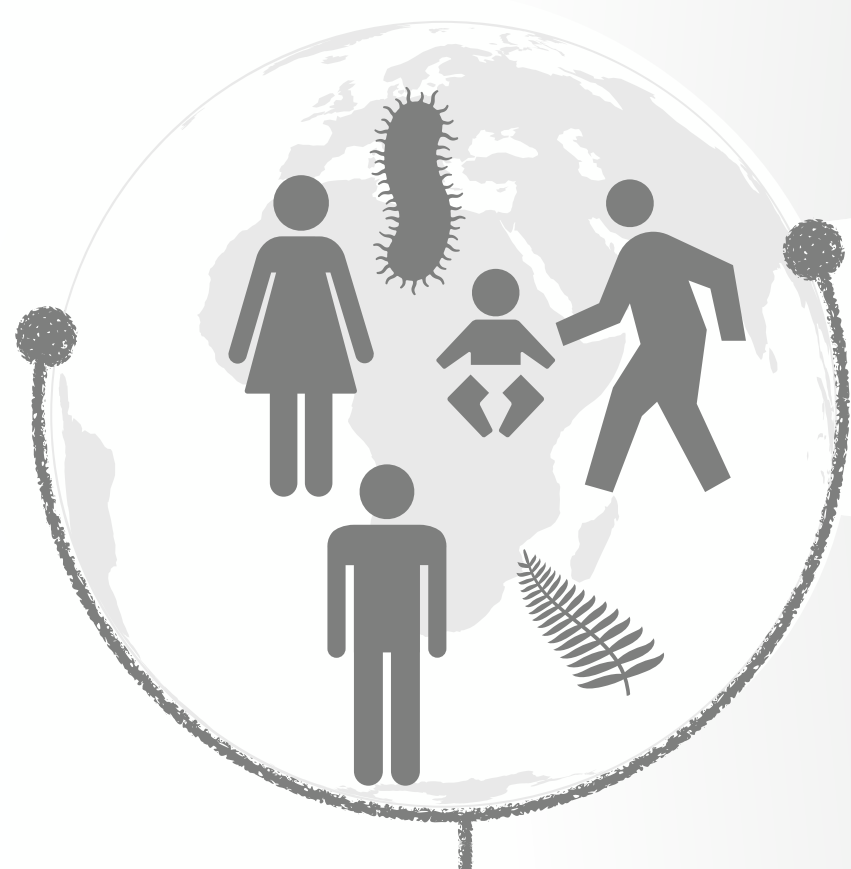
$10^7$



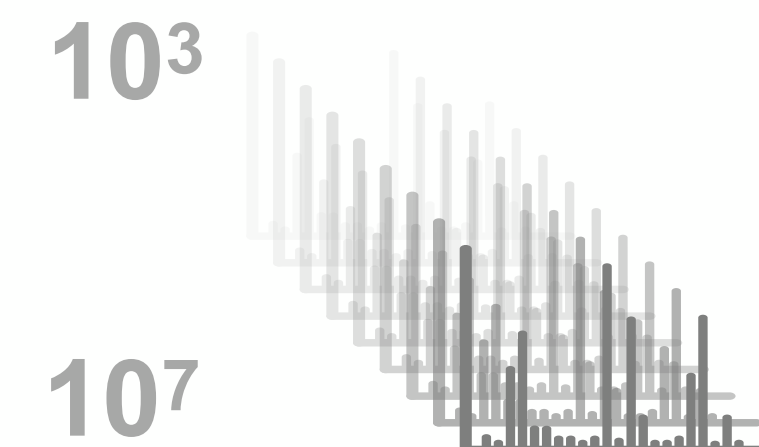
MoNA - MassBank of North America



microscope



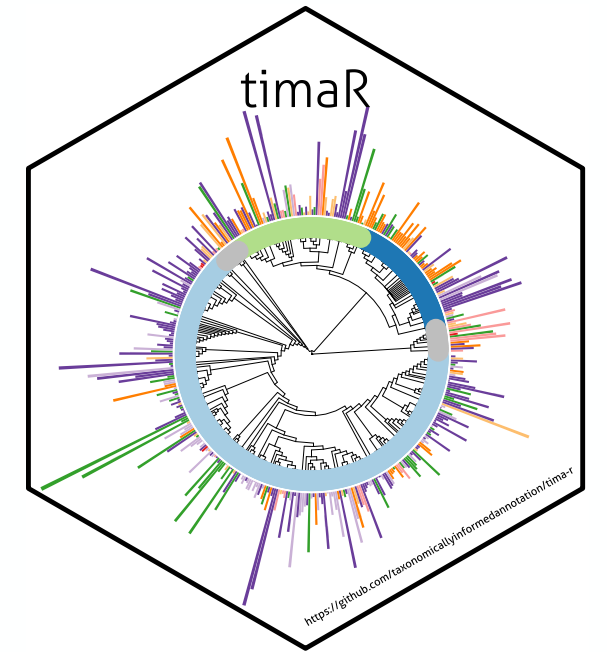
macroscope



MoNA - MassBank of North America



# Taxonomically informed metabolite annotation

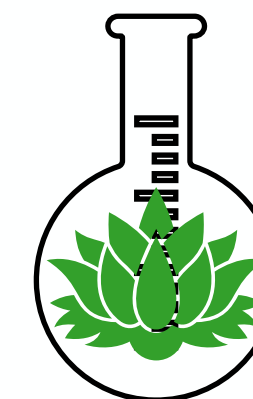


*“Plant taxonomy would be the most useful guide to man in his search for new industrial and medicinal plants”* (de Candolle, 1804)

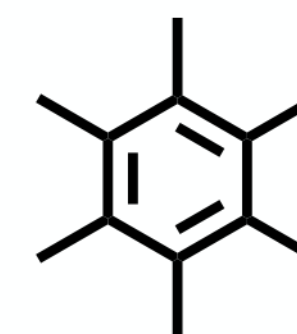
For more: <https://doi.org/10.3389/fpls.2019.01329>



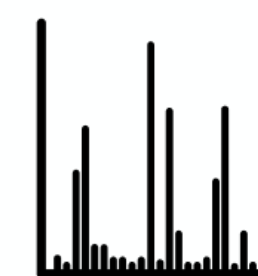
*Metabolomics*



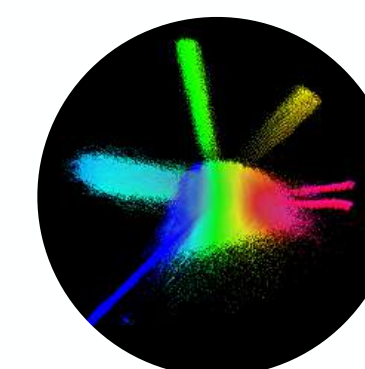
Natural Extract



Structure



MS<sup>2</sup>  
spectrum



Chemical  
Space

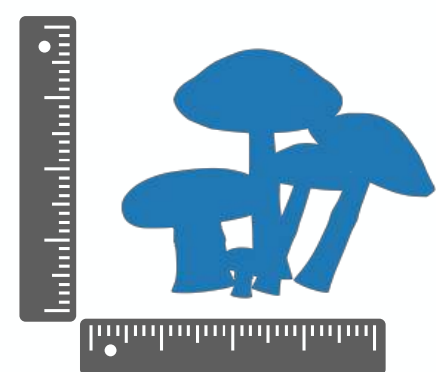


Molecular  
Network

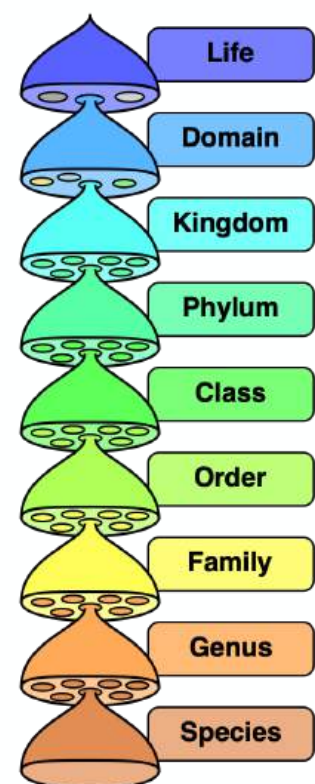
## *Taxonomy*



Biological organism

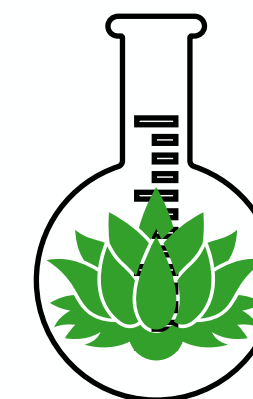


Morphological traits

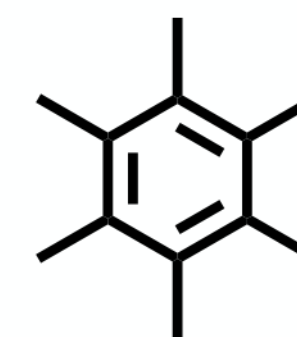


Taxonomy

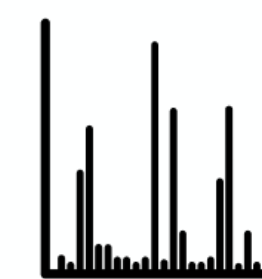
## *Metabolomics*



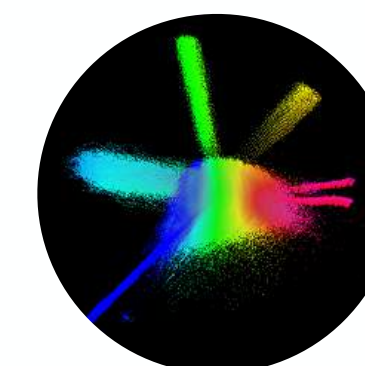
Natural Extract



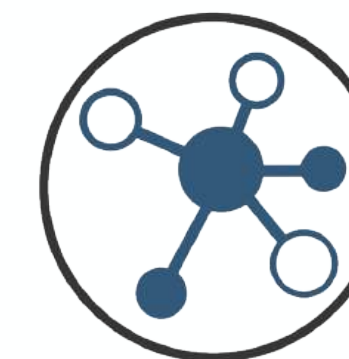
Structure



MS<sup>2</sup>  
spectrum



Chemical  
Space

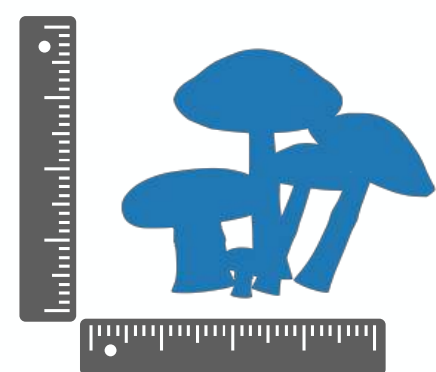


Molecular  
Network

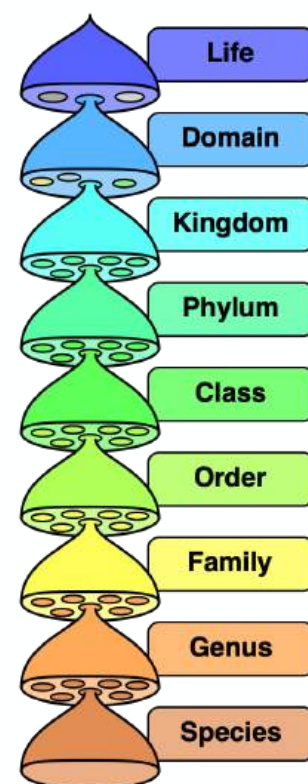
## *Taxonomy*



Biological organism



Morphological traits



Taxonomy

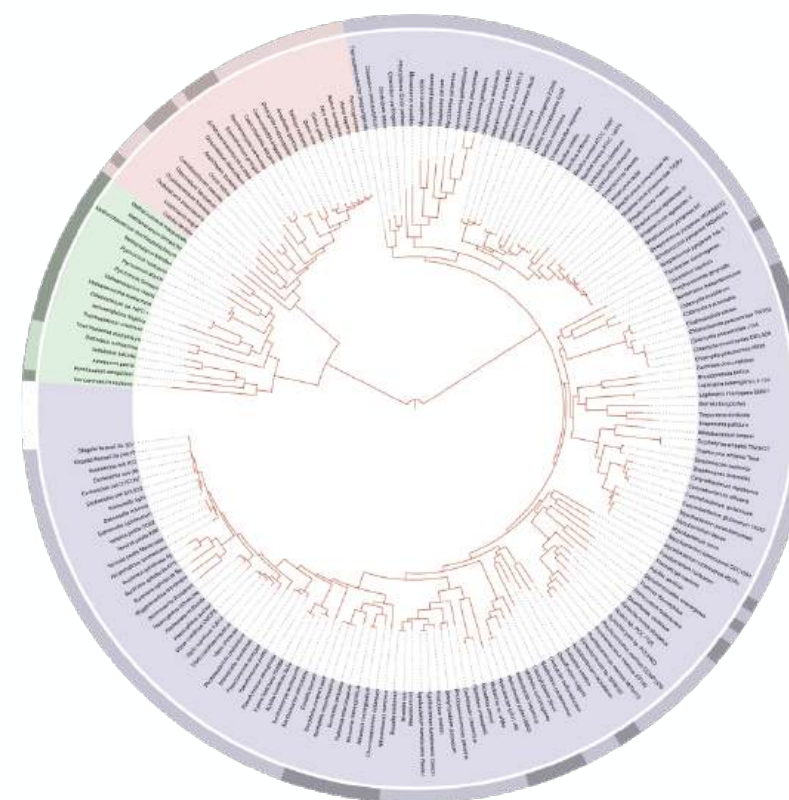
## *Genomics*



Genetic material

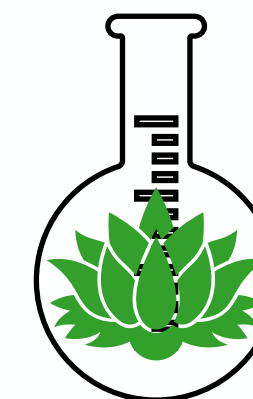
A T G C A A G C T T A  
C G T A C C T A G G C

DNA sequence

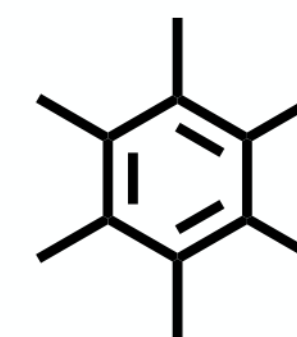


Phylogeny

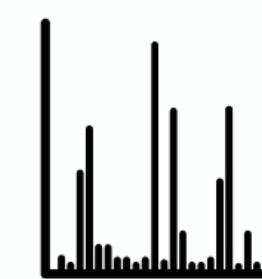
## *Metabolomics*



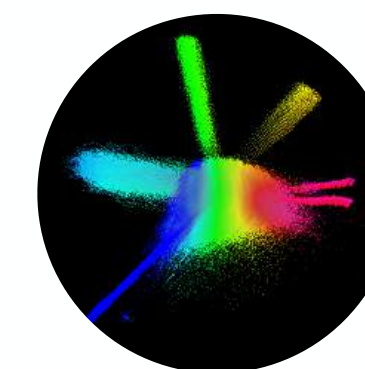
Natural Extract



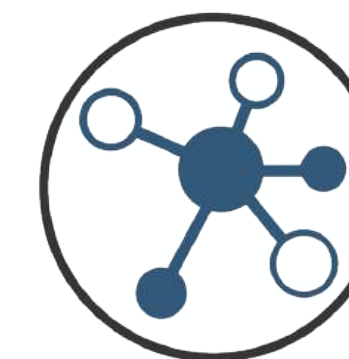
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Chemical  
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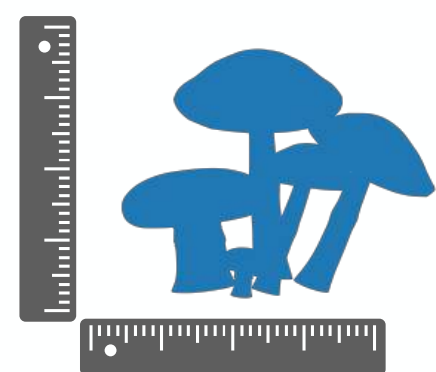
Molecular  
Network



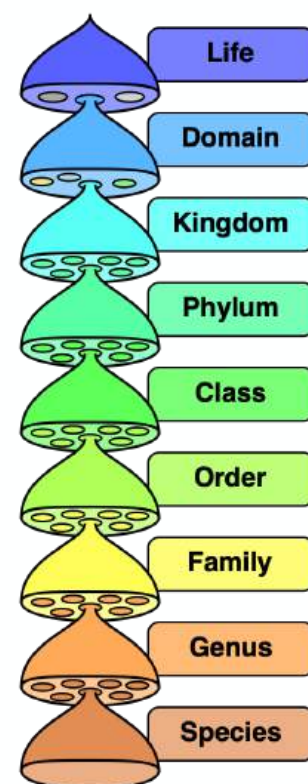
## Taxonomy



Biological organism



Morphological traits



Taxonomy

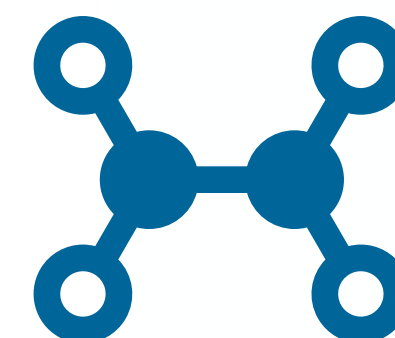


Biological  
Taxon

LOTUS

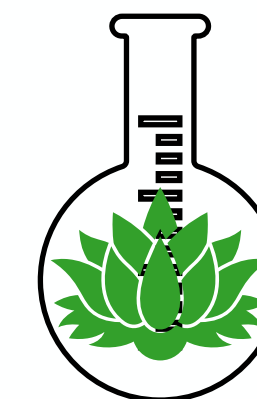


Reference

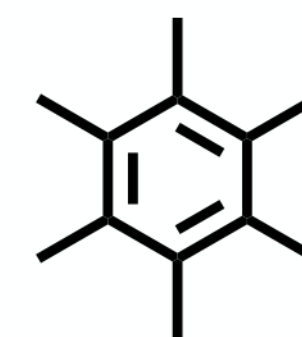


Chemical  
Structure

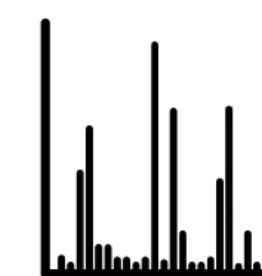
## Metabolomics



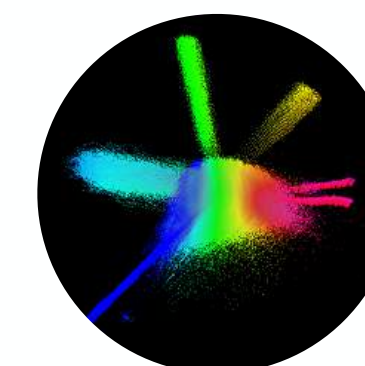
Natural Extract



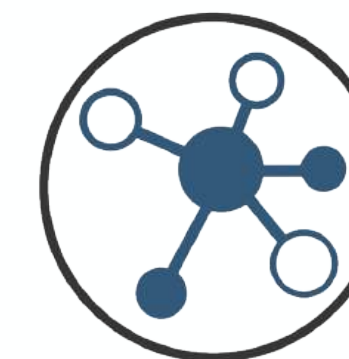
Structure



MS<sup>2</sup>  
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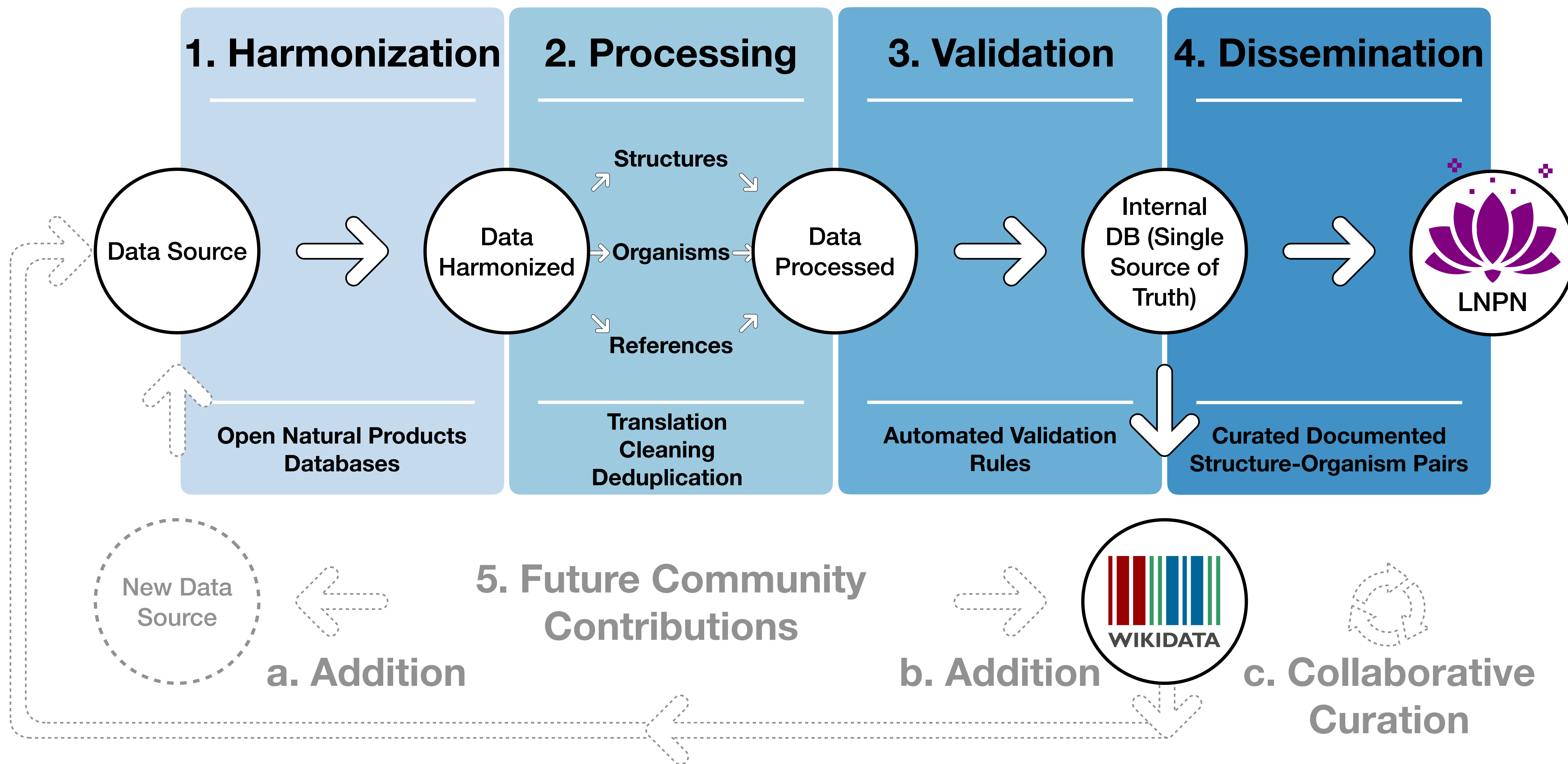


Chemical  
Space



Molecular  
Network

# The LOTUS Initiative





# The LOTUS Initiative



Natural Products Online is an open source project for Natural Products (NPs) storage, search and analysis. This page hosts LOTUS, the natural products occurrence database, one of the biggest and best annotated resources for NPs occurrences available free of charge and without any restriction. LOTUS is a living database which is hosted in parallel at Wikidata and here. The Wikidata version allows for community curation and addition of novel data. The current version allows a more user friendly experience (such as structural search, taxonomy oriented query, flat table and structures exports).

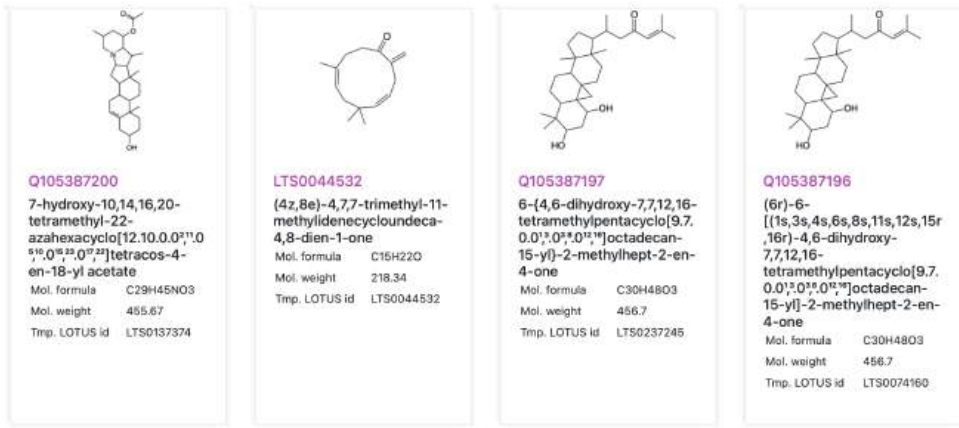
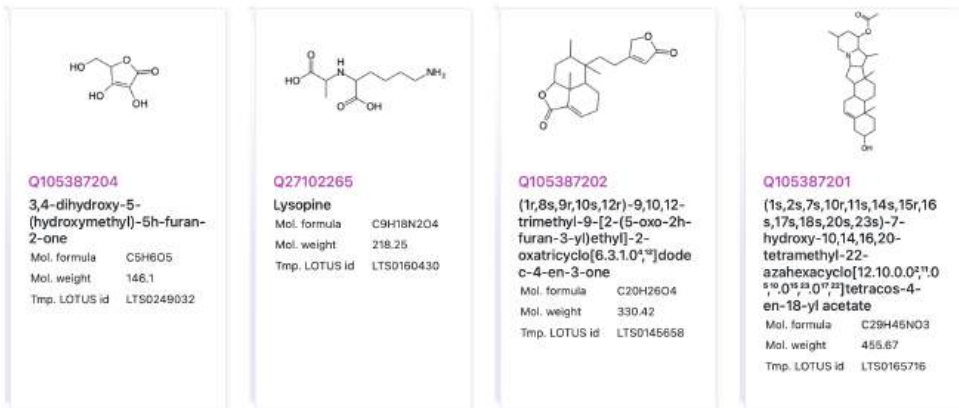
## Component Browser

Cards

Table

There are 276518 natural products in the database

1 2 3 4 5 6 7 8 9 10 ... 11521



## PubChem Classification Browser

Browse PubChem data using a classification of interest, or search for PubChem records (phenylpropionates, or Gene Ontology: DNA repair). More...

Select classification

LOTUS - the natural products occurrence database

Search

Classification description (from LOTUS - the natural products occurrence database)

Biological and chemical tree provided by the LOTUS (natural products occurrence database)

Created on 09/10/2022 09:05:48 More...

Data type counts to display

None Compound Taxonomy

Display zero count nodes?

Yes No

## Browse LOTUS - the natural products occurrence database Tree

- LOTUS Tree

214,481

Biological Tree

192,579

Archaea

72

Bacteria

17,467

Eukaryota

175,387

Chemical Tree

214,481

Alkaloids

28,065

Amino acids and Peptides

13,012

Carbohydrates

2,144

Fatty acids

10,266

Polyketides

27,981

Shikimates and Phenylpropanoids

42,177

Terpenoids

97,499

Not classified

7,328
- zenodo

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The LOTUS Initiative

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September 16, 2022 (v4)

Dataset

Open Access

View

The LOTUS Initiative for Open Natural Products Research: TMAP

Rutz, Adriano, Gaudry, Arnaud

TMAP of the compounds present on Wikidata curated in the frame of the LOTUS Initiative: https://doi.org/10.7554/eLife.70780

Uploaded on September 16, 2022

3 more version(s) exist for this record

September 16, 2022 (v5)

Dataset

Open Access

View

The LOTUS Initiative for Open Natural Products Research: frozen dataset

Rutz, Adriano, Bisson, Jonathan, Allard, Pierre-Marie

Dataset used in the frame of the LOTUS Initiative: https://doi.org/10.7554/eLife.70780

Uploaded on September 16, 2022

5 more version(s) exist for this record

September 16, 2022 (v5)

Dataset

Open Access

View

The LOTUS Initiative for Open Natural Products Research: waste to recycle

Rutz, Adriano, Bisson, Jonathan, Allard, Pierre-Marie

Dataset not uploaded to Wikidata. Generated in the frame of the LOTUS Initiative: https://doi.org/10.7554/eLife.70780

Shared for further curation

Uploaded on September 16, 2022

4 more version(s) exist for this record

September 16, 2022 (v7)

Dataset

Open Access

View

The LOTUS Initiative for Open Natural Products Research: biological and chemical trees

Rutz, Adriano, Bisson, Jonathan, Allard, Pierre-Marie

Biological and chemical trees made from frozen metadata (10.5281/zenodo.5794106) (for example, for PubChem)

Uploaded on September 16, 2022

6 more version(s) exist for this record

September 16, 2022 (v9)

Dataset

Open Access

View

The LOTUS Initiative for Open Natural Products Research: frozen dataset union wikidata (with metadata)

Rutz, Adriano, Bisson, Jonathan, Allard, Pierre-Marie

Dataset present on Wikidata used in the frame of the LOTUS Initiative: https://doi.org/10.7554/eLife.70780

Uploaded on September 16, 2022

8 more version(s) exist for this record

September 16, 2022 (v6)

Dataset

Open Access

View

The LOTUS Initiative for Open Natural Products Research: metadata

Rutz, Adriano, Bisson, Jonathan, Allard, Pierre-Marie

Metadata of each of the three objects (structures, organisms, references) used in the frame of the LOTUS Initiative: https://doi.org/10.7554/eLife.70780

Uploaded on September 16, 2022

3 more version(s) exist for this record

Community

The LOTUS Initiative

A Zenodo community to gather files, datasets and outcomes of the LOTUS Initiative.

See our website https://lotus.nprod.net/ and article https://doi.org/10.7554/eLife.70780 for more details.

Curated by: the-lotus-initiative

Curation policy: After examination by initial members of the LOTUS Initiative (see https://lotus.nprod.net/people/) we will accept all contributions which will help natural products knowledge management and sharing.

Created: October 28, 2021

Harvesting API: OAI-PMH interface

Want your upload to appear in this community?

- Click the button above to upload a record directly to this community.
  - To add one of your existing records to the community, edit the record, add this community under the "Communities" section, save, and finally publish.
  - The community curator will then be notified to either accept or reject your upload (see community curation policy below).
  - If your upload is rejected by the curator, it will still be available on Zenodo, just not in this community.

## Extraction

SELECT ?desired items

WHERE {

VALUES ?classes {

Chemical compound

}

?item instance of ?classes.

{

?item found in taxon ?stmt.

?stmt found in taxon ?taxon.

OPTIONAL {

?stmt occurrence ?ref.

?ref stated in ?art.

}

}

SERVICE wikibase:label {

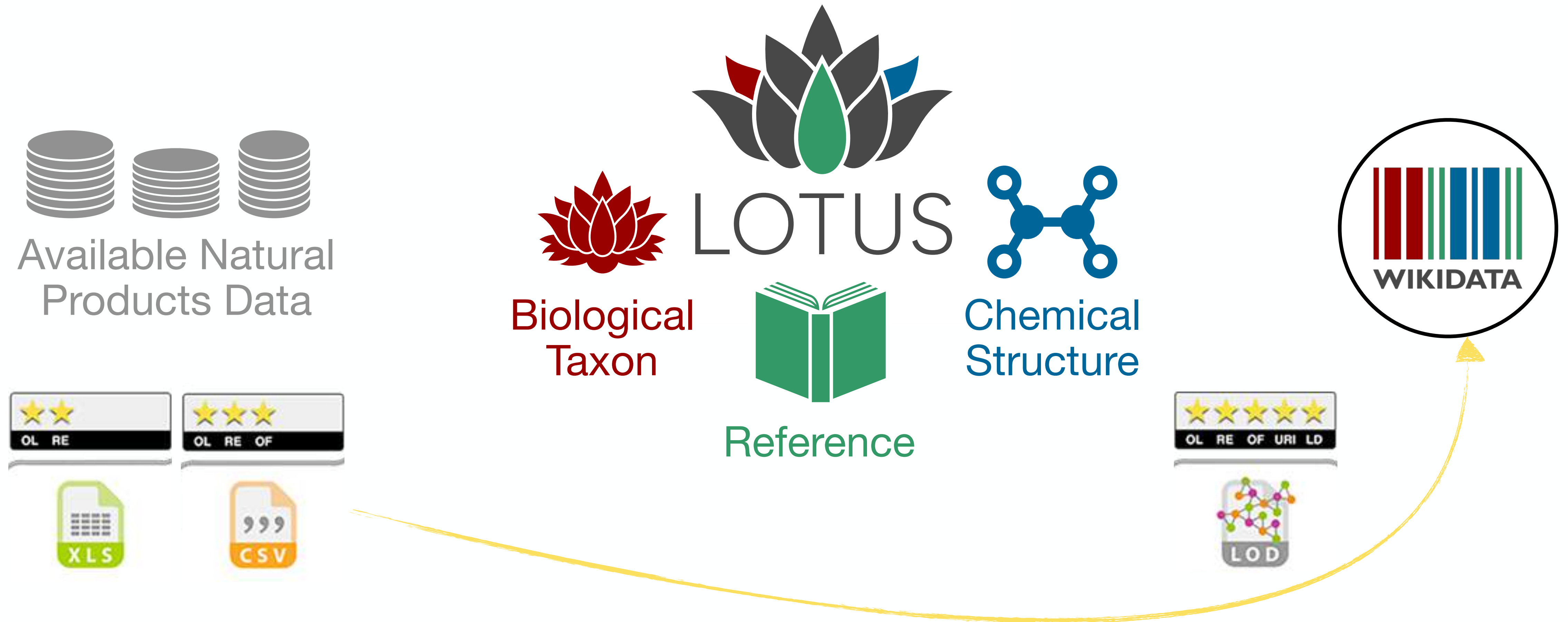
language

"[AUTO\_LANGUAGE],en". }

}



# The LOTUS Initiative



# Give back to *Caesar*...



FEATURE ARTICLE



SCIENCE FORUM

## Wikidata as a knowledge graph for the life sciences

**Abstract** Wikidata is a community-maintained knowledge base that has been assembled from repositories in the fields of genomics, proteomics, genetic variants, pathways, chemical compounds, and diseases, and that adheres to the FAIR principles of findability, accessibility, interoperability and reusability. Here we describe the breadth and depth of the biomedical knowledge contained within Wikidata, and discuss the open-source tools we have built to add information to Wikidata and to synchronize it with source databases. We also demonstrate several use cases for Wikidata, including the crowdsourced curation of biomedical ontologies, phenotype-based diagnosis of disease, and drug repurposing.

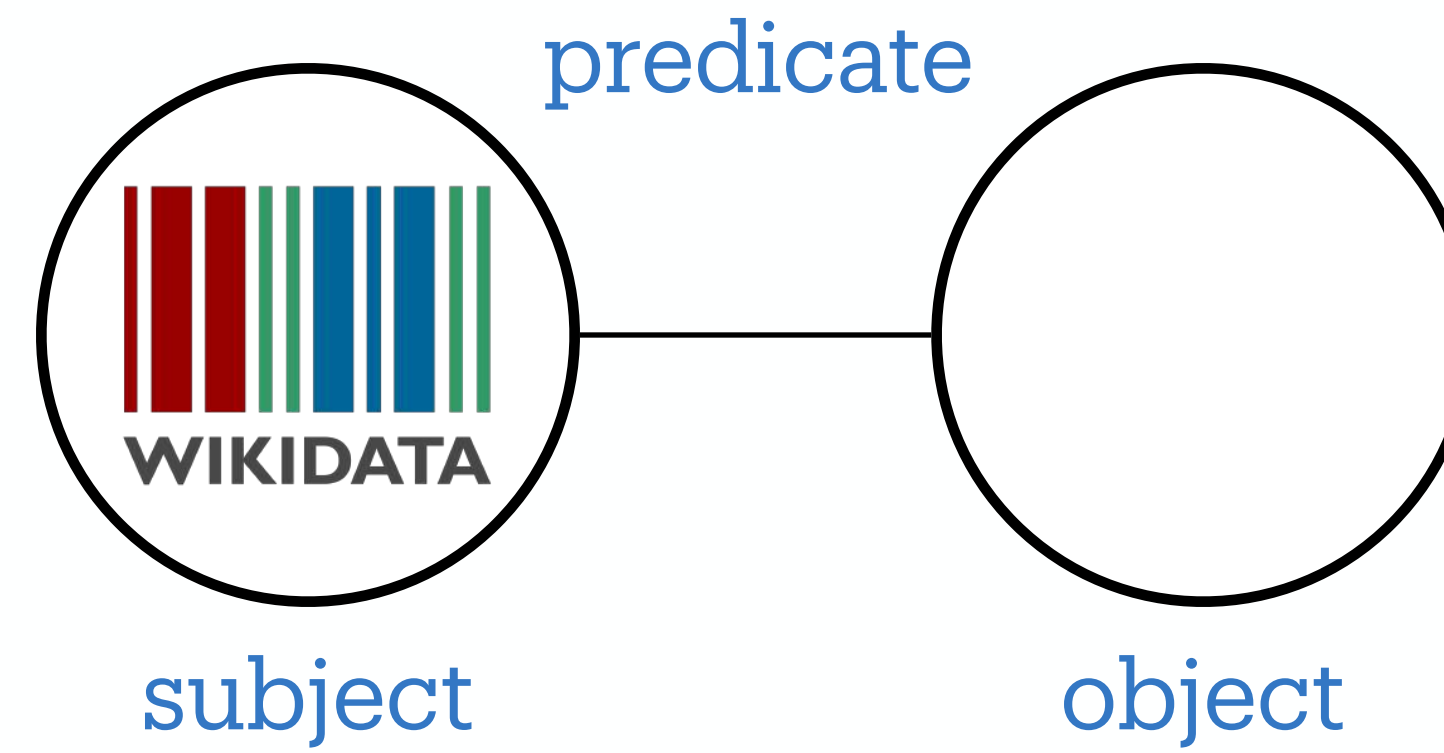
**ANDRA WAAGMEESTER<sup>†</sup>, GREGORY STUPP<sup>†</sup>, SEBASTIAN BURGSTALLER-MUEHLBACHER, BENJAMIN M GOOD, MALACHI GRIFFITH, OBI L GRIFFITH, KRISTINA HANSPERS, HENNING HERMJAKOB, TOBY S HUDSON, KEVIN HYBISKE, SARAH M KEATING, MAGNUS MANSKE, MICHAEL MAYERS, DANIEL MIETCHEN, ELVIRA MITRAKA, ALEXANDER R PICO, TIMOTHY PUTMAN, ANDERS RIUTTA, NURIA QUERALT-ROSINACH, LYNN M SCHRIML, THOMAS SHAFEE, DENISE SLENTER, RALF STEPHAN, KATHERINE THORNTON, GINGER TSUENG, ROGER TU, SABAH UL-HASAN, EGON WILLIGHAGEN, CHUNLEI WU AND ANDREW I SU\***

# Wikidata

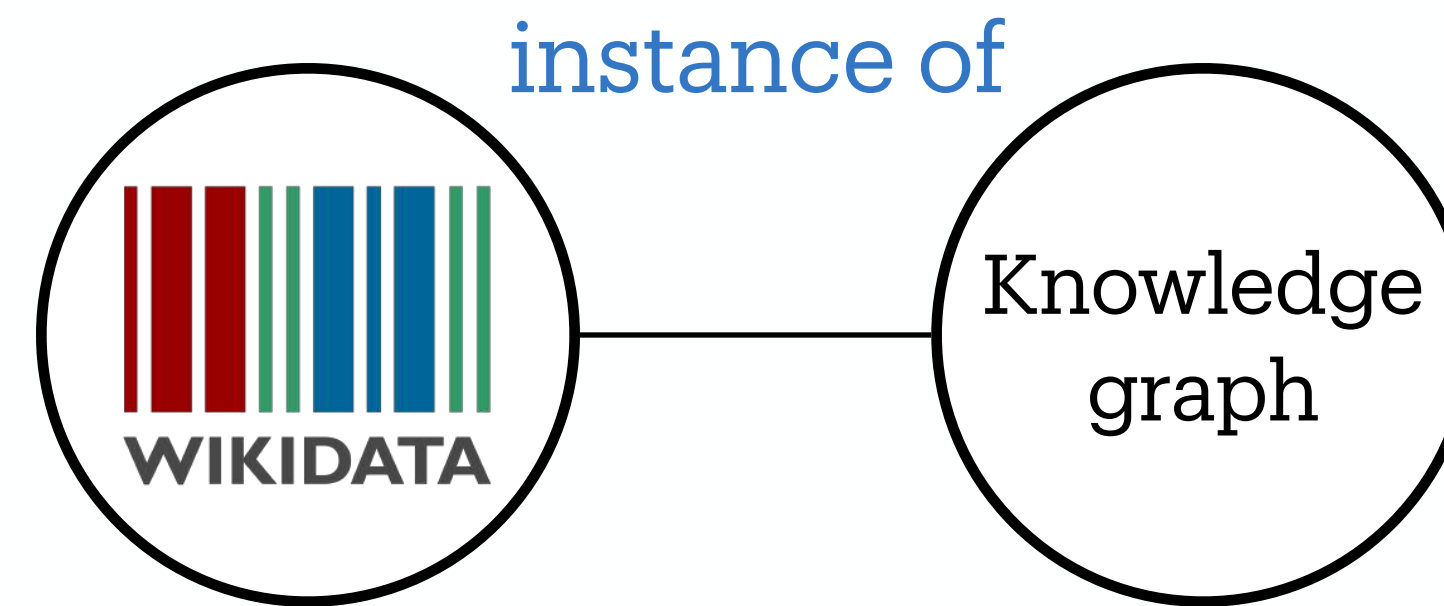




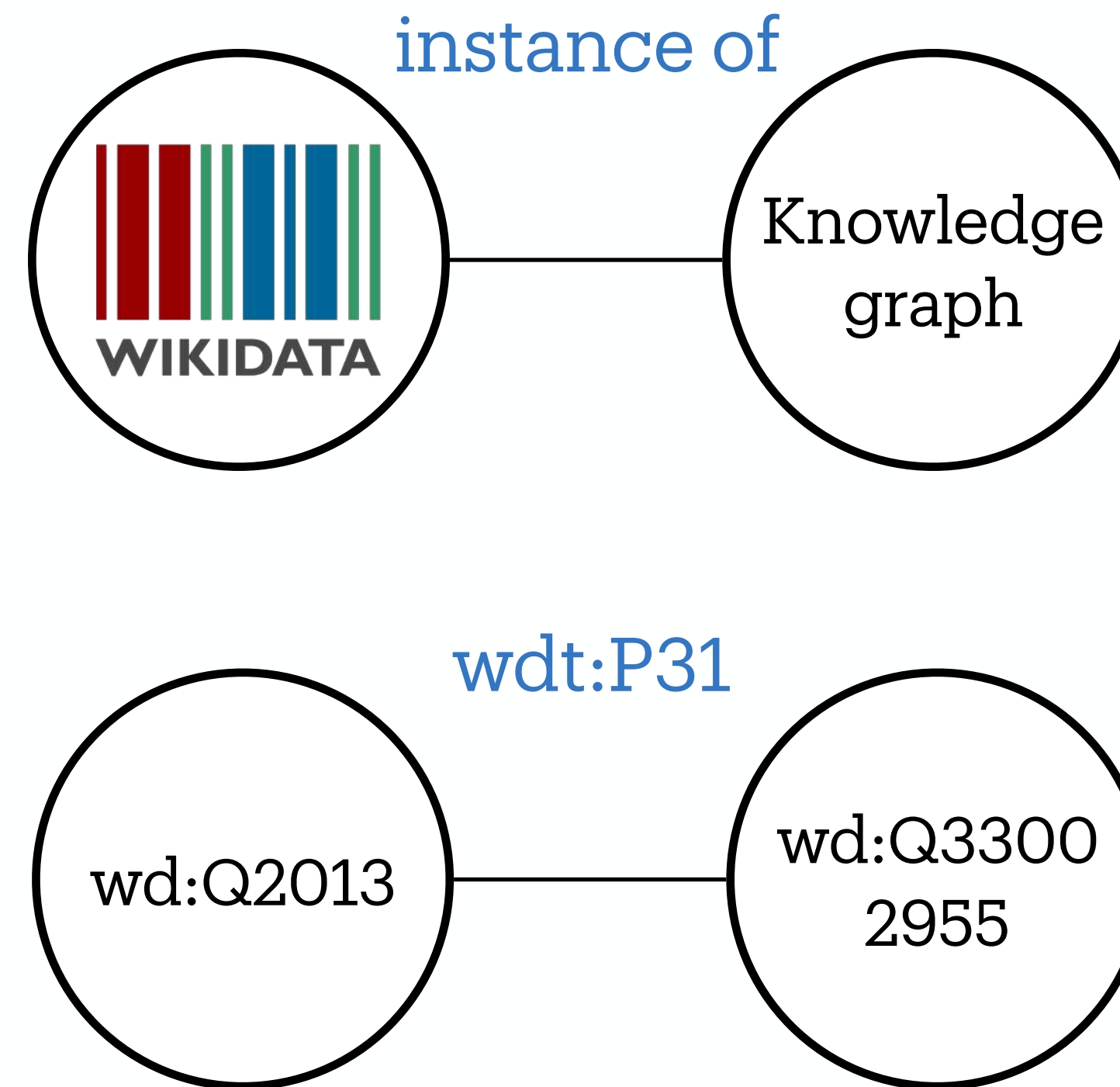
# Wikidata



# Wikidata

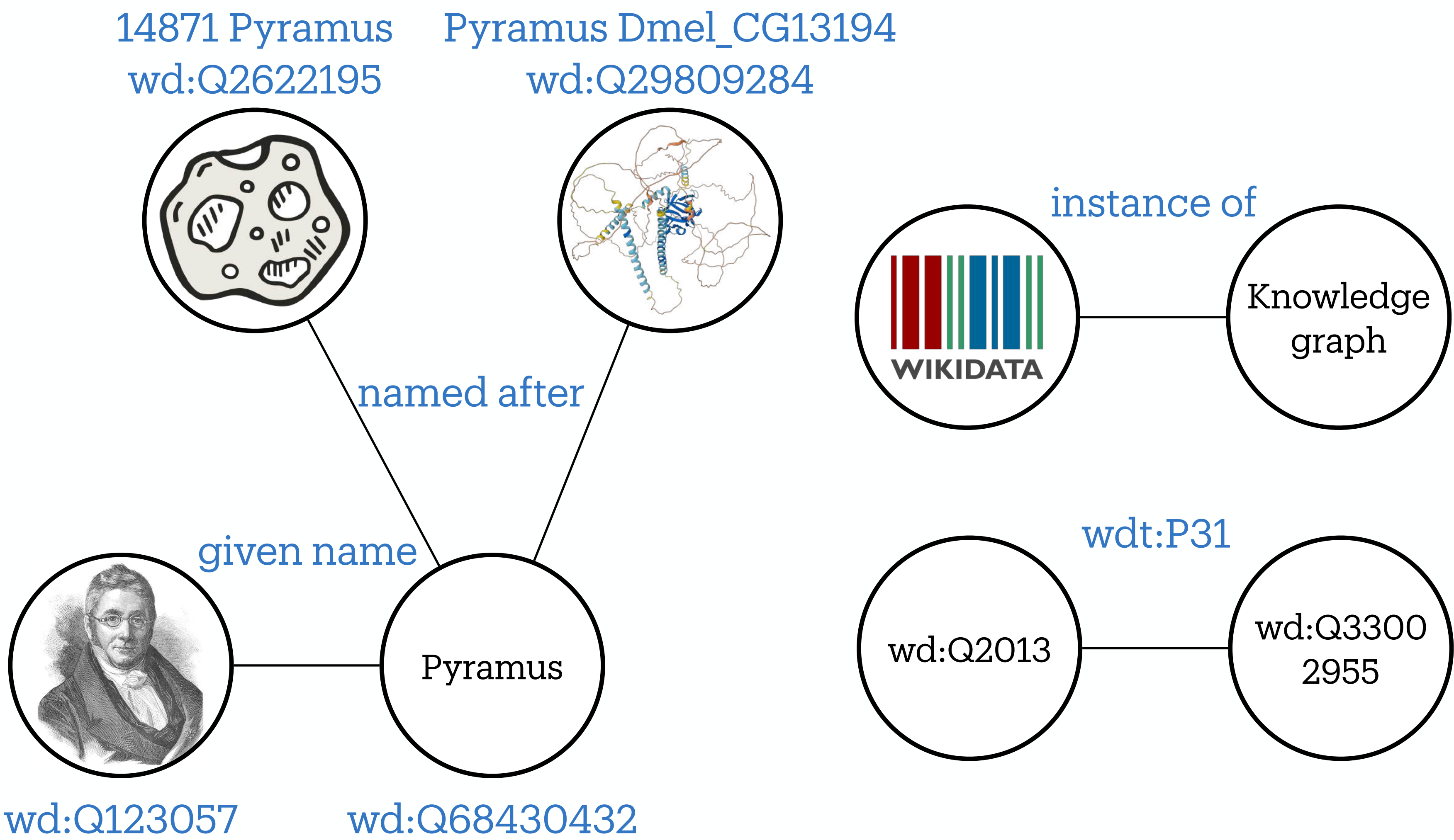


# Wikidata



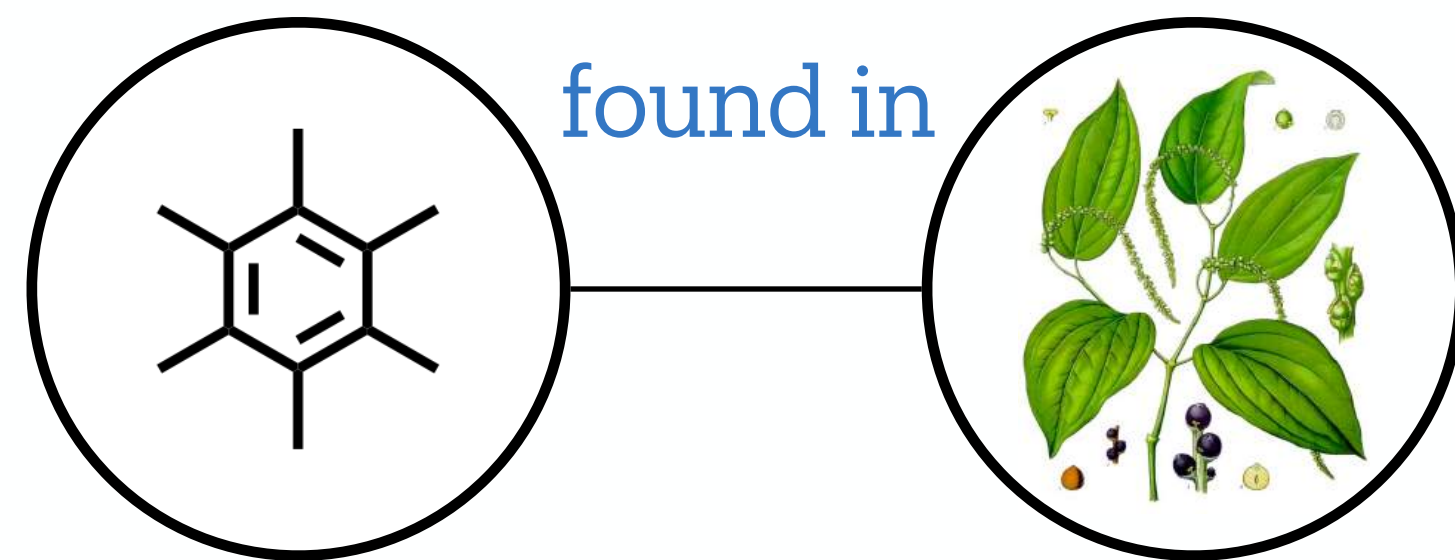


# Wikidata



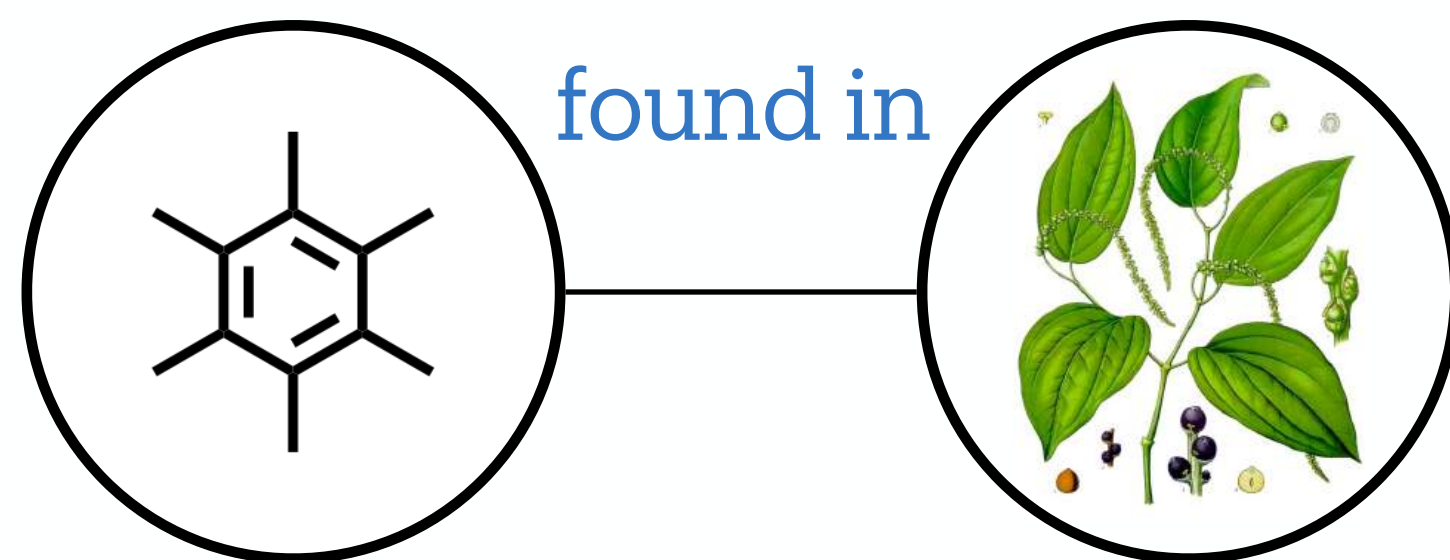
# SPARQL

« Hey Wiki, what are the compounds found in *Piper nigrum* L.? »



# SPARQL

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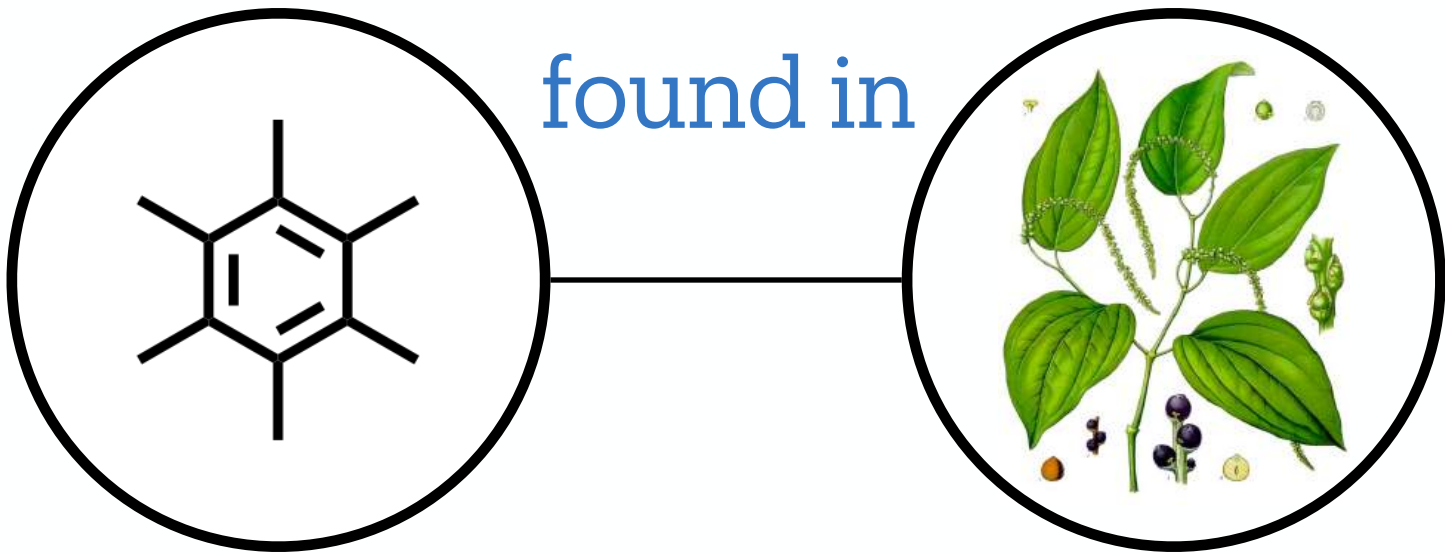


```
SELECT DISTINCT
?structure ?
structure_smiles ?
structure_inchikey WHERE
{
  VALUES ?organism {
    wd:Q43084 # Piper
nigrum
  }
  ?organism_child
(wdt:P171*) ?organism.
  ?structure wdt:P233 ?
structure_smiles;
  wdt:P235 ?
structure_inchikey;
  (p:P703/ps:P703) ?
organism_child.
}
```

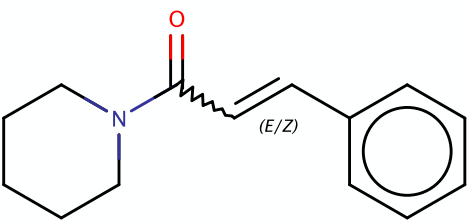
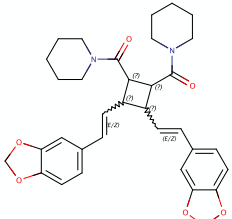


# SPARQL

« Hey Wiki, what are the compounds found in *Piper nigrum* L.? »



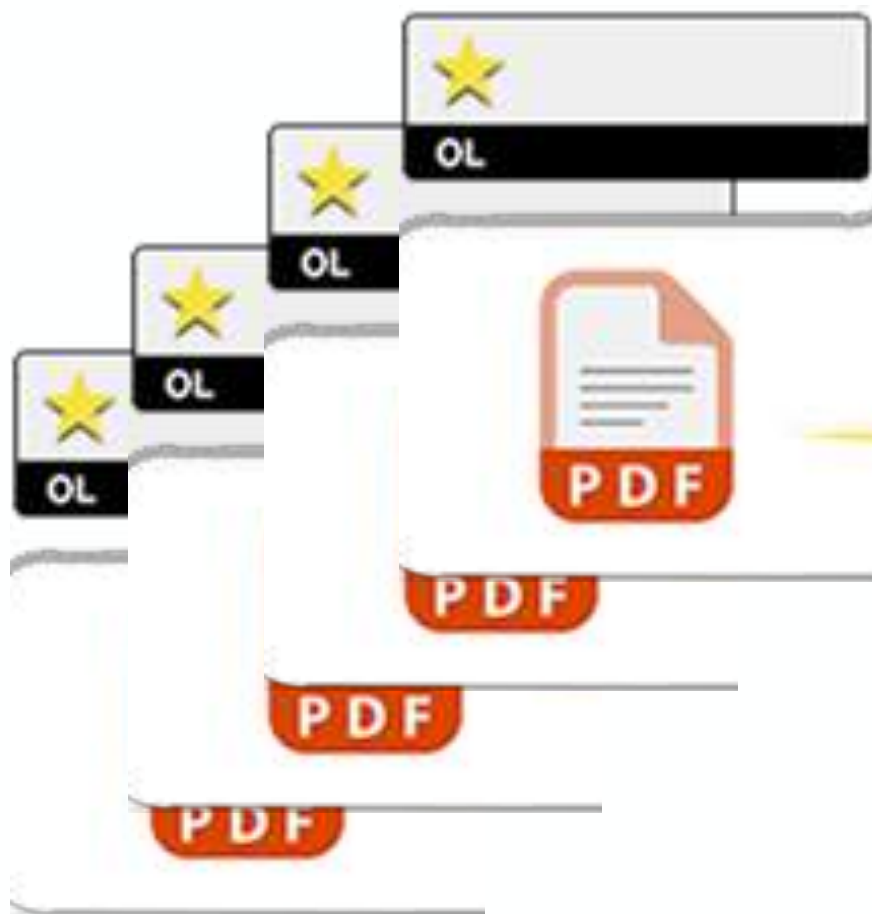
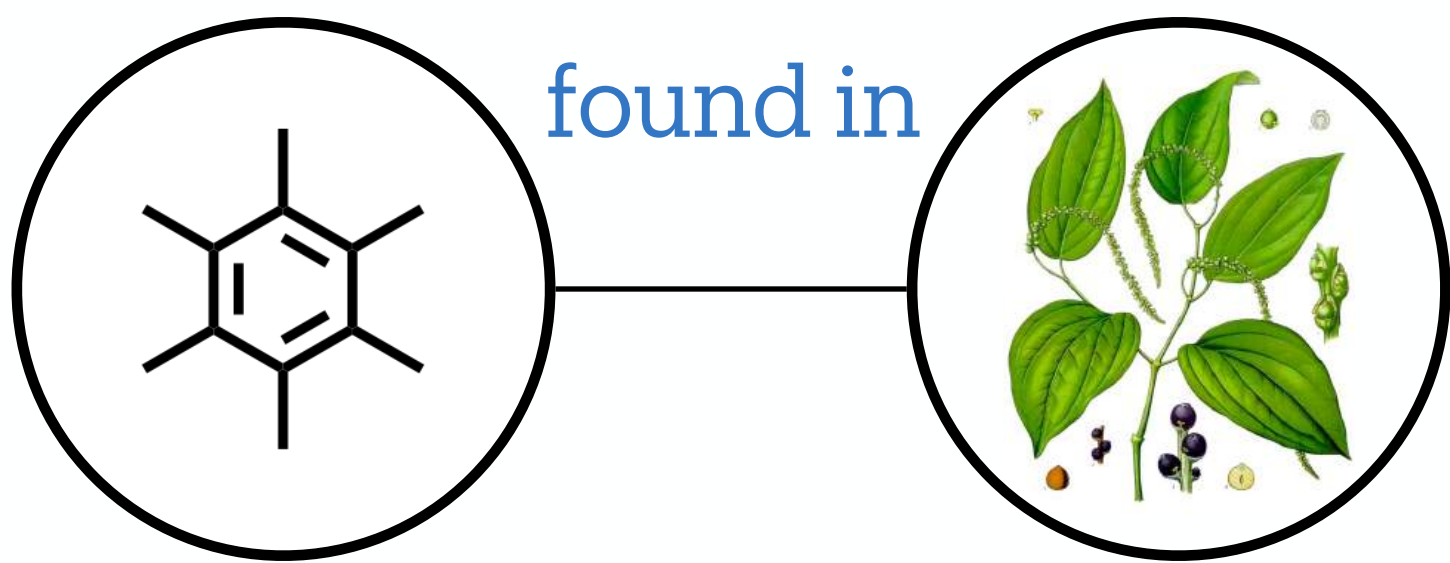
```
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?structure ?
structure_smiles ?
structure_inchikey WHERE
{
  VALUES ?organism {
    wd:Q43084 # Piper
nigrum
  }
  ?organism_child
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  ?structure wdt:P233 ?
structure_smiles;
  wdt:P235 ?
structure_inchikey;
  (p:P703/ps:P703) ?
organism_child.
}
```

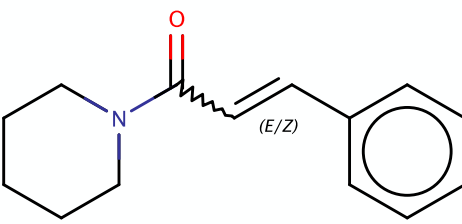
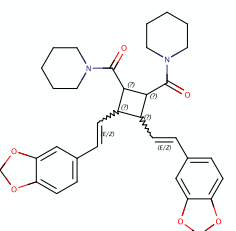
| row | structure                     | structure_smiles                                                                      | structure_inchikey          |
|-----|-------------------------------|---------------------------------------------------------------------------------------|-----------------------------|
| 1   | <a href="#">wd:Q105143500</a> |  | KNOXUMZPTHELAO-UHFFFAOYSA-N |
| ... | ...                           | ...                                                                                   | ...                         |
| 419 | <a href="#">wd:Q105224845</a> |  | QOESZPJYTVEXNN-UHFFFAOYSA-N |

419 results in 0.5 second

# SPARQL

« Hey Wiki, what are the compounds found in *Piper nigrum* L.? »



| row | structure                     | structure_smiles                                                                      | structure_inchikey          |
|-----|-------------------------------|---------------------------------------------------------------------------------------|-----------------------------|
| 1   | <a href="#">wd:Q105143500</a> |  | KNOXUMZPTHELAO-UHFFFAOYSA-N |
| ... | ...                           | ...                                                                                   | ...                         |
| 419 | <a href="#">wd:Q105224845</a> |  | QOESZPJYTVEXNN-UHFFFAOYSA-N |

```
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wdt:P235 ?
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}
```

419 results in 0.5 second

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« Hey Wiki, what are the compounds  
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# SPARQL

« Hey Wiki, what are the compounds  
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3,398 results in 0.9 second



# SPARQL

« Hey Wiki, what are the compounds found in *Ranunculaceae*? »

3,398 results in 0.9 second

« Hey Wiki, what are the compounds found in *Ranunculales*? »

6,350 results in 1.4 seconds

# SPARQL

« Hey Wiki, what are the compounds found in *Ranunculaceae*? »

3,398 results in 0.9 second

« Sorry Wiki, I forgot, how many species belong to *Ranunculales*? »

11,669 results in 1.1 seconds

« Hey Wiki, what are the compounds found in *Ranunculales*? »

6,350 results in 1.4 seconds

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« Hey Wiki, what are the compounds found in *Ranunculales*? »

6,350 results in 1.4 seconds

« Ok Wiki, among the compounds found in the *Ranunculales* how many were already described as bitter? »

59 results in 1.4 seconds



# SPARQL

« Hey Wiki, what are the compounds found in *Ranunculaceae*? »

3,398 results in 0.9 second

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# Some more examples

« Hey Wiki, I saw this review yesterday, could you do the same? »



Contents lists available at [ScienceDirect](#)

**Microbiological Research**

journal homepage: [www.elsevier.com/locate/micres](http://www.elsevier.com/locate/micres)



Actinobacteria in natural products research: Progress and prospects

Polpass Arul Jose<sup>a,\*</sup>, Anjisha Maharshi<sup>a,2</sup>, Bhavanath Jha<sup>a,b,\*\*</sup>

<sup>a</sup> Marine Biotechnology and Ecology Division, CSIR- Central Salt and Marine Chemicals Research Institute, G. B. Marg, Bhavnagar, Gujarat, 364002, India

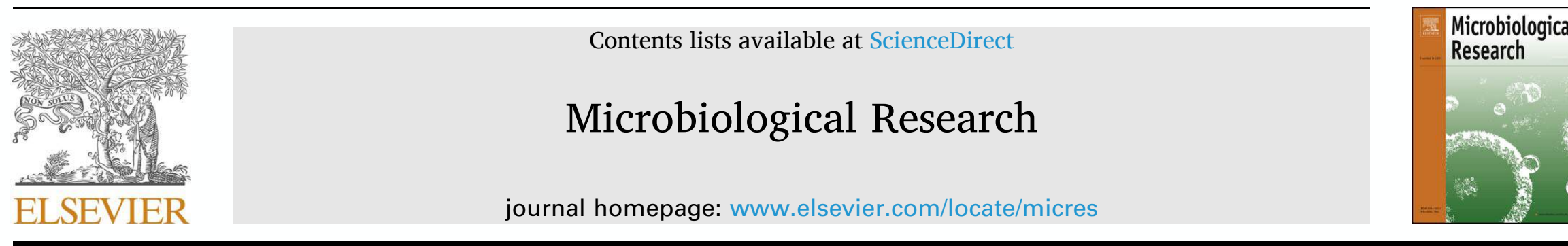
<sup>b</sup> Academy of Scientific and Innovative Research (AcSIR), CSIR, India



<https://doi.org/10.1016/j.micres.2021.126708>

# Some more examples

« Hey Wiki, I saw this review yesterday, could you do the same? »



Which compounds with known bioactivities were isolated from Actinobacteria (Q26262282), between 2014 and 2019, with related organisms and references?

Actinobacteria in natural products research: Progress and prospects

Polpass Arul Jose<sup>a,\*</sup>, Anjisha Maharshi<sup>a,2</sup>, Bhavanath Jha<sup>a,b,\*</sup>

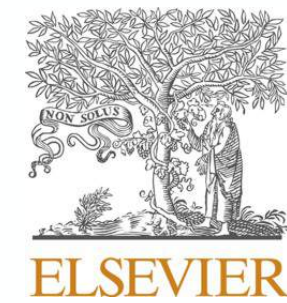
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## Actinobacteria in natural products research: Progress and prospects

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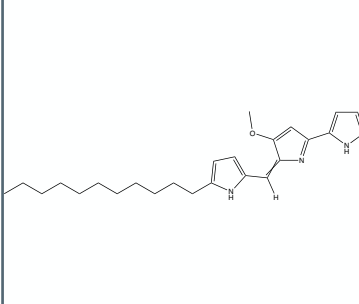
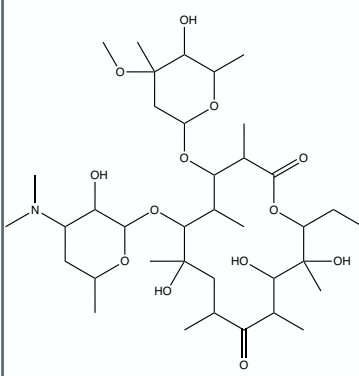
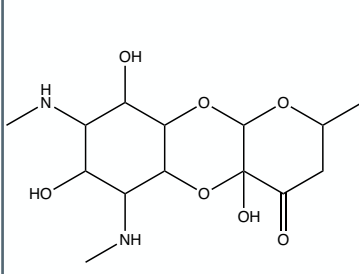
**Table 1**  
Novel bioactive compounds isolated from Actinobacteria during last five years (2014-2019).

| Organism                                                   | Compound                          | Bioactivity <sup>#</sup> | Reference                                     |
|------------------------------------------------------------|-----------------------------------|--------------------------|-----------------------------------------------|
| <i>Arthrobacter</i> sp. PGVB1                              | Arthroamide                       | f (anti-quorum sensing)  | J Nat Prod. 2015;78:2827–2831                 |
| <i>Actinoallomurus</i> sp. ID145113 and 145,206            | Paramagnetoquinones A–C (1–3)     | (1/2) a; (3) a           | J Nat Prod. 2017;80:819–827                   |
| <i>Actinoalloteichus hymeniacidonis</i> 179DD-027          | Dokdolipid B                      | c                        | Mar Drugs 2019;17:237                         |
| <i>Actinomadura atramentaria</i> NBRC 14,695 <sup>**</sup> | Cinnamycin B                      | a                        | J Ind Microbiol Biotechnol. 2016;43:1159–65   |
| <i>Actinomadura</i> sp. BCC 35,430 <sup>*</sup>            | Actinomadurone                    | b                        | Tetrahedron Lett. 2017;58:3223–3225           |
| <i>Actinomadura</i> sp. K13–0306                           | Sagamilactam                      | c, f (antiparasitic)     | J Antibiot. 2016;69:818–824                   |
| <i>Actinomadura</i> sp. K4S16                              | Nonthmicin (1) & ecteinamycin (2) | (1,2) c                  | Org Lett. 2017;19:1406–1409                   |
| <i>Actinomadura</i> sp. KC191 <sup>*</sup>                 | Actinomadurol                     | a                        | J Nat Prod. 2016;79:1886–1890                 |
| <i>Actinosynnema pretiosum</i> HGF052::asm18 <sup>**</sup> | Actinosynneptide A (1) & B (2)    | (1,2) c                  | Appl Microbiol Biotechnol. 2016;101:2273–2279 |
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Which compounds with known bioactivities were isolated from Actinobacteria (Q26262282), between 2014 and 2019, with related organisms and references?

508 results in 4.8 seconds

| Taxon name                         | Structure                                                                             | Activity                                                           | Reference                                     |
|------------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------------------|-----------------------------------------------|
| <i>Streptomyces coelicolor</i>     |  | Antibiotic   immunosuppressive drug                                | <a href="#">10.1007/S11306-016-1025-6</a>     |
| <i>Saccharopolyspora erythraea</i> |  | Antibiotic   protein synthesis inhibitors   gastrointestinal agent | <a href="#">10.1021/ACSSYNBIO.8B00372</a>     |
| <i>Streptomyces venezuelae</i>     |  | Antibiotic   protein synthesis inhibitors                          | <a href="#">10.1016/J.JBIOTEC.2014.01.028</a> |



# Some more examples

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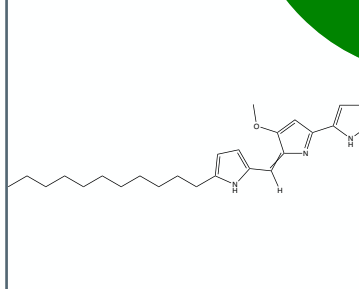
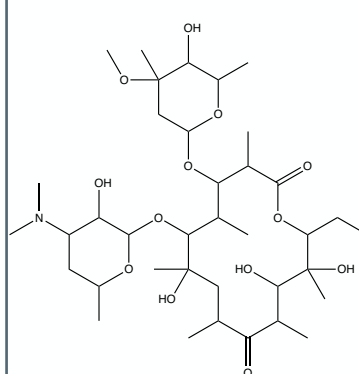
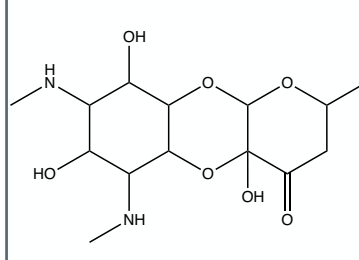
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# Available multiomics knowledge

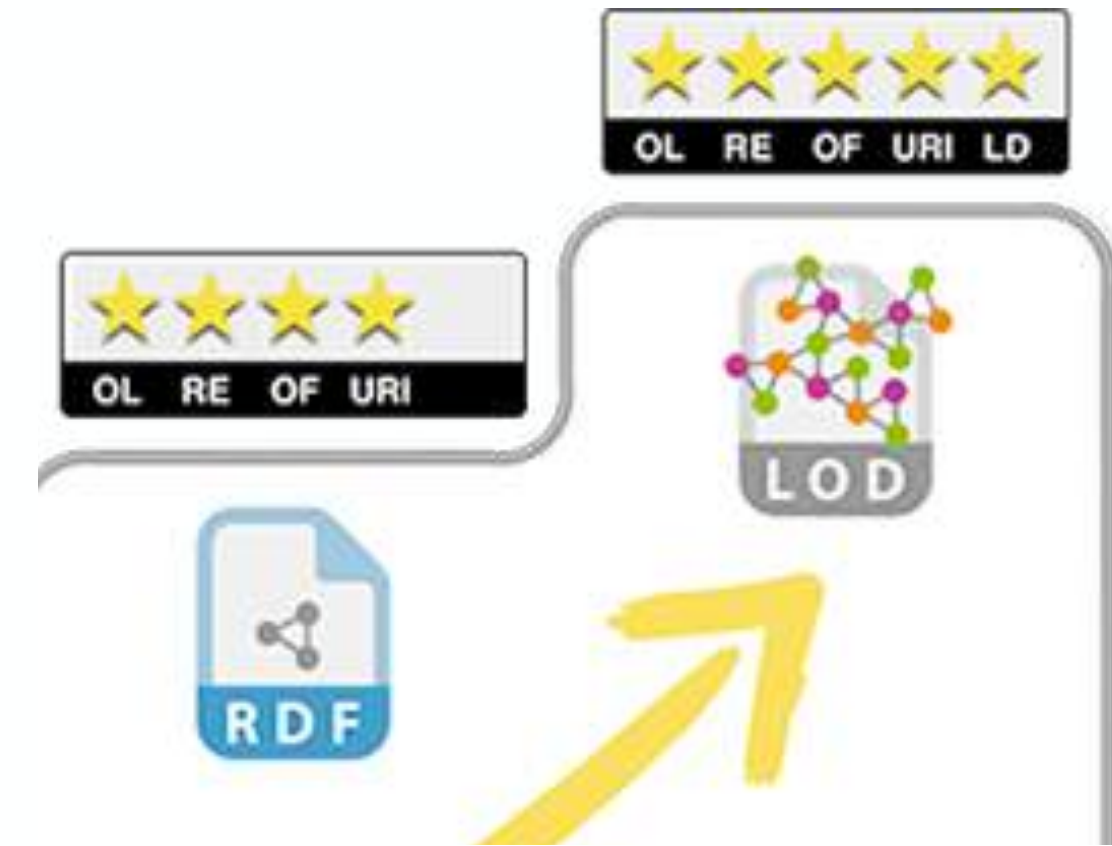
«Hey Wiki, for how many taxa do you have information about at least 1 gene, 1 metabolite and 1 protein?»

15 results in 4.7 seconds

|    | ?taxon                    | ?taxon_name               | ?genome | ?metabolome | ?proteome |
|----|---------------------------|---------------------------|---------|-------------|-----------|
| 1  | <a href="#">Q10758650</a> | Equus caballus            | 3       | 1           | 1         |
| 2  | <a href="#">Q130888</a>   | Drosophila melanogaster   | 25,180  | 5           | 11,314    |
| 3  | <a href="#">Q150162</a>   | Trypanosoma cruzi         | 21,465  | 1           | 22,939    |
| 4  | <a href="#">Q158695</a>   | Arabidopsis thaliana      | 28      | 1,007       | 5         |
| 5  | <a href="#">Q15978631</a> | Homo sapiens              | 67,527  | 1,877       | 25,433    |
| 6  | <a href="#">Q169444</a>   | Danio rerio               | 31,432  | 1           | 14,067    |
| 7  | <a href="#">Q184224</a>   | Rattus norvegicus         | 52,931  | 20          | 15,805    |
| 8  | <a href="#">Q184774</a>   | Gallus gallus             | 1       | 7           | 3         |
| 9  | <a href="#">Q1950752</a>  | Leishmania donovani       | 1       | 2           | 3,963     |
| 10 | <a href="#">Q2236682</a>  | Schizosaccharomyces pombe | 2       | 8           | 4,061     |
| 11 | <a href="#">Q25419</a>    | Escherichia coli          | 2       | 949         | 6         |
| 12 | <a href="#">Q58697</a>    | Sus scrofa                | 1       | 4           | 2         |
| 13 | <a href="#">Q719725</a>   | Saccharomyces cerevisiae  | 3       | 108         | 5         |
| 14 | <a href="#">Q83310</a>    | Mus musculus              | 73,593  | 95          | 19,644    |
| 15 | <a href="#">Q91703</a>    | Caenorhabditis elegans    | 47,274  | 765         | 17,673    |

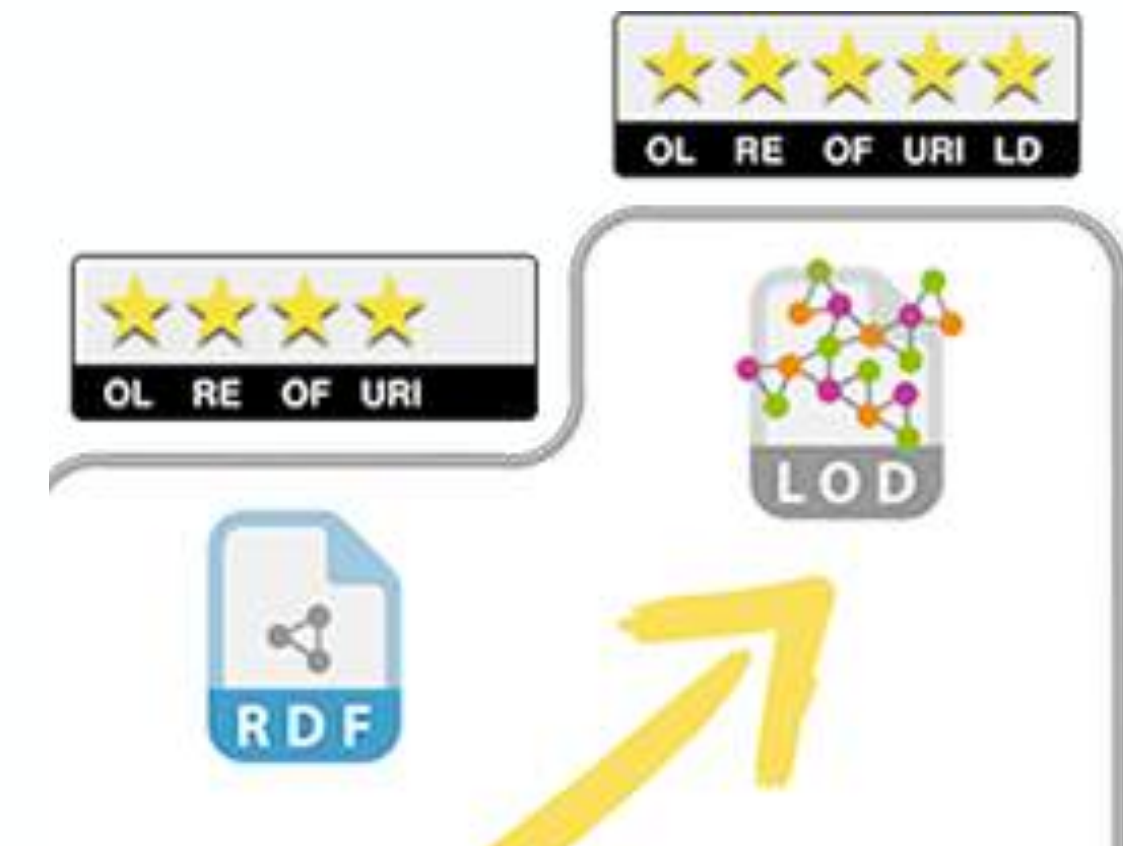
# Linked Open Data-enabled federation

« Hey Wiki, which genera contain compounds structurally similar to the ones reported as pungent in black pepper, and how many? List the top 5 only »



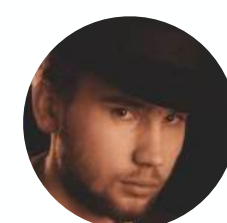
# Linked Open Data-enabled federation

« Hey Wiki, which genera contain compounds structurally similar to the ones reported as pungent in black pepper, and how many? List the top 5 only »



5 results in 1.1 seconds

| row | taxon                       | taxon_name       | count |
|-----|-----------------------------|------------------|-------|
| 1   | <a href="#">wd:Q159536</a>  | <i>Piper</i>     | 35    |
| 2   | <a href="#">wd:Q163563</a>  | <i>Zingiber</i>  | 31    |
| 3   | <a href="#">wd:Q133290</a>  | <i>Syzygium</i>  | 4     |
| 4   | <a href="#">wd:Q2607361</a> | <i>Aframomum</i> | 4     |
| 5   | <a href="#">wd:Q27583</a>   | <i>Bellis</i>    | 3     |



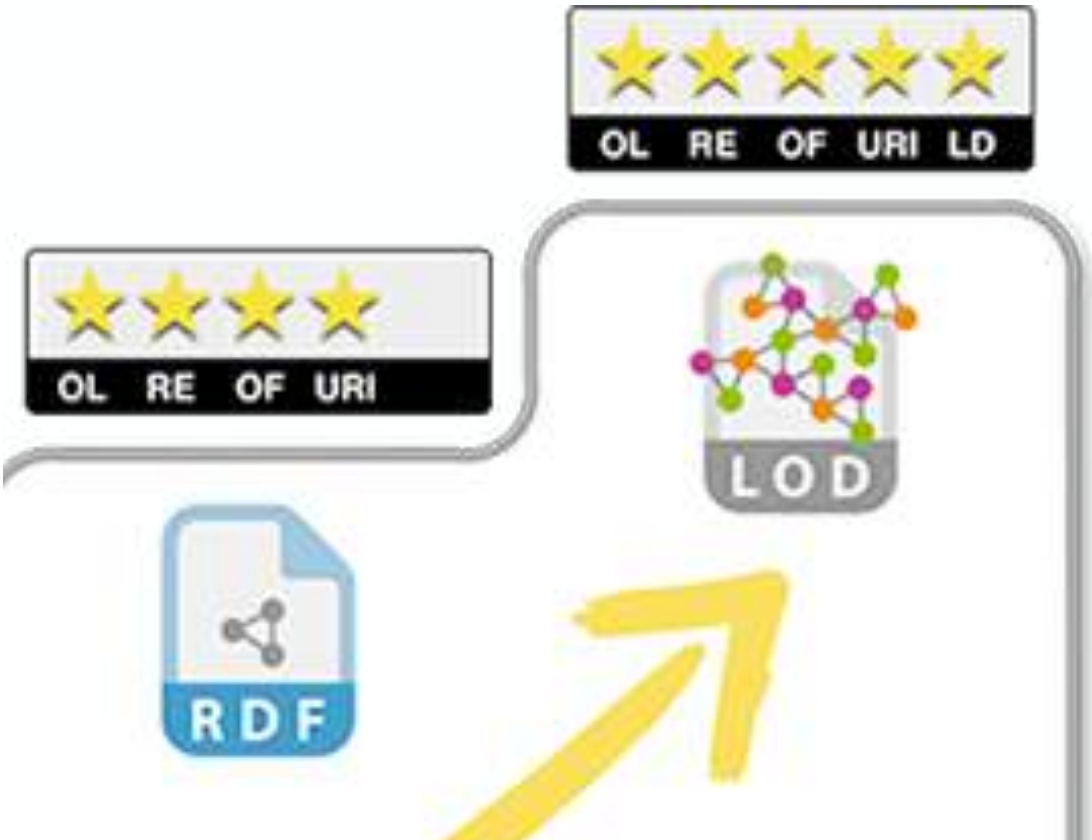
<https://doi.org/10.1186/s13321-018-0282-y>

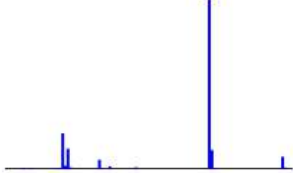
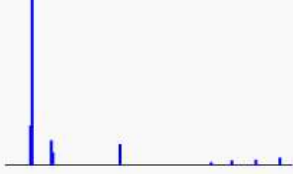
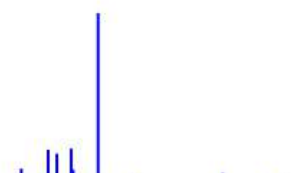




# Linked Open Data-enabled federation

« Hey Wiki, list mass spectra of things similar to positive bioassay results »



| SPECTRUM |                                                                                                                      | MSVALUE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----------|----------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1        |  <a href="#">mona:AU202803_MS</a>   | "188.1043:100"^^ <a href="#">ms:spectrum</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2        |  <a href="#">mona:AU154503_MS</a>  | "188.1428:100 189.1459:8.509666 216.1378:8.315502 217.1415:1.020664 337.2283:14.375924 338.2312:2.634483"^^ <a href="#">ms:spectrum</a>                                                                                                                                                                                                                                                                                                                                                      |
| 3        |  <a href="#">mona:AU154504_MS</a> | "117.0566:0.848243 117.0691:0.759028 120.0799:0.752058 130.0639:0.60778 132.08:20.82343 133.0831:2.088895 134.0956:11.968105 135.0988:1.113101 146.0954:5.500687 147.0992:0.595931 150.0903:1.632363 160.1112:1.145163 188.1428:100 189.1457:11.144257 190.1489:0.575021 216.1374:7.227144 217.1413:0.968823"^^ <a href="#">ms:spectrum</a>                                                                                                                                                  |
| 4        |  <a href="#">mona:JP005261_MS</a> | "56:23.3 57:99.99 69:14.7 70:7.68 112:12.4 169:2 182:3.1 183:0.82 197:3.5 212:4.7"^^ <a href="#">ms:spectrum</a>                                                                                                                                                                                                                                                                                                                                                                             |
| 5        |  <a href="#">mona:AU387304_MS</a> | "117.0214:9.888005 119.0192:2.077807 129.0693:1.399941 141.0703:4.450339 145.0531:20.910698 147.0505:5.157678 152.0614:1.75361 153.0695:5.275567 154.0781:18.611849 155.0837:2.298851 167.0872:1.562039 169.1016:21.52962 170.1051:3.477748 171.0318:9.386973 172.0366:1.296788 173.03:1.886236 182.0716:1.370469 197.0964:100 198.0999:14.839375 200.0597:1.945181 233.0377:2.180961 251.0468:1.164162 327.0796:7.132331 328.0817:2.004126 384.1508:2.564103"^^ <a href="#">ms:spectrum</a> |



<https://doi.org/10.1186/s13321-018-0282-y>

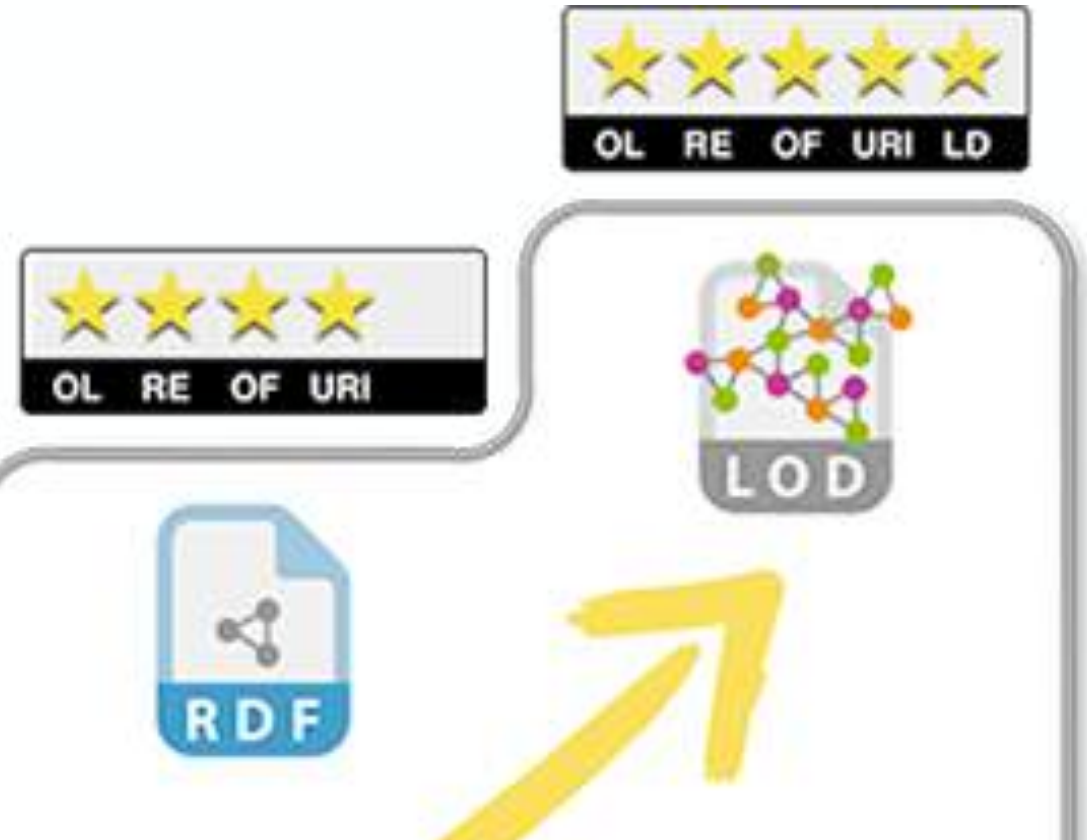




# Linked Open Data-enabled federation



«Hey Wiki, list traits (and their values) of plants producing diterpenoids»



## Query results:

147,817 lines found

8,163ms in total

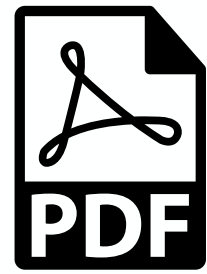
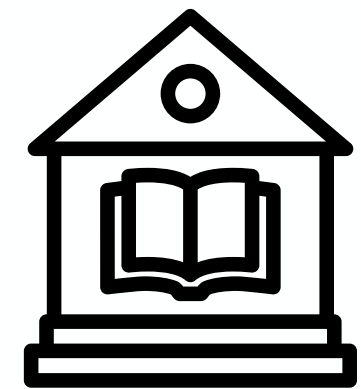
8,156ms for computation

7ms for resolving and sending

Limited to 100 results; show all 147,817 results

|   | ?source_wdx | ?sourceName                     | ?tryDataLab             | ?tryDataVal | ?unit   | ?unitComment   |
|---|-------------|---------------------------------|-------------------------|-------------|---------|----------------|
| 1 | Q1000854    | Salvia miltiorrhiza             | Plant height vegetative | 40-80 cm    | M       | m              |
| 2 | Q1009995    | Plectranthus<br>scutellarioides | Seed dry mass           | 0.28        | MilliGM | g / 1000 seeds |
| 3 | Q1009995    | Plectranthus<br>scutellarioides | Seed dry mass           | 0.2812      | MilliGM | g/1000         |
| 4 | Q1009995    | Plectranthus<br>scutellarioides | Seed dry mass           | 0.29        | MilliGM | g/1000         |

# The Earth Metabolome Initiative (EMI)

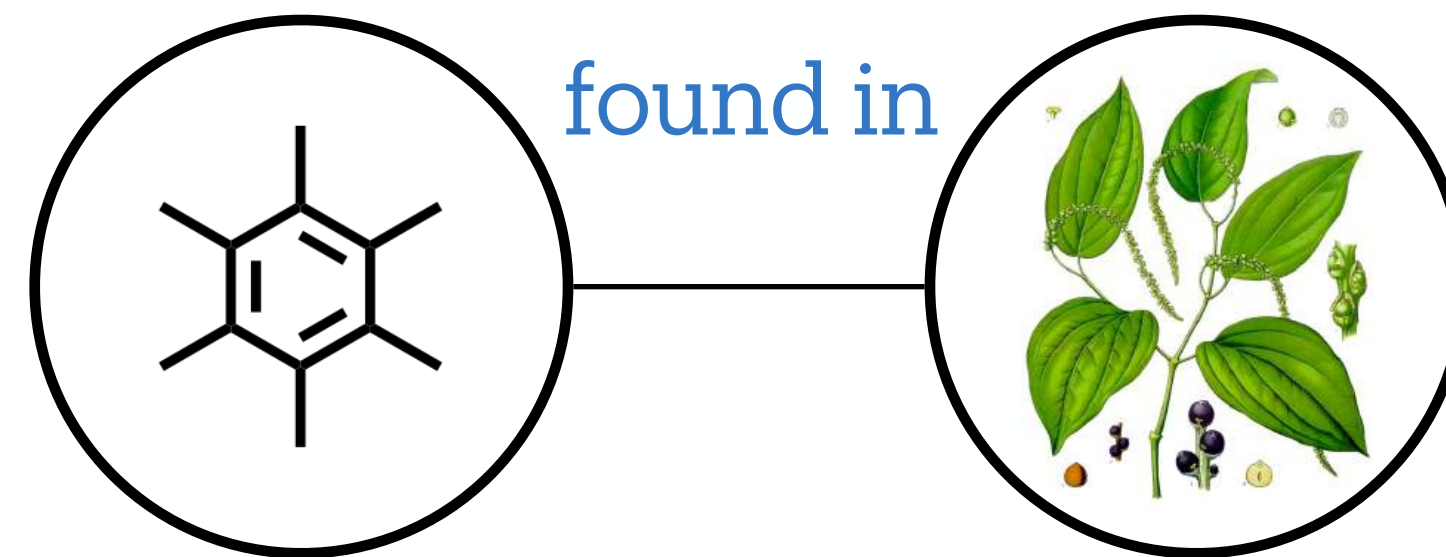
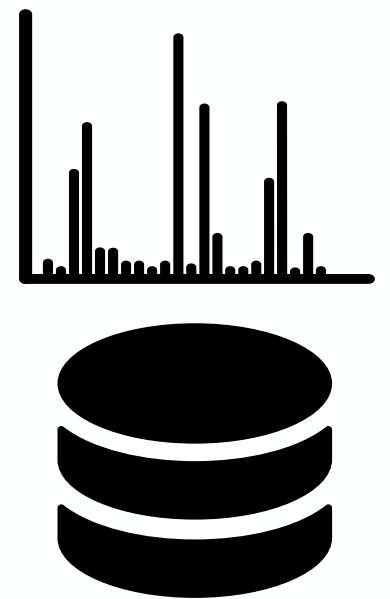


LOTUS

Prior Knowledge



Future Knowledge



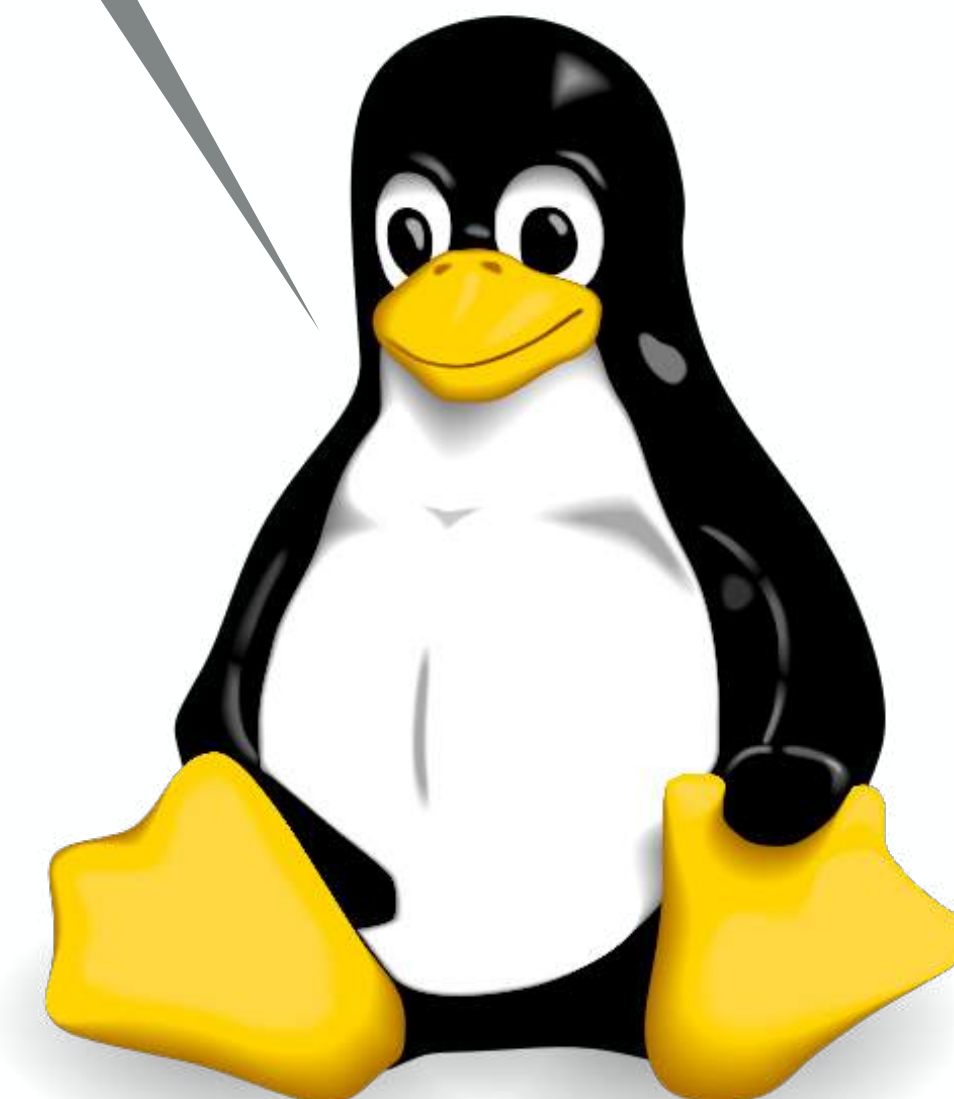
<https://www.earthmetabolome.org/>

# Building knowledge together



## Linus's law

« Given enough eyeballs,  
all bugs are shallow. »





# Community curation



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View history

Search Wikidata

hyperelodione D (Q116482353)

chemical compound

edit

| Language | Label           | Description       | Also known as |
|----------|-----------------|-------------------|---------------|
| English  | hyperelodione D | chemical compound |               |

Statements

instance of

chemical compound

0 references

mass

548.797 dalton

1 reference

based on heuristic

inferred from SMILES

chemical formula

$\text{C}_{36}\text{H}_{52}\text{O}_4$

1 reference

based on heuristic

inferred from SMILES



# Community curation



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- [Community portal](#)
- [Project chat](#)
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| Item | Discussion |
|------|------------|
|------|------------|

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[View history](#)

Search Wikidata

hyperelodione D (Q116482353)

chemical compound

 [edit](#)

| Language | Label           | Description       | Also known as |
|----------|-----------------|-------------------|---------------|
| English  | hyperelodione D | chemical compound |               |

canonical SMILES

C/C(C)=C\CC/C(C)=C/CC1=C2C3(O[H])C(C4(C/C=C(C)/CC/C=C(C)/C)/C1=O)(O[H])C(C(C)=CC4)([H])CC3([H])C(C)(C)O2

reason for deprecated rank

structural reassignment

1 reference

stated in

Cytotoxic polyprenylated  
phloroglucinol derivatives from  
*Hypericum elodeoides* Choisy  
modulating the transactivation of  
RXR $\alpha$

C/C(C)=C/CC1=C2C3(O[H])C(C4(C/C=C(C)/CC/C=C(C)/C)C1=O)(O[H])C(C(C)=CC4)([H])CC3([H])C(CC/C=C(C)/C)(C)O2

reason for preferred rank

structural reassignment

1 reference

stated in

# Bioinspired Total Synthesis of Erectones A and B, and the Revised Structure of Hyperelodione D

# Community curation

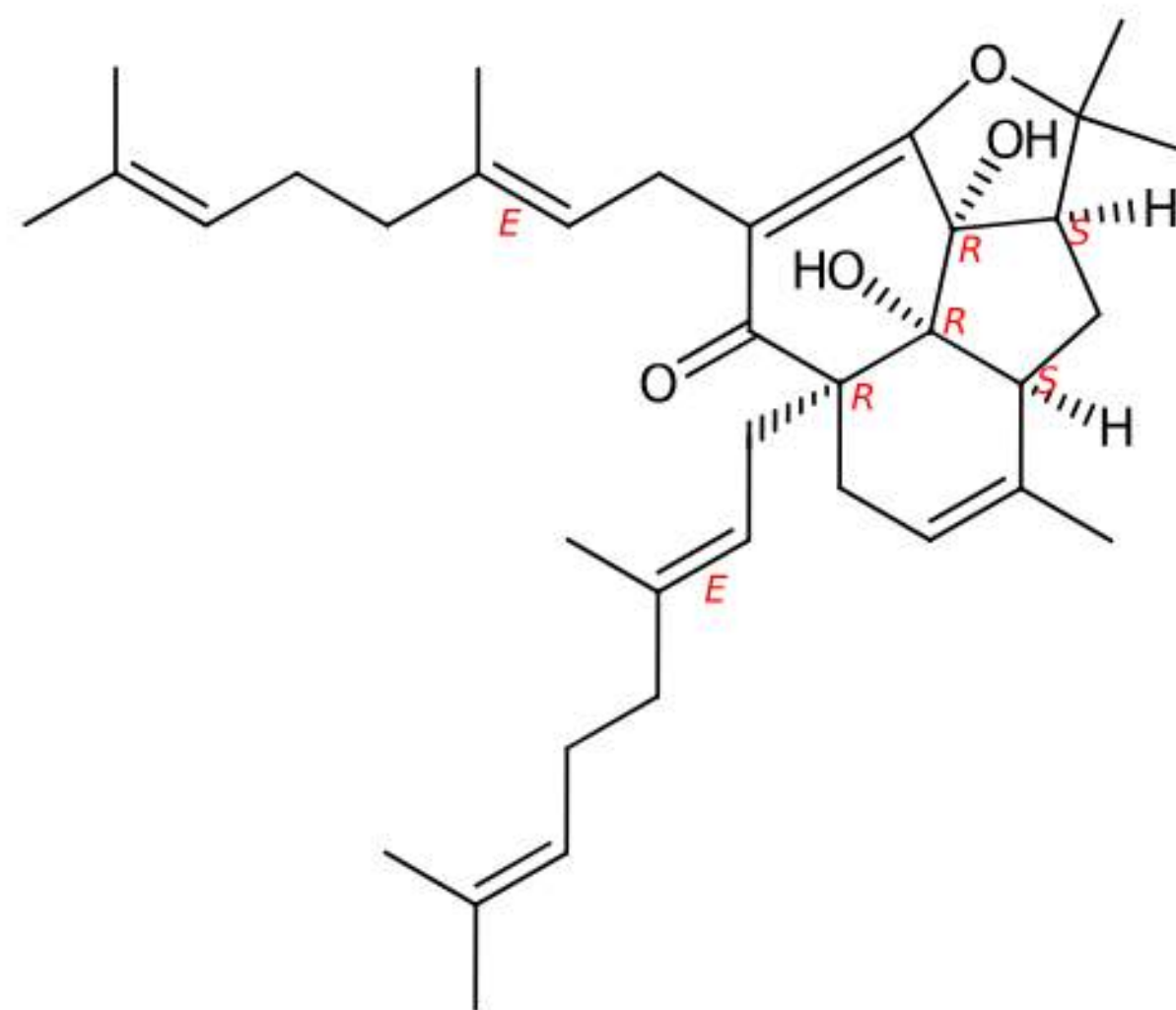
## isomeric SMILES

by AdrianoRutz and MsynBot

 C/C(C)=C\CC/C(C)=C/CC1=C2[C@]3(O[H])[C@@]([C@]4(C/C=C(C)/CC/C=C(C)/C)C1=O)(O[H])[C@@](C(C)=CC4)([H])C[C@]3([H])C(C)(C)O2

reason for deprecated rank

structural reassignment



Author display: <https://www.wikidata.org/wiki/User:Ricordisamoa/WikidataTrust.js>

Chemical depiction: [https://www.wikidata.org/wiki/User:Egon\\_Willighagen/cdkdepict\\_gadget.js](https://www.wikidata.org/wiki/User:Egon_Willighagen/cdkdepict_gadget.js)



# Modern knowledge dissemination

 ✨ SPARQLing NPs all over ✨  
@daily\_lotus@mastodon.social

@daily\_lotus is a bot on a mission:

Every day, it toots a real-world natural product — a molecule found in 🌿 plants, 🍄 fungi, 🐛 animals, or 🦠 bacteria — straight from the depths of Wikidata.

Each post includes:

- ✍ the molecule's name and structure,
- 📷 a pic of the organism it was found in,
- 📖 a reference to the scientific publication documenting the occurrence .

#introduction #OpenScience  
#LinkedOpenData #LOTUS  
#NaturalProducts

🗨 thread 1/n

[https://mastodon.social/@daily\\_lotus](https://mastodon.social/@daily_lotus)

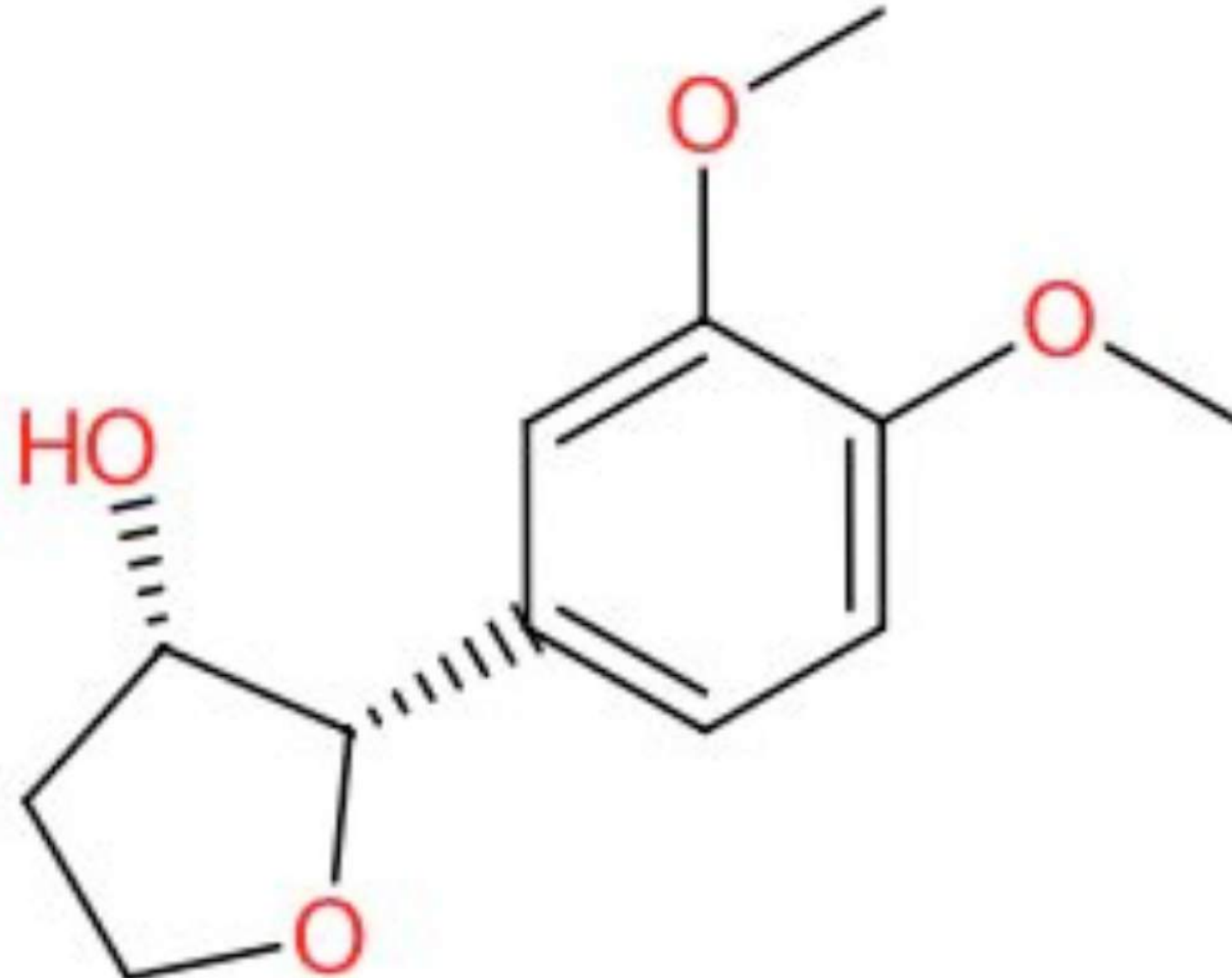
 **The Daily NP**  
2 followers  
3d

✓ (2S,3S)-2-(3,4-dimethoxyphenyl)oxolan-3-ol [<https://lnkd.in/e7yyC-JB>] is a molecule found in a 🌿 plant, Zingiber cassumunar [<https://lnkd.in/eNueSjMJ>]  
📖 according to: [<https://lnkd.in/en6WSF4t>]

✍ This occurrence is available for curation on Wikidata [<https://lnkd.in/e2-BCTG3>]. If you spot an error, feel free to improve it!

#DailyNP #OpenScience #LOTUS #WikiData

🔬 At the Earth Metabolome Initiative we seek for these little treasures in Nature  
<https://lnkd.in/eBkApGGT>  
#EarthMetabolome



<https://www.linkedin.com/showcase/the-daily-np>



# Modern knowledge dissemination

- `{{#invoke:P703|main}}`

no known organism

- `{{#invoke:P703|main|qid=Q2079986|sources=false}}`

6 organisms, including: [Picrasma crenata](#), [Picrasma javanica](#), [Picrasma quassioides](#), [Quassia africana](#) ([war](#)), [Quassia amara](#)

- `{{#invoke:P703|main|qid=Q2079986|numval=1}}`

6 organisms, including: [Picrasma crenata](#)<sup>[1]</sup>

- `{{#invoke:P703|main|qid=Q2079986|numval=999|sources=false}}`

6 organisms: [Picrasma crenata](#), [Picrasma javanica](#), [Picrasma quassioides](#), [Quassia africana](#) ([war](#)), [Quassia amara](#), [Samadera indica](#)

## Preview of references

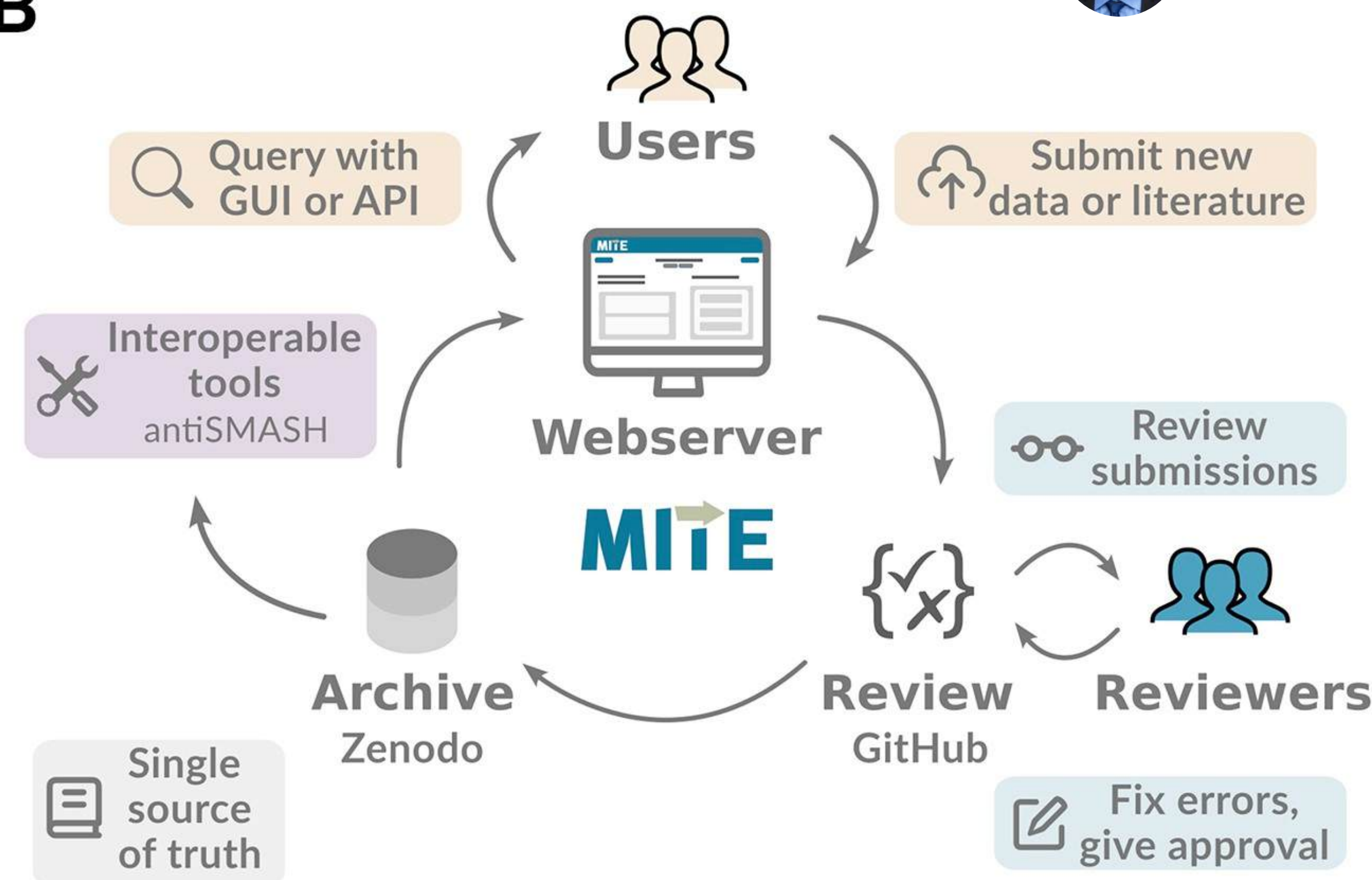
1. <sup>^</sup> Claudio R Novello, Antonio G Ferreira, Luis C Marques, et al.. "Quassinoids from *Picrasma crenata*". *Natural Product Research* **17** (3): pp. 145-148. (2003). doi:10.1080/1478641031000104082. pmid:12737395.



# Minimum Information about a Tailoring Enzyme (MITE)

**A**

The screenshot displays the MITE database interface for entry MITE0000092. The top navigation bar includes links for Repository, Documentation, Utilities, About, and Contact. The entry is titled MITE0000092 with buttons for 'Modify entry' and 'Download entry'. The left sidebar shows 'Reaction Information' with 'Number of Known Reactions: 1' and a 'Select a reaction' dropdown. The main content area shows 'Reaction SMARTS' and 'Reaction Information' tabs. Under 'Reaction SMARTS', there is a chemical reaction scheme (2) showing a substrate being converted to a product with a chlorine or bromine atom. Below this is 'Example 1' (3) showing the 'Substrate' and 'Product 1' with their respective chemical structures and 'Copy SMILES' buttons. The right sidebar shows 'Enzyme Information' with tabs for 'General Information' and 'Changelog'. The 'General Information' tab lists: Accession ID (MITE0000092), Entry Status (Status: active), Enzyme Name (BorH), Enzyme Description (Flavin-dependent halogenase), Database References (UniProtKB: M9QSI0, NCBI GenPept: AGI62217.1, MIBiG: BGC0001335), Cofactors (Organic FAD), and References (doi:10.1002/cbic.201900667).

**B**

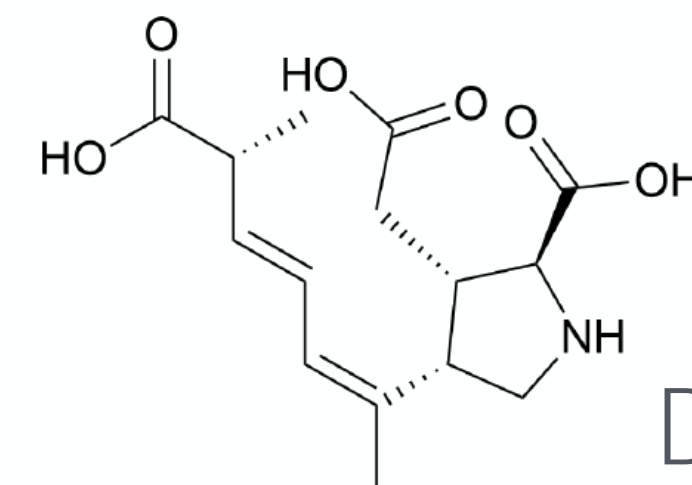
<https://mite.bioinformatics.nl/>

<https://doi.org/10.1093/nar/gkaf969>

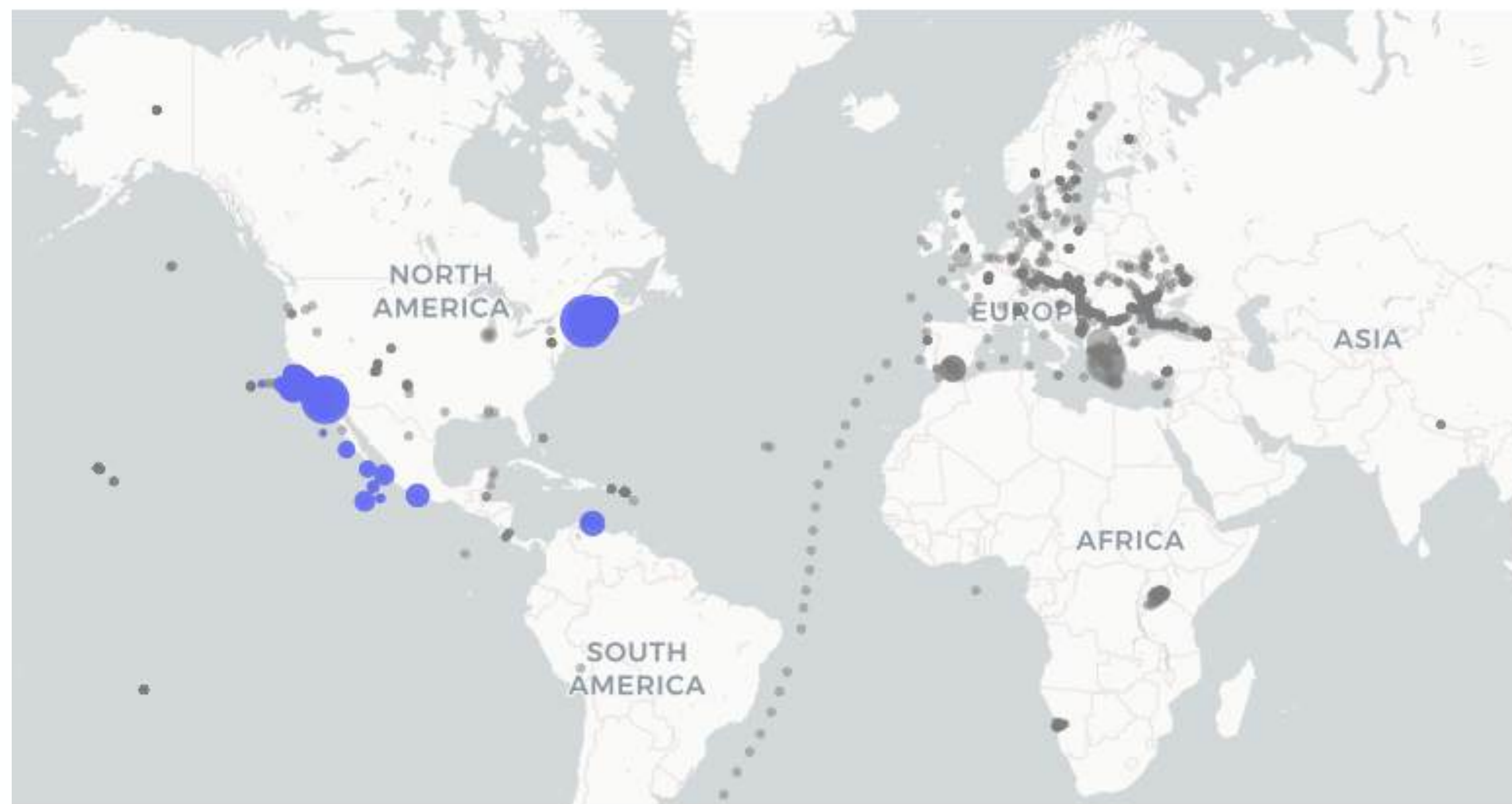


# GeoMASST call

Contribute your environmental **LC-MS DDA** data, make them searchable for the community, and **become a co-author!**



Domoic acid

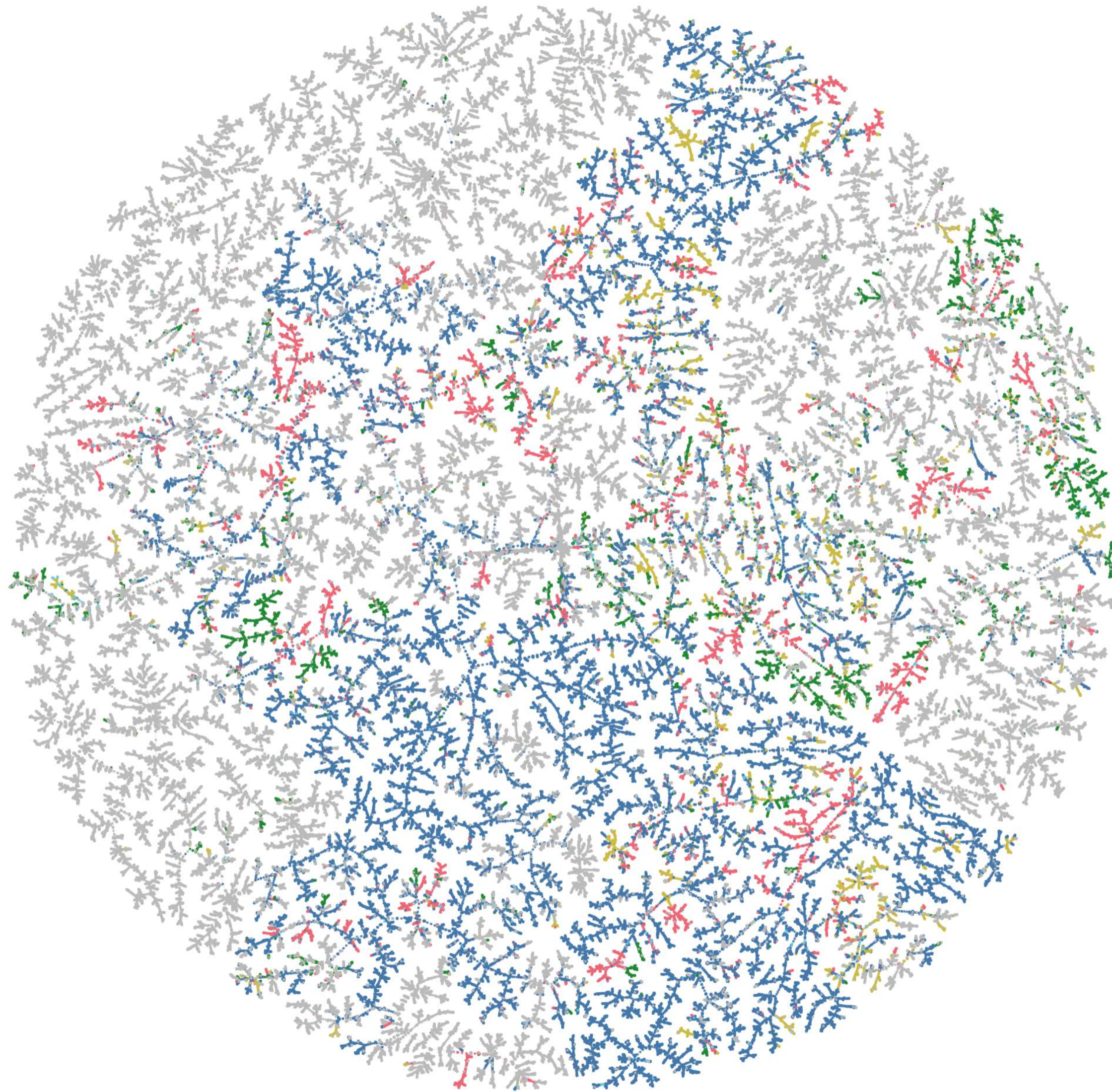


Reach out to [yasin.el-abiead@boku.ac.at](mailto:yasin.el-abiead@boku.ac.at)





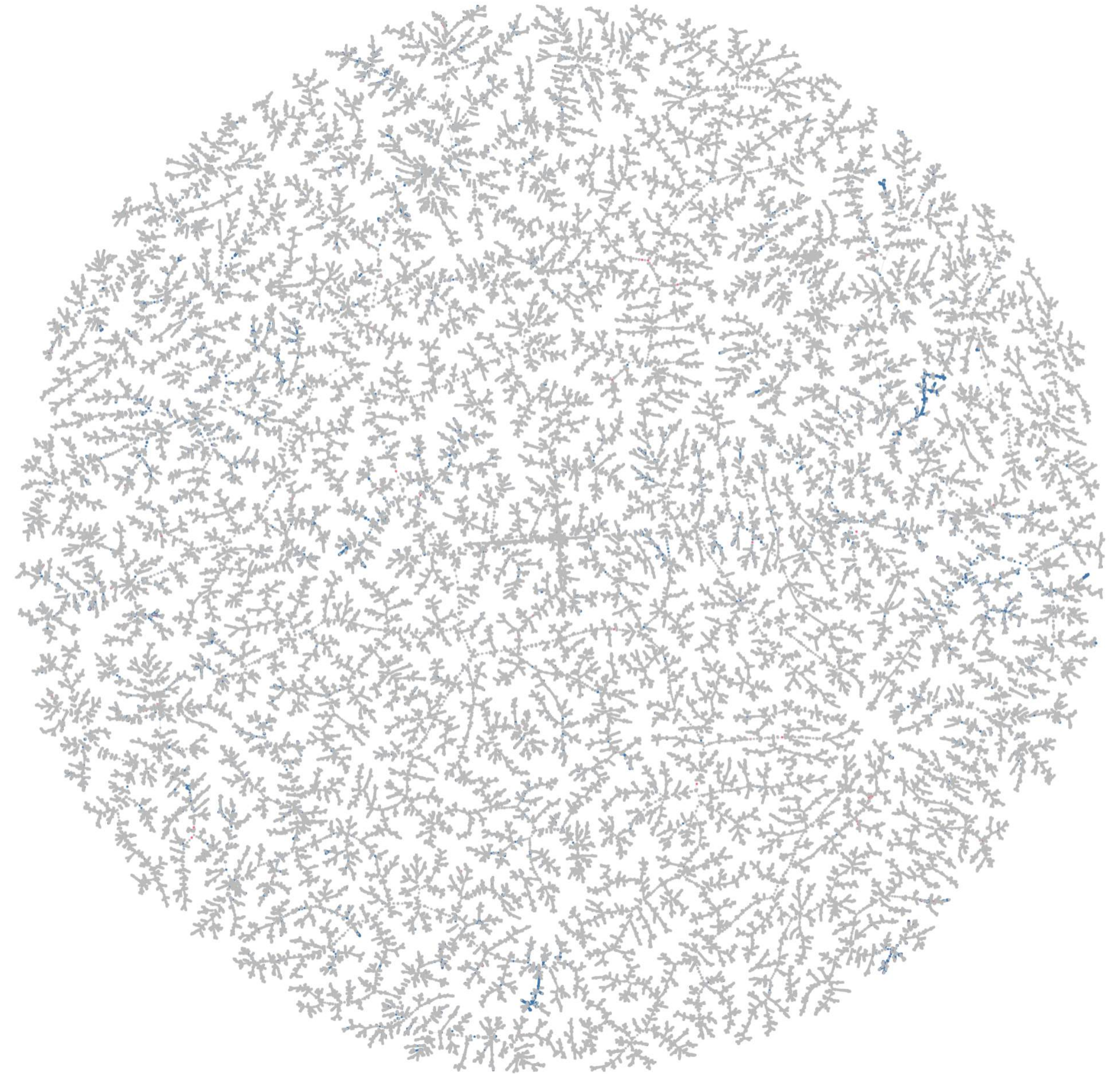
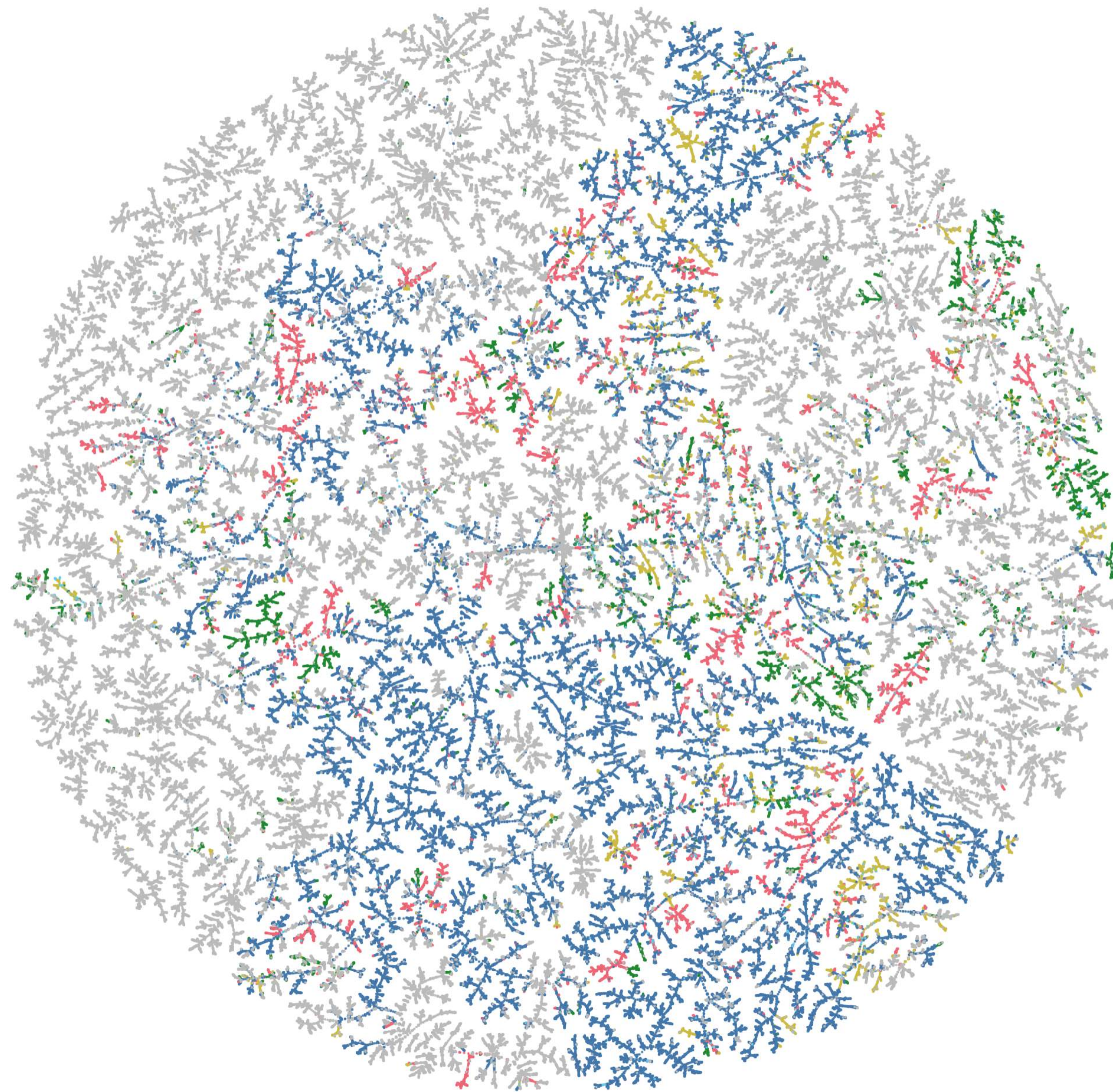
# About available spectral and taxonomical data...



● Plantae ● Fungi ● Bacteria ● Animalia ● Shared



# About available spectral and taxonomical data...

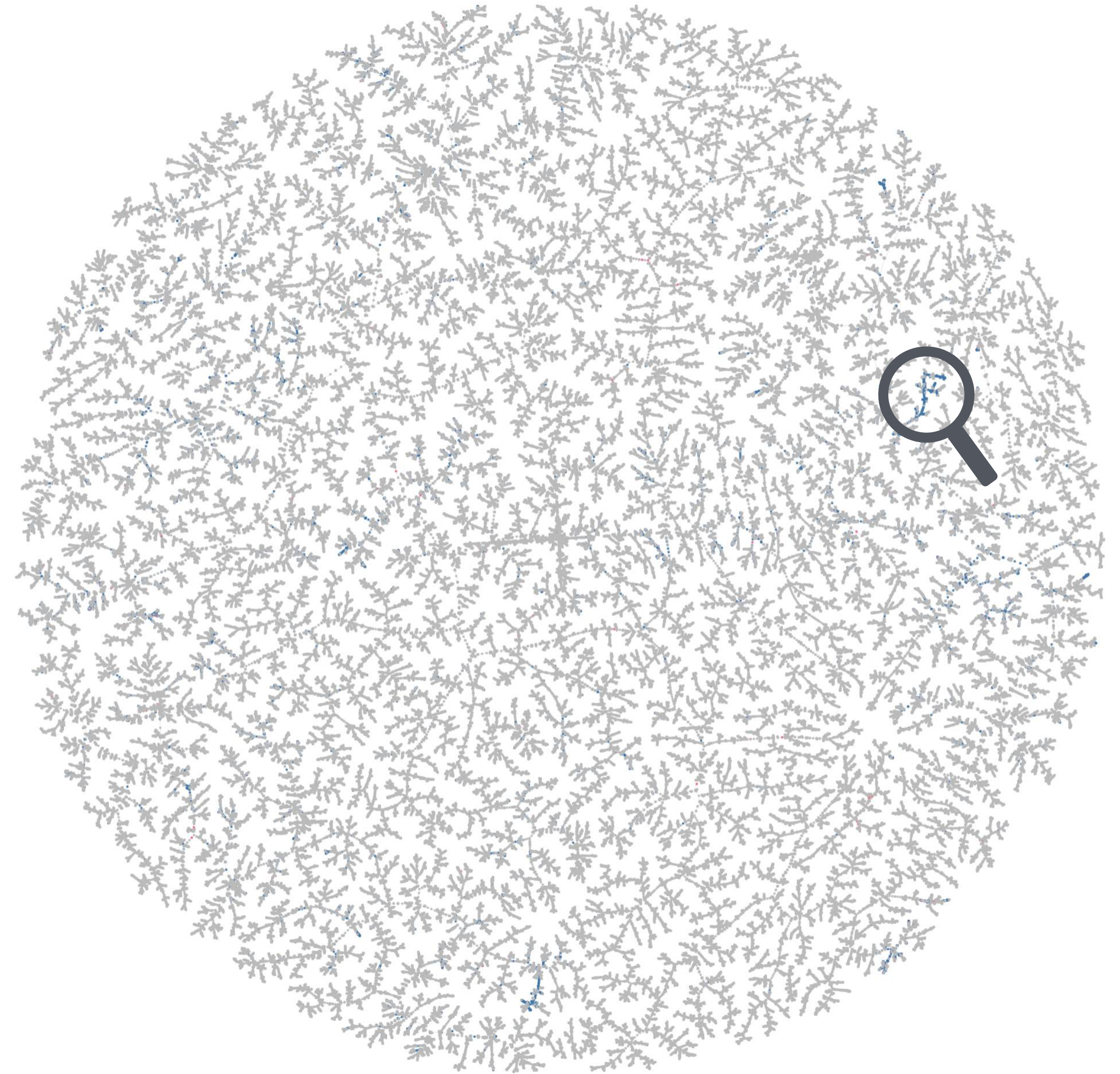
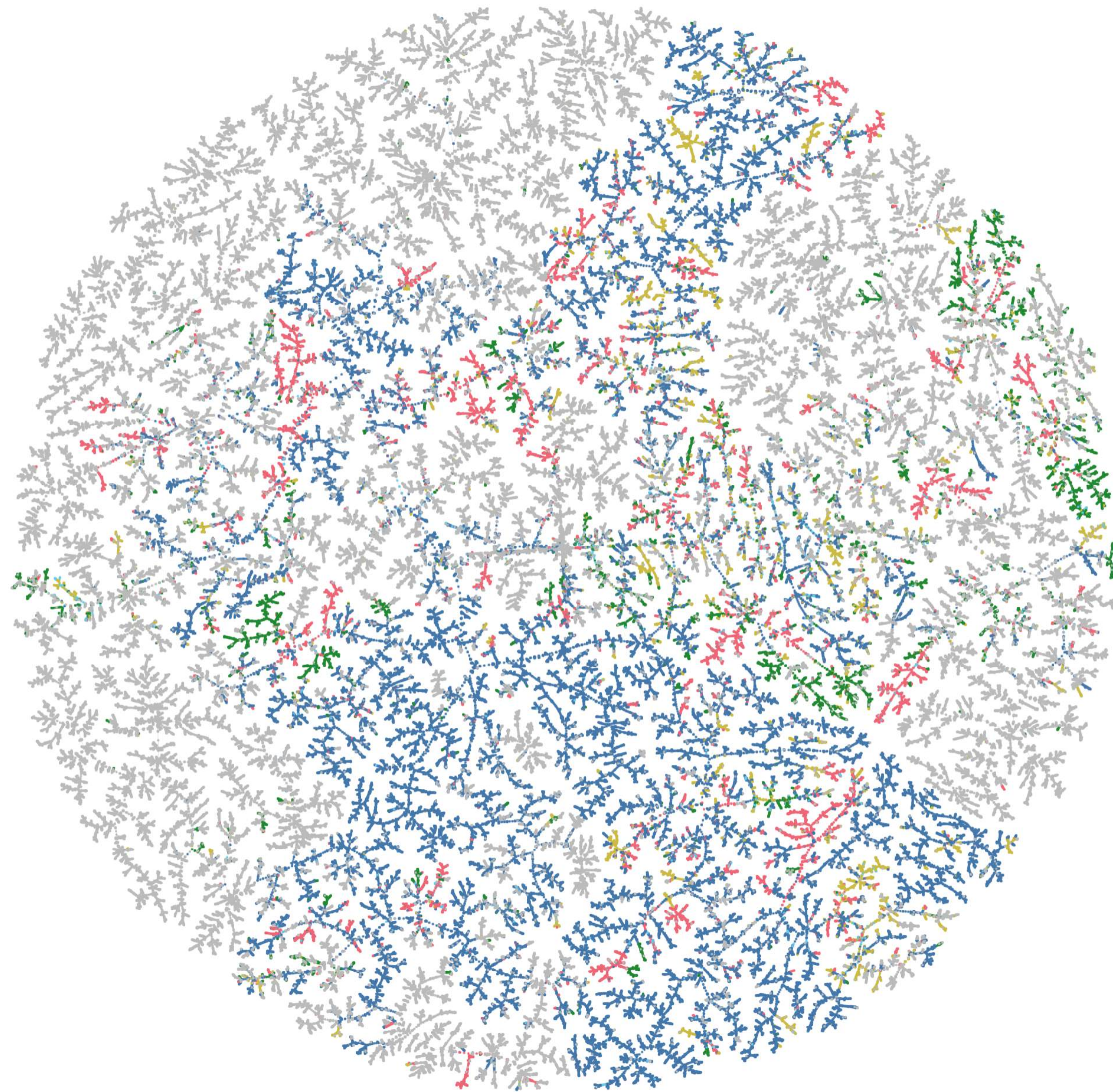


● Plantae ● Fungi ● Bacteria ● Animalia ● Shared

● CID ● HCD ● EAD ● UVPD ● Shared



# About available spectral and taxonomical data...



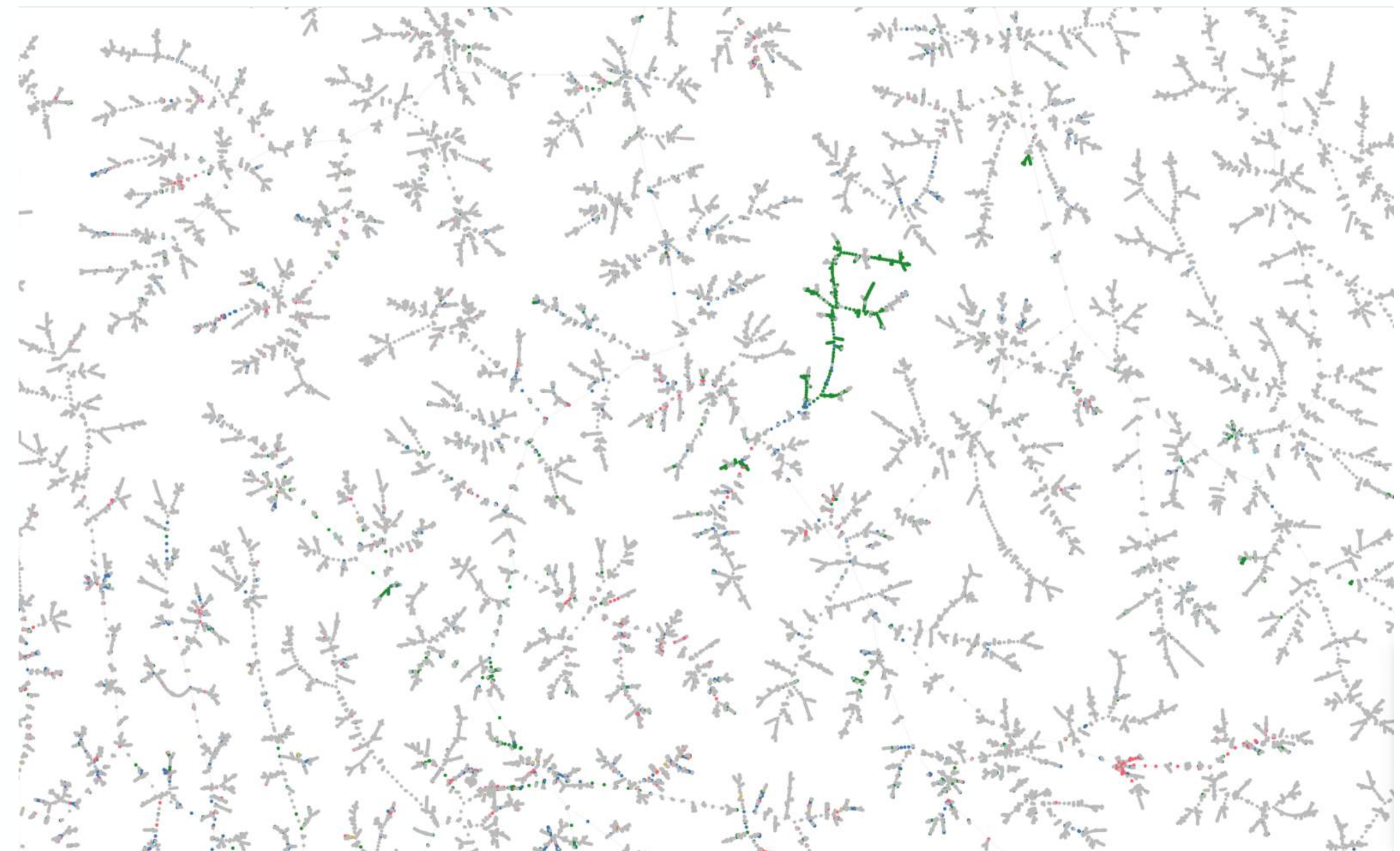
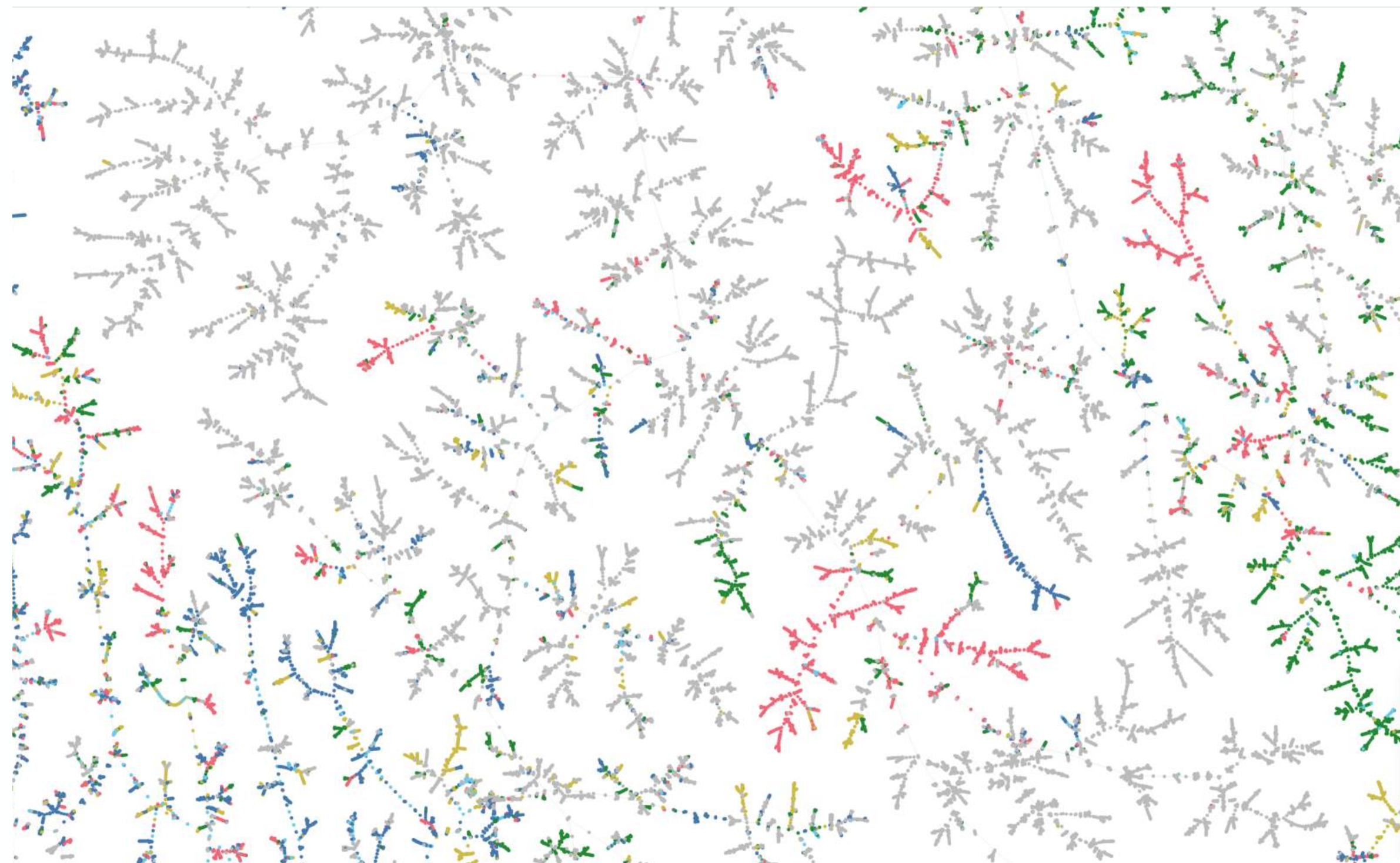
● Plantae ● Fungi ● Bacteria ● Animalia ● Shared

● CID ● HCD ● EAD ● UVPD ● Shared



# About available spectral and taxonomical data...

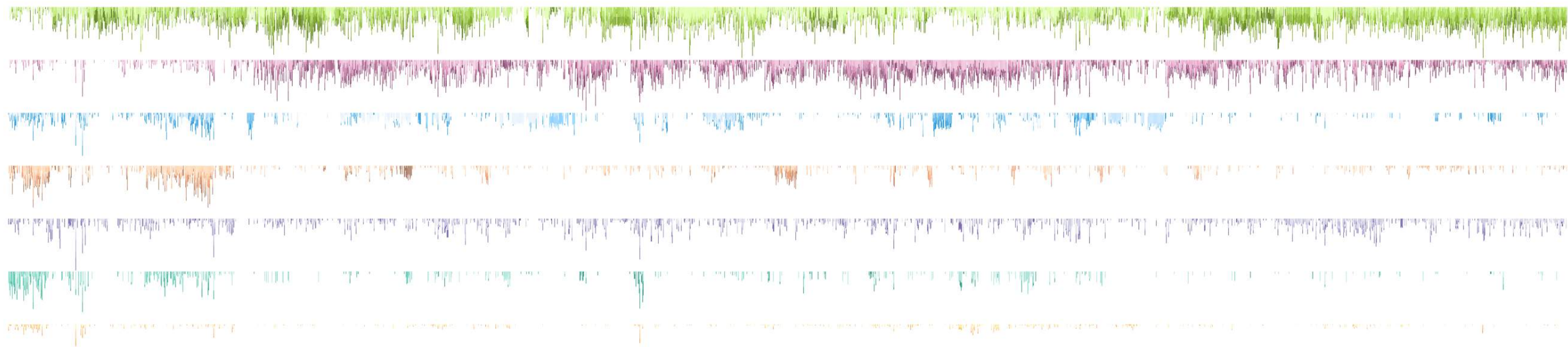
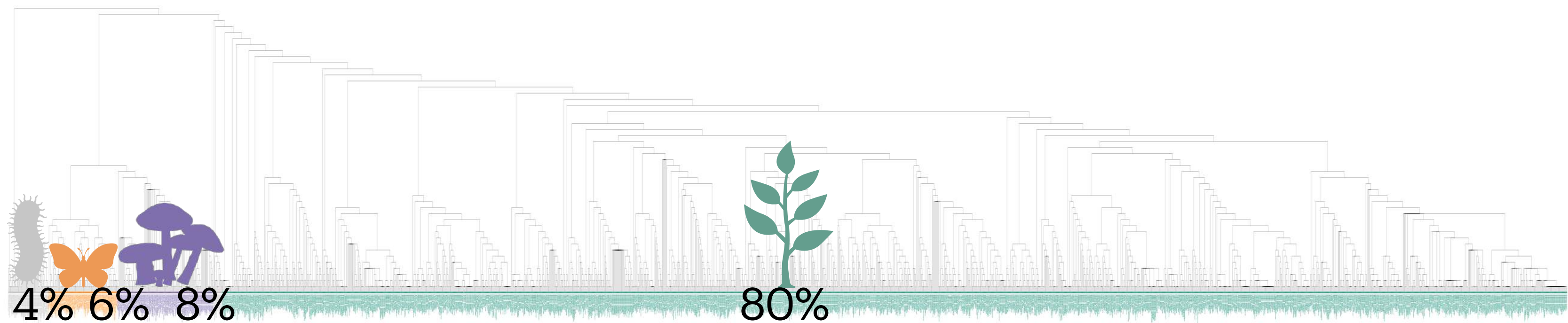
**Only unique 10,404 connectivities “found in taxon”**



● Plantae ● Fungi ● Bacteria ● Animalia ● Shared



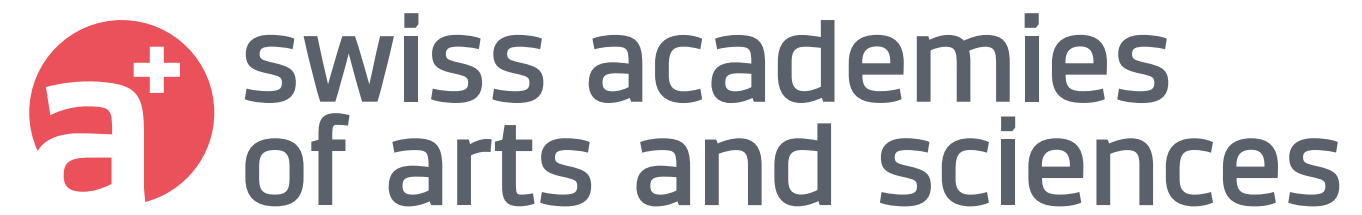
# Current status of the chemically investigated tree of Life



- Terpenoids
- Shikimates & Phenylpropanoids
- Alkaloids
- Polyketides
- Amino acids & Peptides
- Fatty acids
- Carbohydrates



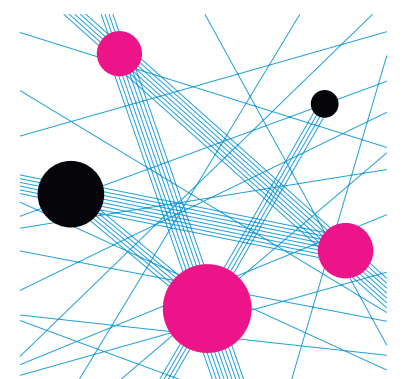
# Acknowledgements



Institute of Molecular Systems Biology



Prof. Nicola Zamboni



Contact



[rutz@imsb.biol.ethz.ch](mailto:rutz@imsb.biol.ethz.ch)



# Additional useful links

- **Wikidata:**

- [https://www.wikidata.org/wiki/Wikidata:WikiProject\\_Chemistry\\_Natural\\_products](https://www.wikidata.org/wiki/Wikidata:WikiProject_Chemistry_Natural_products)
- [https://www.wikidata.org/wiki/Wikidata:Request\\_a\\_query](https://www.wikidata.org/wiki/Wikidata:Request_a_query)

- **Qlever:** <https://qllever.cs.uni-freiburg.de/wikidata>

- **Scholia:** <https://scholia.toolforge.org/chemical>

- **IDSM:** <https://idsm.elixir-czech.cz/sparql>

- **SPARQL examples:**

- **LOTUS:** <https://lotus.nprod.net/lotus-sparql-examples>
- **EMI:** <https://www.earthmetabolome.org/sparql-examples>